

CONVERGENCE OF MUON WITH NEWTON–SCHULZ

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ABSTRACT

We analyze MUON as originally proposed and used in practice—using the momentum orthogonalization with a few NEWTON–SCHULZ steps. The prior theoretical results replace this key step in MUON with an exact SVD-based polar factor. We prove that MUON with NEWTON–SCHULZ converges to a stationary point at the same rate as the SVD-polar idealization, up to a constant factor for a given number q of NEWTON–SCHULZ steps. We further analyze this constant factor and prove that it converges to 1 doubly exponentially in q and improves with the degree of the polynomial used in NEWTON–SCHULZ for approximating the orthogonalization direction. We also prove that MUON removes the typical square-root-of-rank loss compared to its vector-based counterpart, SGD with momentum. Our results explain why MUON with a few low-degree NEWTON–SCHULZ steps matches exact-polar (SVD) behavior at a much faster wall-clock time and explain how much momentum matrix orthogonalization via NEWTON–SCHULZ benefits over the vector-based optimizer. Overall, our theory justifies the practical NEWTON–SCHULZ design of MUON, narrowing its practice–theory gap.

1 INTRODUCTION

Modern deep neural networks comprise billions of parameters and demand highly efficient training procedures. A persistent challenge is that most widely used optimizers—such as stochastic gradient descent (SGD) (Robbins & Monro, 1951) and adaptive methods such as Adam (Kingma & Ba, 2015)—operate on *vectorized* parameters, thereby discarding the native *matrix* structure present in linear layers and attention projections. Optimizers that explicitly respect matrix structure can, in principle, yield search directions that are better aligned with the underlying geometry while remaining computationally efficient at scale.

MUON (Jordan et al., 2024) is an optimizer designed for matrix-structured parameters. At each iteration, instead of following the raw momentum, MUON *orthogonalizes* the mntum matrix and then uses this orthogonalized direction to update the weights. In practice, this orthogonalization is not computed via an exact singular value decomposition (SVD)—which is accurate but expensive—but is approximated efficiently by a small, fixed number of NEWTON–SCHULZ steps. Empirical studies (Jordan et al., 2024; Liu et al., 2025a) have reported strong performance at scale with this SVD-free implementation, making MUON an attractive alternative to vector-based optimizers. Despite recent attempts to analyze the convergence of MUON (Shen et al., 2025; Li & Hong, 2025; Sato et al., 2025), theory still lags behind practice. Existing analyses typically study an *idealized* variant that replaces the NEWTON–SCHULZ step—central to practical MUON—with an exact polar step computed by SVD for analytical convenience. This leaves open whether the *actual* SVD-free orthogonalization used in practice—i.e., a finite number of NEWTON–SCHULZ steps—admits principled nonconvex convergence guarantees, and how the NEWTON–SCHULZ approximation impacts rank dependence and efficiency. Therefore, the following research questions remain open:

Research questions.

- Does MUON with NEWTON–SCHULZ admit nonconvex convergence guarantees, and how do its rates compare to the exact SVD–polar idealization?

MUON与NEWTON–SCHULZ的收敛性

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摘要

我们分析了MUON最初提出并在实践中使用的版本——使用动量正交化和少量NEWTON–SCHULZ步数。先前的理论结果用基于SVD的极化因子替换了MUON中的这个关键步骤。我们证明, 使用NEWTON–SCHULZ的MUON收敛到与SVD-极化理想化相同的驻点速率, 对于给定的 q 个NEWTON–SCHULZ步数, 其收敛速度仅相差一个常数因子。我们进一步分析了这个常数因子, 并证明它在 q 中双指数收敛到1, 并且随着NEWTON–SCHULZ中用于逼近正交化方向的多项式次数的提高而改善。我们还证明, 与基于向量的SGD动量版本相比, MUON消除了典型的秩的平方根损失。我们的结果表明, 使用少量低次NEWTON–SCHULZ步数的MUON为何能在更短的墙上时钟时间内匹配精确极化 (SVD) 行为, 以及NEWTON–SCHULZ的动量矩阵正交化相对于基于向量的优化器有多少优势。总体而言, 我们的理论证明了MUON的实用NEWTON–SCHULZ设计, 缩小了其实践与理论之间的差距。

1 简介

现代深度神经网络包含数十亿个参数, 需要高效的训练流程。一个持续的挑战是, 大多数广泛使用的优化器——如随机梯度下降 (SGD) (Robbins & Monro, 1951) 以及Adam (Kingma & Ba, 2015) 等自适应方法——在向量化参数上运行, 从而丢弃了线性层和注意力投影中存在的原生矩阵结构。那些明确尊重矩阵结构的优化器, 原则上可以产生与底层几何形状更好对齐的搜索方向, 同时在规模上保持计算效率。

MUON (乔丹等人, 2024) 是一种为矩阵结构参数设计的优化器。在每次迭代中, MUON 进行正交化动量矩阵, 然后使用这个正交化方向来更新权重。在实践中, 这种正交化不是通过精确的奇异值分解 (SVD) ——虽然准确但昂贵——来计算的, 而是通过少量固定的NEWTON–SCHULZ步数高效地近似计算的。实证研究 (乔丹等人, 2024; 刘等人, 2025a) 报告了这种无奇异值分解的实现在大规模上的强大性能, 使MUON成为基于向量的优化器的有吸引力的替代方案。尽管最近有尝试分析MUON的收敛性 (沈等人, 2025; 李 & 邓肯, 2025; 佐藤等人, 2025), 但理论仍然落后于实践。现有的分析通常研究一个理想化变体, 该变体用SVD计算的精确极步——这是实际MUON的核心——替换了NEWTON–SCHULZ步, 以便于分析。这留下了实际中使用的实际无奇异值分解正交化——即有限数量的NEWTON–SCHULZ步数——是否具有原则性的非凸收敛保证的问题, 以及NEWTON–SCHULZ近似如何影响秩依赖性和效率。因此, 以下研究问题仍然悬而未决:

研究问题。

- MUON与NEWTON–SCHULZ是否支持非凸收敛保证, 其收敛速度与精确SVD-极理想化相比如何?

- How does the NEWTON–SCHULZ steps q and NEWTON–SCHULZ polynomial degree κ control the accuracy–compute trade-off? In particular, how large is the gap caused by using NEWTON–SCHULZ, and how does that gap become negligible?
- Can we show that MUON converges faster than the vector-counterpart, SGD with momentum? What geometric mechanisms and rank dependence drive this gap?

To address these questions, we analyze MUON as originally proposed (Jordan et al., 2024) and as used in practice: MUON with momentum orthogonalization computed via NEWTON–SCHULZ. For nonconvex objectives, under standard smoothness assumptions, we prove the convergence of MUON to a stationary point, measured by the nuclear norm of the gradient. We establish that the convergence rate in the number of iterations matches the idealized (but not used in practice) SVD-based exact polar variant, up to a constant factor that depends on the polar approximation error ε_q (defined in Definition 3) for the fixed number of NEWTON–SCHULZ steps q . Moreover, we show that the polar approximation error ε_q shrinks doubly exponentially as q grows and decays with larger κ , which is the degree of the polynomial used in NEWTON–SCHULZ steps. Recursive updates using this polynomial allow the optimizer to find the approximated orthogonalization direction of the momentum matrix. Hence, with only a few NEWTON–SCHULZ steps, the convergence rate of MUON quickly tends to the convergence rate of the exact polar step via SVD. Consequently, our results imply that, because a few NEWTON–SCHULZ steps are far cheaper per iteration than SVD, the practical MUON implementation with NEWTON–SCHULZ attains substantially faster wall-clock convergence.

Our main contributions are summarized as follows:

- **The first convergence result of MUON with NEWTON–SCHULZ.** To our knowledge, we present the first nonconvex convergence guarantees for MUON with a finite number of NEWTON–SCHULZ steps (Theorem 1), as originally proposed and practically used. It is important to note that even for convex optimization, the convergence of MUON with NEWTON–SCHULZ has not been shown previously. The key distinction from the existing analyses of MUON is that we do not replace NEWTON–SCHULZ in the original MUON with the exact polar computed by SVD.
- **Analysis of polar approximation error and wall-clock convergence.** We prove that the polar approximation error ε_q due to using NEWTON–SCHULZ instead of SVD, in the MUON optimizer, decays doubly exponentially with the number of NEWTON–SCHULZ steps q and decays with the degree κ of the polynomial required to approximate the orthogonalization direction of the momentum matrix (Definition 2). Thus, even with a few steps of NEWTON–SCHULZ, the convergence of MUON with NEWTON–SCHULZ becomes arbitrarily close to that of the SVD-variant in the number of iterations (Theorems 2 and 4). Hence, given that per-iteration computation is much more efficient for NEWTON–SCHULZ steps compared to SVD, the overall convergence in wall-clock time is much faster for MUON with NEWTON–SCHULZ.
- **Sharper rank dependence in MUON with NEWTON–SCHULZ.** To prove the comparative advantage of MUON against the vector-based counterpart, we demonstrate for the first time that MUON with NEWTON–SCHULZ sharpens the convergence rate by a factor of the square root of the rank of the momentum matrix (see Table 1 and Theorem 3).

2 RELATED WORK

MUON and momentum orthogonalization. Jordan et al. (2024) introduced MUON, which orthogonalizes a momentum matrix via a few NEWTON–SCHULZ steps (SVD-free), and reported strong empirical results at LLM scale (Liu et al., 2025a). Earlier work orthogonalized gradients by SVD before applying momentum (Orthogonal-SGDM; Tuddenham et al. (2022)), whereas MUON applies momentum before orthogonalization and replaces SVD with NEWTON–SCHULZ, resulting in faster computation with only matrix multiplications. More details about MUON are described in 3.3.

Second-order preconditioners vs. MUON. Matrix-aware optimizers such as SHAMPOO (Gupta et al., 2018) SOAP (Vyas et al., 2024) and their variants (An et al., 2025) are *second-order preconditioners*: they maintain layerwise curvature (Kronecker-factored second moments) and periodically apply inverse-root preconditioning. By contrast, *Muon is not a second-order method*; it neither estimates nor inverts curvature but orthogonalizes the momentum matrix via a few Newton–Schulz

- NEWTON–SCHULZ步数 q 和NEWTON–SCHULZ多项式阶 κ 如何控制精度-计算权衡？特别是，使用NEWTON–SCHULZ造成的差距有多大，这个差距如何变得可以忽略？
- 我们能证明MUON比SGD（带动量）的向量对应版本收敛得更快吗？是什么几何机制和秩依赖性导致了这种差距？

为了解决这些问题，我们分析MUON，包括其原始提出版本（乔丹等人，2024）以及实际应用中的版本：带动量正交化的MUON，其正交化通过NEWTON–SCHULZ计算。对于非凸目标，在标准的平滑性假设下，我们证明了MUON收敛到驻点的收敛性，该驻点通过梯度核范数衡量。我们证明，迭代次数中的收敛率与理想化的（但在实践中未使用）基于SVD的精确极化变体相匹配，直到一个取决于极化近似误差 ε_q （定义3中定义）的常数因子，该误差针对固定的NEWTON–SCHULZ步数 q 。此外，我们表明极化近似误差 ε_q 随着 q 的增长呈双指数级缩小，并随着更大的 κ 衰减， κ 是NEWTON–SCHULZ步中使用的多项式的次数。使用此多项式的递归更新允许优化器找到动量矩阵的近似正交化方向。因此，只需少数NEWTON–SCHULZ步，MUON的收敛率就能迅速趋近于通过SVD的精确极化步的收敛率。因此，我们的结果表明，由于少数NEWTON–SCHULZ步远比SVD每迭代更便宜，实际使用NEWTON–SCHULZ的MUON实现实现了实质上更快的时钟收敛。

我们的主要贡献总结如下：

- **MUON与NEWTON–SCHULZ的首次收敛结果MUON与NEWTON–SCHULZ的收敛结果。**据我们所知，我们首次为MUON在有限数量的NEWTON–SCHULZ步数（定理1）下提供了非凸收敛保证，这与最初提出并在实践中使用的方案一致。重要的是要指出，即使对于凸优化，MUON与NEWTON–SCHULZ的收敛性之前也未被证明。与现有MUON分析的关键区别在于，我们并未用SVD计算出的精确极化来替换原始MUON中的NEWTON–SCHULZ。
- **极化近似误差和墙上时钟收敛的分析。**我们证明，由于在MUON优化器中使用NEWTON–SCHULZ而不是SVD，导致的极化近似误差 ε_q 随着NEWTON–SCHULZ步数 q 的增加呈双指数衰减，并且随着逼近动量矩阵正交化方向所需的多项式度数 κ （定义2）的增加而衰减。因此，即使只有少数NEWTON–SCHULZ步数，MUON与NEWTON–SCHULZ的收敛性在迭代次数上也会无限接近SVD变体（定理2和4）。因此，鉴于每次迭代的计算效率相比SVD对NEWTON–SCHULZ要高得多，MUON与NEWTON–SCHULZ的整体收敛速度在墙上时钟时间上要快得多。
- **M中更尖锐的秩依赖性MUON与NEWTON–SCHULZ。**为了证明MUON相对于基于向量的对应方法的比较优势，我们首次证明了MUON结合NEWTON–SCHULZ能够通过动量矩阵的秩的平方根因子来加速收敛率（参见表1和定理3）。

2 相关工作

MUON 和动量正交化。 乔丹等人（2024）提出了MUON，它通过几次NEWTON–SCHULZ步（无奇异值分解）对动量矩阵进行正交化，并在LLM规模上报告了强烈的经验结果（刘等人，2025a）。早期工作在应用动量之前通过SVD正交化梯度（Orthogonal-SGDM; Tuddenham等人（2022）），而MUON在正交化之前应用动量，并将SVD替换为NEWTON–SCHULZ，从而仅通过矩阵乘法实现更快的计算。有关MUON的更多细节在3.3中描述。

二阶预处理器与 MUON 对比。 像 SHAMPOO（Gupta 等人，2018 年）、SOAP（Vyas 等人，2024 年）及其变体（An 等人，2025 年）这样的矩阵感知优化器是二阶预处理器：它们保持逐层的曲率（Kronecker 分解的第二阶矩），并定期应用逆根预处理器。相比之下，*MUON* 不是一个二阶方法；它既不估计也不求逆曲率，而是通过几次 NEWTON–SCHULZ 正交化动量矩阵

iterations (SVD-free, using only matrix multiplications). These methods target different mechanisms—curvature preconditioning vs. projection/normalization—and can be complementary rather than directly comparable.

Practical efficiency of MUON. Large-scale training reports for MUON (Liu et al., 2025a; Shah et al., 2025; Tveit et al., 2025) and communication/memory-aware variants (Ahn & Dion, 2025; Liu et al., 2025b) motivate a theory that is SVD-free and GPU-aligned. Our analysis of NEWTON–SCHULZ, a key step in the MUON optimizer, adopts precisely that stance. The analysis provided in this work can be adapted to other variants of MUON.

Convergence analysis of MUON. Several recent analyses examine MUON but typically either idealize the orthogonalization by assuming an exact SVD polar step (Shen et al., 2025), work under Frobenius-smoothness with dimension-driven constants (Li & Hong, 2025), focus on stability/variant phenomena (Sato et al., 2025), or offer complementary lenses (steepest descent under norms, trust-region views, implicit constraints, or LMO/Frank-Wolfe formulations) without addressing NEWTON–SCHULZ step accuracy in nonconvex rates (Bernstein & Newhouse, 2024; Kovalev, 2025; Chen et al., 2025; Riabinin et al., 2025).

Key distinctions compared to the existing analyses of MUON. Prior work either assumes exact SVD polar steps, measures progress in a geometry that obscures rank benefits, or does not quantify the approximation from NEWTON–SCHULZ in MUON. Our work is the first to analyze how two key parameters—the number of NEWTON–SCHULZ steps and the degree of the polynomial used for finding the orthogonalization direction of the momentum matrix approximately—affect the convergence rate of MUON. The results indicate a nonconvex convergence rate to a stationary point, an explicit and rapidly vanishing constant factor derived from NEWTON–SCHULZ instead of SVD, and sharper rank dependence than SGD with momentum under the same metric. Overall, our work explains why MUON converges quickly (particularly in wall-clock time) as well as why only a few steps of NEWTON–SCHULZ suffice in practice.

3 PRELIMINARIES

3.1 NOTATIONS

For matrix $X \in \mathbb{R}^{m \times n}$, $X^\top$ is its transpose. For $X = (x_{ij}) \in \mathbb{R}^{n \times n}$, $\text{tr}(X) := \sum_{i=1}^n x_{ii}$. We write $\|X\|_*$, $\|X\|_{\text{op}}$, and $\|X\|_F$ for the nuclear, spectral (operator), and Frobenius norms, respectively, and $\langle X, Y \rangle_F := \text{tr}(X^\top Y)$. For a thin SVD $X = U\Sigma V^\top$, the *polar factor* is $\text{Polar}(X) := UV^\top$, a partial isometry with $\|\text{Polar}(X)\|_{\text{op}} \leq 1$ and $\langle X, \text{Polar}(X) \rangle_F = \|X\|_*$ (See Appendix A.1). For two sequences $\{a_n\}_{n=1}^\infty$ and $\{b_n\}_{n=1}^\infty$, $a_n = \mathcal{O}(b_n)$ implies that there exists a constant $C > 0$ such that $a_n \leq Cb_n$ holds for all $n \geq 1$. We use $\mathbb{E}[\cdot]$ for expectations over all algorithmic randomness.

3.2 PROBLEM SETTING: NONCONVEX OPTIMIZATION

We consider the stochastic optimization of a matrix-valued parameter: $W \in \mathbb{R}^{m \times n}$

$$\min_{W \in \mathbb{R}^{m \times n}} f(W) = \mathbb{E}_\xi[f(W; \xi)],$$

where the objective f is nonconvex with $f^* := \inf_W f(W) > -\infty$. We denote by $r := \min\{m, n\}$ the maximal possible rank of the matrix. At iteration t , a mini-batch $\xi_t = (\xi_{t,1}, \dots, \xi_{t,B})$ is drawn with $\{\xi_{t,i}\}_i$ i.i.d., and $\{\xi_t\}_t$ are independent across $t = 1, \dots, T$. At $t = 0$, the model parameter $W_0 \in \mathbb{R}^{m \times n}$ is initialized, and we define the initial sub-optimality as $D := f(W_0) - f^*$. In this paper, the following assumptions are made for the convergence analysis:

Assumption 1 (Lipschitz smoothness). *The objective function $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is continuously differentiable and L -Lipschitz smooth, i.e., for all $X, Y \in \mathbb{R}^{m \times n}$,*

$$\|\nabla f(X) - \nabla f(Y)\|_* \leq L \|X - Y\|_{\text{op}}.$$

We use smoothness with respect to the operator norm (and the nuclear norm as its dual).

Assumption 2 (Bounded variance). *We assume $\nabla f(W; \xi)$ is an unbiased stochastic estimator of the true gradient $\nabla f(W)$ and has bounded variance for all W and a single sample ξ , i.e.*

$$\mathbb{E}[\nabla f(W; \xi)] = \nabla f(W), \quad \mathbb{E}[\|\nabla f(W; \xi) - \nabla f(W)\|_F^2] \leq \sigma^2.$$

For a mini-batch of size B , the variance is at most σ^2/B .

迭代（无奇异值分解的，仅使用矩阵乘法）。这些方法针对不同的机制——曲率预处理器与投影/归一化——并且可以互补，而不是直接可比。

MUON的实用效率。 MUON (Liu等人, 2025a; Shah等人, 2025; Tveit等人, 2025) 和通信/内存感知变体 (Ahn & Dion, 2025; Liu等人, 2025b) 的大规模训练报告, 促使人们提出一种无奇异值分解 (SVD-free) 且与GPU兼容的理论。我们对NEWTON–SCHULZ的分析——MUON优化器的一个关键步骤——正是基于这种立场。本工作提供的新分析可以推广到MUON的其他变体。

MUON的收敛性分析。 最近有若干分析研究MUON, 但通常要么通过假设精确的SVD极步来理想化正交化 (Shen等人, 2025), 要么在Frobenius平滑性下使用维度驱动的常数 (Li & Hong, 2025), 要么关注稳定性/变体现象 (Sato等人, 2025), 要么提供互补视角 (如范数下的最速下降法、信任域视图、隐式约束或LMO/Frank-Wolfe公式), 但并未解决NEWTON–SCHULZ步在非凸情况下的精度问题 (Bernstein & Newhouse, 2024; Kovalev, 2025; Chen等人, 2025; Riabinin等人, 2025)。

与现有对MUON的对比分析相比的关键区别。 先前工作要么假设精确的SVD极步数, 要么在一个掩盖秩收益的几何空间中测量进度, 要么没有量化MUON中NEWTON–SCHULZ的近似效果。我们的工作首次分析两个关键参数——用于寻找动量矩阵正交化方向的近似多项式的阶数以及NEWTON–SCHULZ步数——如何影响MUON的收敛速度。结果表明, 向稳定点的收敛率是非凸的, 来自NEWTON–SCHULZ而不是SVD的显式且快速消失的常数因子, 以及在相同指标下比带动量的SGD更强的秩依赖性。总体而言, 我们的工作解释了为什么MUON收敛速度快 (尤其是在墙上时钟时间上), 以及为什么实践中只需少数几步NEWTON–SCHULZ就足够了。

3 预备知识

3.1 符号

对于矩阵 $X \in \mathbb{R}^{m \times n}$, $X^\top$ 是其转置。对于 $X = (x_{ij}) \in \mathbb{R}^{n \times n}$, $\text{tr}(X) := \sum_{i=1}^n x_{ii}$ 。我们分别用 $\|X\|_*$, $\|X\|_{\text{op}}$ 和 $\|X\|_F$ 表示核范数、谱 (算子) 范数和Frobenius范数, $\langle X, Y \rangle_F := \text{tr}(X^\top Y)$ 。对于一个薄 SVD $X = U\Sigma V^\top$, 极因子 $\text{Polar}(X) := UV^\top$ 是一个部分等距映射, 满足 $\|\text{Polar}(X)\|_{\text{op}} \leq 1$ 和 $\langle X, \text{Polar}(X) \rangle_F = \|X\|_*$ (参见附录A.1)。对于两个序列 $\{a_n\}_{n=1}^\infty$ 和 $\{b_n\}_{n=1}^\infty$, $a_n = \mathcal{O}(b_n)$ 意味着存在一个常数 $C > 0$, 使得对于所有 $n \geq 1$, $a_n \leq Cb_n$ 都成立。我们用 $\mathbb{E}[\cdot]$ 表示对所有算法随机性的期望。

3.2 问题设定: 非凸优化

我们考虑一个矩阵值参数的随机优化: $W \in \mathbb{R}^{m \times n}$

$$\min_{W \in \mathbb{R}^{m \times n}} f(W) = \mathbb{E}_\xi[f(W; \xi)],$$

其中目标函数 f 关于 f^* 是非凸的: $= \inf_W f(W) > -\infty$ 。我们用 $r := \min\{m, n\}$ 表示矩阵的最大可能秩。在迭代 t 时, 抽取一个大小为 $\xi_t = (\xi_{t,1}, \dots, \xi_{t,B})$ 的小批量 $\{\xi_{t,i}\}_i$ i.i.d., 且 $\{\xi_t\}_t$ 在 $t = 1, \dots, T$ 中相互独立。在 $t = 0$ 时, 模型参数 $W_0 \in \mathbb{R}^{m \times n}$ 被初始化, 我们定义初始次优性为 $D := f(W_0) - f^*$ 。在本文中, 为了收敛性分析, 我们做了以下假设:

假设1 (Lipschitz平滑性)。目标函数 $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 是连续可微的且 L -Lipschitz平滑的, 即对于所有 $X, Y \in \mathbb{R}^{m \times n}$,

$$\|\nabla f(X) - \nabla f(Y)\|_* \leq L \|X - Y\|_{\text{op}}.$$

我们使用算子范数 (以及核范数作为其共轭) 的平滑性。

假设2 (有界方差)。我们假设 $\nabla f(W; \xi)$ 是真实梯度 $\nabla f(W)$ 的无偏随机估计量, 并且对所有 W 和单个样本 ξ 的方差有界, 即

$$\mathbb{E}[\nabla f(W; \xi)] = \nabla f(W), \quad \mathbb{E}[\|\nabla f(W; \xi) - \nabla f(W)\|_F^2] \leq \sigma^2.$$

对于大小为 B 的小批量, 其方差最多为 σ^2/B 。

Assumptions 1 and 2 are standard for analyzing first-order methods in stochastic optimization (Boyd & Vandenberghe, 2004; Nemirovski et al., 2009; Candes & Recht, 2012; Shapiro et al., 2021; Reddi et al., 2018; Zou et al., 2019). L -smoothness is typically defined using a norm and its dual, and the specific operator-nuclear geometry chosen here is a natural fit when working with matrix parameters and polar or orthogonal updates (Nesterov, 2013; Beck, 2017; Jaggi, 2013). The bounded-variance assumption, where the mini-batch variance is σ^2/B , is a foundational concept for algorithms like SGD in both convex and nonconvex scenarios (Bottou et al., 2018; Ghadimi & Lan, 2013).

In this paper, the metric for the convergence rate is defined as follows:

Definition 1 (ϵ -stationary point). *We call $W \in \mathbb{R}^{m \times n}$ an ϵ -stationary point (in the nuclear norm) if $\mathbb{E}[\|\nabla f(W)\|_*] \leq \epsilon$. Equivalently, we say an algorithm attains ϵ -stationarity in T steps if*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \epsilon.$$

Note that when working with functions that have matrix inputs, using the nuclear norm to find a stationary point provides a more restrictive and precise condition than using the standard Frobenius norm. In other words, if a point satisfies the stationarity condition for the nuclear norm, it is guaranteed to also satisfy the condition for the Frobenius norm.

3.3 MUON ALGORITHM AND NEWTON–SCHULZ ORTHOGONALIZATION

Algorithm 1 MUON (with the illustration of NEWTON–SCHULZ orthogonalization)

Require: learning rate $\eta > 0$, momentum $\beta \in [0, 1)$, NEWTON–SCHULZ steps $q \in \mathbb{N}$, NEWTON–SCHULZ polynomial p_κ (degree κ), batch size B , total iteration T .

- 1: **Initialize:** $M_0 \leftarrow 0$, $W_0 \in \mathbb{R}^{m \times n}$
- 2: **for** $t = 1$ to T **do**
- 3: $G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_{t-1}; \xi_{t,i})$ ▷ Compute (batch) gradients
- 4: $M_t \leftarrow \beta M_{t-1} + G_t$
- 5: $X_{t,0} \leftarrow M_t / \alpha_t$ with $\alpha_t = \max\{1, \|M_t\|_F\}$ ▷ Pre-NEWTON–SCHULZ scaling
- 6: **for** $j = 1$ to q **do**
- 7: $X_{t,j} \leftarrow p_\kappa(X_{t,j-1} X_{t,j-1}^\top) X_{t,j-1}$ ▷ NEWTON–SCHULZ steps (Lines 6-9)
- 8: **end for**
- 9: $O_t \leftarrow X_{t,q}$
- 10: $W_t \leftarrow W_{t-1} - \eta O_t$ ▷ Update parameters
- 11: **end for**

For clarity of exposition, we present pseudocode for MUON with an explicit illustration of the NEWTON–SCHULZ steps (Lines 5–9) in Algorithm 1. Note that this is not a new algorithm; it is the original method of Jordan et al. (2024), here written in a more general mini-batch form with a step-by-step illustration of NEWTON–SCHULZ. Rather than orthogonalizing via SVD, MUON approximates the orthogonalization direction using only matrix multiplications; the key mechanism enabling this is the NEWTON–SCHULZ-based orthogonalization.

The key advantages of using NEWTON–SCHULZ for orthogonalization are as follows:

- NEWTON–SCHULZ makes MUON inversion-free and SVD-free. SVD is computationally expensive and makes each iteration costly. In contrast, the NEWTON–SCHULZ approach relies solely on matrix multiplications, yielding substantially better per-iteration efficiency—especially for large parameter matrices.
- NEWTON–SCHULZ is an iterative method that allows for precise control over the degree of orthogonality by adjusting the number of iterations. The number of NEWTON–SCHULZ steps provides a direct trade-off between computational cost and the degree of orthogonality, offering valuable flexibility.

3.4 NEWTON–SCHULZ POLYNOMIAL

In the MUON algorithm, at every iteration, a scaled momentum matrix X is orthogonalized via NEWTON–SCHULZ steps. First, the matrix $XX^\top$ is formed and is then passed to a polynomial

假设1和假设2是分析随机优化中一阶方法的标准方法 (Boyd & Vandenberghe, 2004; Nemirovski et al., 2009; Candes & Recht, 2012; Shapiro et al., 2021; Reddi et al., 2018; Zou et al., 2019)。 L -平滑性通常使用范数及其对偶来定义，此处选择的特定算子核几何在处理矩阵参数和极或正交更新时是一个自然的选择 (Nesterov, 2013; Beck, 2017; Jaggi, 2013)。有界方差假设，即小批量方差为 σ^2/B ，是像SGD这样算法在凸和非凸场景中的基础概念 (Bottou et al., 2018; Ghadimi & Lan, 2013)。

本文中，收敛速度的度量定义为如下：

定义1(ϵ -稳定点)。若 $W \in \mathbb{R}^{m \times n}$ 为 ϵ -稳定点（在核范数意义下），则 $\mathbb{E}[\|\nabla f(W)\|_*] \leq \epsilon$ 。等价地，我们称算法在 T 步内达到 ϵ -稳定性，当

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \epsilon.$$

注意，当处理具有矩阵输入的函数时，使用核范数寻找稳定点会提供比标准Frobenius范数更严格和精确的条件。换句话说，若某点满足核范数的稳定条件，则它必定也满足Frobenius范数的稳定条件。

3.3 MUON算法与NEWTON–SCHULZ正交化

算法1 MUON（包含NEWTON–SCHULZ正交化的示意图）

要求：学习率 $\eta > 0$, 动量 $\beta \in [0, 1)$, NEWTON–SCHULZ步数 $q \in \mathbb{N}$, NEWTON–SCHULZ多项式 p_κ (度数 κ), 批量大小 B , 总迭代次数 T .

- 1: **初始化:** $M_0 \leftarrow 0$, $W_0 \in \mathbb{R}^{m \times n}$
- 2: **for** $t = 1$ 到 T **do**
- 3: $G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_{t-1}; \xi_{t,i})$ ▷ 计算（批量）梯度
- 4: $M_t \leftarrow \beta M_{t-1} + G_t$
- 5: $X_{t,0} \leftarrow M_t / \alpha_t$ 使用 $\alpha_t = \max\{1, \|M_t\|_F\}$ ▷ NEWTON–SCHULZ预缩放
- 6: **for** $j = 1$ 到 q **do**
- 7: $X_{t,j} \leftarrow p_\kappa(X_{t,j-1} X_{t,j-1}^\top) X_{t,j-1}$ ▷ NEWTON–SCHULZ步数（第6-9行）
- 8: **结束for循环**
- 9: $O_t \leftarrow X_{t,q}$
- 10: $W_t \leftarrow W_{t-1} - \eta O_t$ ▷ 更新参数
- 11: **结束for循环**

为清晰起见，我们在算法1中展示了MUON的伪代码，并明确说明了NEWTON–SCHULZ步数（第5-9行）。请注意，这并非新算法；它是乔丹等人（2024）提出的方法，此处以更通用的形式呈现，并逐步说明了NEWTON–SCHULZ。MUON并非通过SVD进行正交化，而是仅使用矩阵乘法来近似正交化方向；实现这一机制的关键是NEWTON–SCHULZ正交化。

使用NEWTON–SCHULZ进行正交化的主要优势如下：

- NEWTON–SCHULZ使MUON无反演操作且无奇异值分解。奇异值分解计算成本高，导致每次迭代代价高昂。相比之下，NEWTON–SCHULZ方法仅依赖矩阵乘法，显著提高了每次迭代的效率——尤其对于大型参数矩阵而言。
- NEWTON–SCHULZ 是一种迭代方法，通过调整迭代次数，可以精确控制正交性的程度。NEWTON–SCHULZ 步数提供了计算成本与正交性程度之间的直接权衡，提供了宝贵的灵活性。

3.4 NEWTON–SCHULZ 多项式

在MUON算法中，每次迭代时，都会通过NEWTON–SCHULZ步数对缩放后的动量矩阵 X 进行正交化。首先，会形成矩阵 $XX^\top$ ，然后将其传递给一个多项式函数 X ，该函数的次数为 $XX^\top$ 。通过这个多项式的递归更新，使矩阵几乎正交，即。我们首先在定义2中定义了这个函数，并陈述了NEWTON–SCHULZ中使用的这个多项式的性质。

function p_κ with degree κ . Recursive updates by this polynomial make the matrix X nearly orthogonal, i.e., $XX^\top = I$. We first define this function p_κ in Definition 2 and state the properties of this polynomial used in NEWTON–SCHULZ.

Definition 2 (NEWTON–SCHULZ polynomial). *For degree $\kappa \in \mathbb{N}$, the NEWTON–SCHULZ polynomial is the Taylor truncation of $1/\sqrt{\lambda}$ at $\lambda = 1$, i.e.,*

$$p^{(s)}(1) = \left. \frac{d^s}{d\lambda^s} \lambda^{-1/2} \right|_{\lambda=1}$$

for $s = 1, \dots, \kappa$. The explicit form of the NEWTON–SCHULZ polynomial for degree κ is

$$p_\kappa(\lambda) = \sum_{s=0}^{\kappa} c_s (1-\lambda)^s, \quad c_s = \frac{(2s)!}{4^s (s!)^2} > 0.$$

Equivalently, with reparametrization $u = 1 - \lambda \in [0, 1]$, $p_\kappa(1-u) = \sum_{s=0}^{\kappa} c_s u^s$.

Proposition 1 (Properties of p_κ). *For $\lambda \in [0, 1]$:*

- **Positivity.** $p_\kappa(\lambda) > 0$ and $p_\kappa(\lambda) \geq 1$ with equality iff $\lambda = 1$.
- **Monotonicity of τ .** Let $\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$, then we have τ non-decreasing on $[0, 1]$ and $\tau(1) = 1$.

Consequently, for any symmetric $A \succeq 0$ with spectrum in $[0, 1]$, the NEWTON–SCHULZ update $A \mapsto p_\kappa(A)Ap_\kappa(A)$ satisfies $\|p_\kappa(A)Ap_\kappa(A)\|_{\text{op}} \leq 1$: NEWTON–SCHULZ steps preserve the unit spectral ball (see Appendix A.3). Moreover, the property of the function τ is used when proving how fast does one step of NEWTON–SCHULZ make the momentum orthogonal (Lemma 2).

In order to quantify the degree of orthogonality of the output matrix $X_{t,q}$ after q steps of NEWTON–SCHULZ, and to measure the approximation error derived from NEWTON–SCHULZ compared to the exact-polar method via SVD under operator-norm, we define the following:

Definition 3 (Orthogonality residual and polar approximation error). *For fixed t , let Π_t be the orthogonal projector onto $\text{range}(M_t)$. With $\{X_{t,j}\}_{j=0}^q$ from Algorithm 1, define the orthogonality residual $\delta_{t,j}$ and the polar approximation error $\varepsilon_{t,q}$ by*

$$\delta_{t,j} := \|\Pi_t - X_{t,j}X_{t,j}^\top\|_{\text{op}} \in [0, 1), \quad \varepsilon_{t,q} := \|X_{t,q} - \text{Polar}(M_t)\|_{\text{op}}.$$

Define $\varepsilon_q := \sup_t \varepsilon_{t,q}$ and $\delta_0 := \sup_t \delta_{t,0}$.

Remark 1. In MUON, there is a scaling step for the momentum matrix before applying it to the recursive update by the NEWTON–SCHULZ polynomial (Line 5 in Algorithm 1). This scaling ensures $\|X_{t,0}\|_{\text{op}} \leq 1$, which is required to apply the NEWTON–SCHULZ polynomial properties described in Proposition 1. In parallel, the initial residual is strictly less than 1, i.e., $\delta_{t,0} \in [0, 1)$ for every iteration t (see Appendix D).

4 MAIN RESULTS

We begin by stating our two main theorems: (i) a nonconvex convergence rate for MUON with a finite number of NEWTON–SCHULZ steps (Theorem 1); and (ii) an explicit bound on the multiplicative constant induced by NEWTON–SCHULZ, together with its (doubly) exponential decay in q (Theorem 2). We then provide brief proof sketches for both theorems in Sections 4.3 and 4.4. For comparisons, Section 4.5 also presents convergence rates for the idealized MUON with an exact SVD-based polar step and for SGD with momentum—the vector-based baseline—stated under the same nuclear-norm stationarity metric.

4.1 CONVERGENCE OF MUON (WITH NEWTON–SCHULZ)

Theorem 1 (Convergence of MUON with NEWTON–SCHULZ). *Suppose Assumptions 1 and 2 hold, and run MUON (Algorithm 1) with initialization $W_0 \in \mathbb{R}^{m \times n}$. Choose the stepsize and momentum as $\eta = \sqrt{\frac{(1-\beta)D}{TL}}$ and $\beta = 1 - \min\left\{\frac{\sqrt{LDB}}{\sigma\sqrt{rT}}, 1\right\}$ where $r = \min\{m, n\}$. Then there exists a factor $\chi_q > 0$, depending only on the number q of NEWTON–SCHULZ steps, such that*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \chi_q \cdot \mathcal{O}\left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT}\right)^{1/4}\right)$$

这个多项式函数 p_κ 的次数为 κ 。通过这个多项式的递归更新，使矩阵 X 几乎正交，即 $XX^\top = I$ 。我们首先在定义2中定义了这个函数 p_κ ，并陈述了NEWTON–SCHULZ中使用的这个多项式的性质。

定义2 (NEWTON–SCHULZ polynomial). 对于度数 $\kappa \in \mathbb{N}$ ，NEWTON–SCHULZ 多项式是 $1/\sqrt{\lambda}$ 在 $\lambda = 1$ 处的泰勒级数截断，即，

$$p^{(s)}(1) = \left. \frac{d^s}{d\lambda^s} \lambda^{-1/2} \right|_{\lambda=1}$$

对于 $s = 1, \dots, \kappa$ 。度数 κ 的NEWTON–SCHULZ 多项式的显式形式是

$$p_\kappa(\lambda) = \sum_{s=0}^{\kappa} c_s (1-\lambda)^s, \quad c_s = \frac{(2s)!}{4^s (s!)^2} > 0.$$

等价地，使用重新参数化 $u = 1 - \lambda \in [0, 1]$, $p_\kappa(1-u) = \sum_{s=0}^{\kappa} c_s u^s$ 。

命题1 (性质 p_κ)。对于 $\lambda \in [0, 1]$:

- **Positivity.** $p_\kappa(\lambda) > 0$ and $p_\kappa(\lambda) \geq 1$ with equality iff $\lambda = 1$.
- **单调性 τ .** 设 $\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$ ，则我们有 τ 在 $[0, 1]$ 上非递减，且 $\tau(1) = 1$ 。

因此，对于任何谱在 $[0, 1]$ 中的对称 $A \succeq 0$ ，NEWTON–SCHULZ更新 $A \mapsto p_\kappa(A)Ap_\kappa(A)$ 满足：NEWTON–SCHULZ步骤保持单位谱球（参见附录A.3）。此外，在证明NEWTON–SCHULZ一步如何使动量正交时，使用了函数 τ 的性质（引理2）。

为了量化输出矩阵 X_t 在经过 q 步NEWTON–SCHULZ后的正交程度，并通过SVD在算子范数下测量NEWTON–SCHULZ与精确极化方法之间的近似误差，我们定义如下：

定义3 (正交残差和极化近似误差)。对于固定的 t ，设 Π_t 是到 $\text{range}(M_t)$ 的正交投影。根据算法1中的 $\{X_{t,j}\}_{j=0}^q$ ，定义正交残差 δ_t 和极化近似误差 ε_t 。

$$\delta_{t,j} := \|\Pi_t - X_{t,j}X_{t,j}^\top\|_{\text{op}} \in [0, 1), \quad \varepsilon_{t,q} := \|X_{t,q} - \text{Polar}(M_t)\|_{\text{op}}.$$

定义 $\varepsilon_q := \sup_t \varepsilon_{t,q}$ 和 $\delta_0 := \sup_t \delta_{t,0}$ 。

备注 1. 在 MUON 中，动量矩阵在应用于 NEWTON–SCHULZ 多项式（算法1的第5行）的递归更新之前有一个缩放步骤。这个缩放确保 $\|X_{t,0}\|_{\text{op}} \leq 1$ ，这是应用命题1中描述的 NEWTON–SCHULZ 多项式性质所必需的。同时，初始残差严格小于1，即 $\delta_{t,0} \in [0, 1)$ 对于每次迭代 t （参见附录D）。

4 主要结果

我们首先陈述我们的两个主要定理：(i) MUON 在有限个 NEWTON–SCHULZ 步数下的非凸收敛率（定理1）；以及 (ii) NEWTON–SCHULZ 引入的乘法常数的显式界限及其在 q 中的（双重）指数衰减（定理2）。然后我们在第4.3节和第4.4节中分别提供这两个定理的简要证明。为了比较，第4.5节还展示了具有精确 SVD 基于极步骤的理想化 MUON 的收敛率，以及带有动量的 SGD（基于向量的基线）的收敛率——在相同的核范数平稳性度量下陈述。

4.1 MUON 与 NEWTON–SCHULZ 的收敛

定理1 (μ 子与牛顿–舒尔茨的收敛性)。假设满足假设1和假设2，并运行 μ 子 (算法1)，初始化为 $W_0 \in \mathbb{R}^{m \times n}$ 。选择步长和动量为 $\eta = \sqrt{\frac{(1-\beta)D}{TL}}$ 和 $\beta = 1 - \min\left\{\frac{\sqrt{LDB}}{\sigma\sqrt{rT}}, 1\right\}$ ，其中 $r = \min\{m, n\}$ 。那么存在一个因子 $\chi_q > 0$ ，它仅取决于 NEWTON–SCHULZ 步数 q ，使得

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \chi_q \cdot \mathcal{O}\left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT}\right)^{1/4}\right)$$

Consequently, MUON with NEWTON–SCHULZ attains ϵ -stationarity with an iteration complexity of $T = \mathcal{O}\left(\max\left\{\frac{\chi_q^2 LD}{\epsilon^2}, \frac{\chi_q^2 r^2 \sigma^2}{B\epsilon^2}, \frac{\chi_q^4 r \sigma^2 LD}{B\epsilon^4}\right\}\right)$ iterations.

Discussions of Theorem 1. Theorem 1 guarantees that MUON with NEWTON–SCHULZ converges to an ϵ -stationary point. To the best of our knowledge, this convergence guarantee is the first result for MUON with NEWTON–SCHULZ. Moreover, as shown later by comparison with the SVD-based polar variant, the iteration complexity of MUON with NEWTON–SCHULZ matches the exact-polar rate up to a multiplicative factor χ_q that depends on the polar-approximation error ε_q (e.g., Table 1). Crucially, we show later in Theorem 2 that $\chi_q \rightarrow 1$ converges at an exponential rate in the number of NEWTON–SCHULZ steps q , so the convergence gap (in the number of iterations) to the ideal SVD-polar rate can be made arbitrarily small. Since each NEWTON–SCHULZ step is substantially cheaper than an SVD, these results provide the first theoretical explanation for the superior practical performance observed for the original (SVD-free) MUON.

4.2 DECAY RATE OF ε_q AND CONVERGENCE RATE OF $\chi_q \rightarrow 1$

We now quantify how fast χ_q approaches 1. For a given number q of NEWTON–SCHULZ steps, we can show that the polar-approximation error ε_q decays *doubly exponentially* in q (with faster decay for larger κ). Since χ_q is controlled by ε_q , it follows that $\chi_q \rightarrow 1$ is at the same rate. The following theorem formalizes this result.

Theorem 2 (Upper-bounds on ε_q and χ_q). *For the NEWTON–SCHULZ polynomial with degree κ and for any t , $\delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}$. Hence, the bound of the polar approximation error ε_q and the factor χ_q occurred by NEWTON–SCHULZ is*

$$\varepsilon_q \leq 1 - \sqrt{1 - \delta_0^{(\kappa+1)^q}} \leq \delta_0^{(\kappa+1)^q}, \quad \chi_q = \frac{1}{1 - \varepsilon_q} \leq \frac{1}{\sqrt{1 - \delta_0^{(\kappa+1)^q}}},$$

where $\delta_0 := \sup_t \delta_{t,0} < 1$.

Discussion of Theorem 2. The theorem shows that ε_q is bounded by $\delta_0^{(\kappa+1)^q}$. Hence, ε_q vanishes *doubly exponentially* in q (and improves with larger κ). Therefore, $\chi_q \rightarrow 1$ at the same doubly exponential rate. Together with Theorem 1, this result implies that the iteration-complexity gap between NEWTON–SCHULZ and the idealized SVD-polar update becomes negligible after only a few NEWTON–SCHULZ steps.

Practical implication. A finite number of NEWTON–SCHULZ steps yields iteration complexity essentially indistinguishable from exact SVD updates (up to the factor $\chi_q \rightarrow 1$ doubly exponentially fast), while dramatically reducing per-iteration cost by using only matrix multiplications.

4.3 PROOF SKETCH FOR THEOREM 1.

We briefly outline the main ideas. First, introduce the scaled momentum $N_t = (1 - \beta)M_t$ (faithfully following the original update rule), so the EMA becomes $N_t \leftarrow \beta N_{t-1} + (1 - \beta)G_t$. Next, apply the *descent lemma* (Lemma 6) to the update $W_t \leftarrow W_{t-1} - \eta O_t$, which yields a term of the form $\langle \nabla f(W_{t-1}), O_t \rangle_F$. Decompose this inner product as

$$\langle \nabla f(W_{t-1}), O_t \rangle_F = \langle N_t, O_t \rangle_F + \langle \nabla f(W_{t-1}) - N_t, O_t \rangle_F,$$

thereby isolating the momentum mismatch. Prior analyses typically stop here and average over iterations. In contrast, we further split $\langle N_t, O_t \rangle_F$ as

$$\langle N_t, O_t \rangle_F = \langle N_t, P_t \rangle_F + \langle N_t, O_t - P_t \rangle_F,$$

separating the exact polar factor P_t from the NEWTON–SCHULZ orthogonalizer (the output matrix of the NEWTON–SCHULZ routine). As we define the *polar approximation error* ε_q as the discrepancy between the exact polar factor $P_t = \text{Polar}(N_t) = \text{Polar}(M_t)$ and the actual step O_t produced by q steps of NEWTON–SCHULZ, we can control this part with respect to ε_q . This yields a one-step descent inequality for MUON that explicitly includes the NEWTON–SCHULZ-induced error ε_q . Averaging this inequality over $t = 1, \dots, T$ and choosing η and β as specified in Theorem 1 produces the stated convergence rate. Full details appear in Appendix B.

因此， μ 子与牛顿-舒尔茨在 ϵ -平稳状态下达到，其迭代复杂度为 $T = \mathcal{O}\left(\max\left\{\frac{\chi_q^2 LD}{\epsilon^2}, \frac{\chi_q^2 r^2 \sigma^2}{B\epsilon^2}, \frac{\chi_q^4 r \sigma^2 LD}{B\epsilon^4}\right\}\right)$ 次迭代。

关于定理1的讨论。 定理1保证了MUON结合NEWTON–SCHULZ能够收敛到一个 ϵ -不动点。据我们所知，这是MUON结合NEWTON–SCHULZ的第一个收敛性结果。此外，正如后面通过与基于SVD的极化变体进行比较所展示的，MUON结合NEWTON–SCHULZ的迭代复杂度与精确极化率在乘性因子 χ_q 上相匹配，该因子取决于极化近似误差 ε_q （例如，表1）。关键在于，我们在定理2中证明， $\chi_q \rightarrow 1$ 在NEWTON–SCHULZ步数 q 上以指数速率收敛，因此与理想SVD-极化率的收敛差距（迭代次数上的差距）可以任意小。由于每个NEWTON–SCHULZ步的成本远低于SVD，这些结果为原始（无奇异值分解的）MUON所观察到的优异实际性能提供了第一个理论解释。

4.2 ε_q 衰变率与 $\chi_q \rightarrow 1$ 收敛率

我们现在量化 χ_q 接近1的速度。对于给定的NEWTON–SCHULZ步数 q ，我们可以证明极化近似误差 ε_q 在 q 中呈双指数衰减（对于更大的 κ 衰减更快）。由于 χ_q 受控于 ε_q ，因此 $\chi_q \rightarrow 1$ 以相同的速率变化。以下定理将这一结果形式化。

定理2 (关于 ε_q 和 χ_q 的上界)。**对于度数**为的NEWTON–SCHULZ多项式**以及任何**,, $q \leq \delta_{t,0}^{(\kappa+1)^q}$ 。因此，极化近似误差 ε_q 的上界以及由NEWTON–SCHULZ产生的因子 χ_q 为

$$\varepsilon_q \leq 1 - \sqrt{1 - \delta_0^{(\kappa+1)^q}} \leq \delta_0^{(\kappa+1)^q}, \quad \chi_q = \frac{1}{1 - \varepsilon_q} \leq \frac{1}{\sqrt{1 - \delta_0^{(\kappa+1)^q}}},$$

其中 $\delta_0 := \sup_t \delta_{t,0} < 1$ 。

定理2的讨论。 该定理表明 ε_q 被 $\delta_0^{(\kappa+1)^q}$ 界定。因此， ε_q 在 q 中呈双指数消失双指数（随着更大的 κ 而改善）。因此， $\chi_q \rightarrow 1$ 以相同的双指数速率变化。结合定理1，这一结果意味着NEWTON–SCHULZ与理想化SVD-极化更新之间的迭代复杂度差距在仅经过少数NEWTON–SCHULZ步数后变得可以忽略不计。

实际意义。 有限次的 NEWTON–SCHULZ 步数即可获得迭代复杂度，其与精确 SVD 更新（在因子 $\chi_q \rightarrow 1$ 的双指数速度下）本质上无法区分，同时通过仅使用矩阵乘法大幅降低了每次迭代的成本。

4.3 定理 1 的证明概要。

我们简要概述主要思想。首先，引入缩放动量 $N_t = (1 - \beta)M_t$ （严格遵循原始更新规则），因此 EMA 变为 $N_t \leftarrow \beta N_{t-1} + (1 - \beta)G_t$ 。接下来，将下降引理（引理 6）应用于更新 $W_t \leftarrow W_{t-1} - \eta O_t$ ，得到形式为 $\langle \nabla f(W_{t-1}), O_t \rangle_F$ 的项。将此内积分解为

$$\langle \nabla f(W_{t-1}), O_t \rangle_F = \langle N_t, O_t \rangle_F + \langle \nabla f(W_{t-1}) - N_t, O_t \rangle_F,$$

从而隔离出动量不匹配。先前分析通常在此停止并迭代求平均。相比之下，我们进一步将 $\langle N_t, O_t \rangle_F$ 分解为

$$\langle N_t, O_t \rangle_F = \langle N_t, P_t \rangle_F + \langle N_t, O_t - P_t \rangle_F,$$

将精确极因子 P_t 与 NEWTON–SCHULZ 正交化器（即 NEWTON–SCHULZ 程序的输出矩阵）分离。由于我们将极化近似误差 ε_q 定义为精确极因子 $P_t = \text{Polar}(N_t) = \text{Polar}(M_t)$ 与 NEWTON–SCHULZ 实际产生的步骤 O_t q 步之间的差异，因此我们可以相对于 ε_q 控制这部分。这为 MUON 产生了一个包含 NEWTON–SCHULZ 诱导误差 ε_q 的一步下降不等式。将此不等式在 $t = 1, \dots, T$ 上平均，并根据定理1中指定的 η 和 β ，即可得到所述收敛率。完整细节见附录B。

Table 1: Comparison of convergence rates.

Method	Convergence rate
SGD with momentum (Theorem 3)	$\mathcal{O}\left(\sqrt{\frac{rLD}{T}} + \left(\frac{r^2\sigma^2LD}{BT}\right)^{1/4}\right)$
MUON with SVD (Theorem 4)	$\mathcal{O}\left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2LD}{BT}\right)^{1/4}\right)$
MUON (Theorem 1)	$\chi_q \cdot \mathcal{O}\left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2LD}{BT}\right)^{1/4}\right)$

$D = f(W_0) - f^*$, $r = \min\{m, n\}$, iterations T , batch size B , Lipschitz constant L , variance bound σ^2 . The factor χ_q converges to 1 at exponential rate in step q . (Theorem 2): $\chi_q \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}$, where q is the number of NEWTON-SCHULZ step and κ is the degree of the NEWTON-SCHULZ polynomial (Def. 2).

4.4 PROOF SKETCH FOR THEOREM 2

We outline how Theorem 2 follows; full proofs appear in Appendix D. Recall that in Theorem 1, the convergence rate contains the polar-approximation error ε_q and the multiplicative factor χ_q depending on ε_q . To bound the resulting multiplicative factor χ_q , we relate ε_q to an *orthogonality residual* that quantifies how close the NEWTON-SCHULZ iterate is to having orthonormal columns. Concretely, letting $X_{t,q}$ be the output matrix after q NEWTON-SCHULZ steps applied to the (scaled) momentum at iteration t , define $\delta_{t,q} := \|I - X_{t,q}X_{t,q}^\top\|_{\text{op}}$.

The next lemma provides the spectral link between the residual and the polar-approximation error:

Lemma 1 (Orthogonality residual vs. Polar approximation error). *Let $\lambda_{\min}^+$ be the smallest positive eigenvalue of $X_{t,q}X_{t,q}^\top$ restricted to $\text{range}(M_t)$ (set $\lambda_{\min}^+ = 1$ if $\text{rank}(M_t) = 0$). Then*

$$\delta_{t,q} = 1 - \lambda_{\min}^+, \quad \varepsilon_{t,q} = 1 - \sqrt{\lambda_{\min}^+} = 1 - \sqrt{1 - \delta_{t,q}}.$$

Next, we describe how a single NEWTON-SCHULZ step transforms the residual through the degree- κ polynomial p_κ :

Lemma 2 (Residual update). *For NEWTON-SCHULZ polynomial p_κ , the orthogonality residual $\delta_{t,j}$ is updated by NEWTON-SCHULZ per step as*

$$\delta_{t,j+1} = \phi(\delta_{t,j}),$$

where $\phi(u) := 1 - (1 - u)[p_\kappa(1 - u)]^2 = 1 - \tau(1 - u)$.

To prove Lemma 2, we introduce $\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$, note that τ is non-decreasing on $[0, 1]$ and satisfies $\tau(1) \leq 1$ (Proposition 1), and then translate this monotonicity to the residual map ϕ .

Finally, we obtain a contraction bound and its multi-step consequence:

Lemma 3 (Residual decay by NEWTON-SCHULZ polynomial). *For NEWTON-SCHULZ polynomial p_κ , $\phi(u) \leq u^{\kappa+1}$ on $[0, 1]$ where ϕ is a function defined in Lemma 2. Hence, for every t and all $j \geq 0$,*

$$\delta_{t,j+1} \leq \delta_{t,j}^{\kappa+1}, \quad \delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}.$$

4.5 COMPARISONS WITH SGD WITH MOMENTUM AND MUON WITH SVD

To enable a fair comparison with the original MUON with NEWTON-SCHULZ method, we also establish convergence guarantees for two baselines: (i) *SGD with momentum* and (ii) the *idealized MUON with an exact polar step computed by SVD*. All results are derived under the same smoothness and bounded-variance assumptions, and progress is measured by the nuclear norm of the gradient, so the rates are directly comparable.

The convergence bound in Theorem 3 for *SGD with momentum* (Theorem 3) is, to our knowledge, new and may be of independent interest. It provides a clean reference point for vector-based updates under the same stationarity metric.

表1: 收敛速度比较

方法	收敛速度
带动量的SGD (定理3)	$\mathcal{O}\left(\sqrt{\frac{rLD}{T}} + \left(\frac{r^2\sigma^2LD}{BT}\right)^{1/4}\right)$
带SVD的μ子 (定理4)	$\mathcal{O}\left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2LD}{BT}\right)^{1/4}\right)$
MUON (定理1)	$\chi_q \cdot \mathcal{O}\left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2LD}{BT}\right)^{1/4}\right)$

$D = f(W_0) - f^*$, $r = \min\{m, n\}$, 迭代 T , 批量大小 B , 利普希茨常数 L , 方差界限 σ^2 因数 χ_q 在步骤 q 以指数速率收敛到 1。 (定理2) : $\chi_q \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}$, 其中 q 是 NEWTON-SCHULZ 步数的数量, κ 是 NEWTON-SCHULZ 多项式的次数 (定义2)。

4.4 定理2的证明概要

我们概述了定理2的推导过程; 完整证明见附录D。回想在定理1中, 收敛率包含极化近似误差 ε_q 和依赖于 ε_q 的乘法因子 χ_q 。为界定所得乘法因子 χ_q , 我们将 ε_q 与一个量化 NEWTON-SCHULZ迭代接近正交列的程度正交残差联系起来。具体而言, 设 $X_{t,q}$ 为在迭代 t 的 (缩放) 动量上应用 q 步 NEWTON-SCHULZ 后的输出矩阵, 定义 $\delta_{t,q} = \|I - X_{t,q}X_{t,q}^\top\|_{\text{op}}$ 。

The 下一个引理提供了残差与极化近似误差之间的谱联系:

引理1 (正交残差与极化近似误差)。 设 $\lambda_{\min}^+$ 是 $X_{t,q}X_{t,q}^\top$ 在 $\text{range}(M_t)$ (如果 $\text{rank}(M_t) = 0$, 则设 $\lambda_{\min}^+ = 1$) 的最小正特征值。那么

$$\delta_{t,q} = 1 - \lambda_{\min}^+, \quad \varepsilon_{t,q} = 1 - \sqrt{\lambda_{\min}^+} = 1 - \sqrt{1 - \delta_{t,q}}.$$

接下来, 我们描述单个 NEWTON-SCHULZ 步骤如何通过度- κ 多项式 p_κ 转换残差:

引理2 (残差更新)。对于 NEWTON-SCHULZ 多项式 p_κ , 正交残差 $\delta_{t,j}$ 按步由 NEWTON-SCHULZ 更新,

$$\delta_{t,j+1} = \phi(\delta_{t,j}),$$

其中 $\phi(u) := 1 - (1 - u)[p_\kappa(1 - u)]^2 = 1 - \tau(1 - u)$ 。

为证明引理2, 我们引入 $\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$, 注意到 τ 在 $[0, 1]$ 上单调不减且满足 $\tau(1) \leq 1$ (命题1), 然后将这一单调性转化为残差映射 ϕ 。

最后, 我们得到一个收缩界及其多步推论:

引理3 (残差通过 NEWTON-SCHULZ 多项式衰减)。对于 NEWTON-SCHULZ 多项式 p_κ , $\phi(u) \leq u^{\kappa+1}$ 在 $[0, 1]$ 上, 其中 ϕ 是引理 2 中定义的函数。因此, 对于每个 t 和所有 $j \geq 0$,

$$\delta_{t,j+1} \leq \delta_{t,j}^{\kappa+1}, \quad \delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}.$$

4.5 与SGD带M动量和 MUON与SVD的比较

为了与原始的MUON with NEWTON-SCHULZ方法进行公平比较, 我们还为两个基线建立了收敛保证: (i) 带动量的SGD和(ii) 理想化的MUON 通过SVD计算得到的精确极步。所有结果都是在相同的平滑性和有界方差假设下推导的, 并且通过梯度的核范数来衡量进展, 因此这些比率可以直接比较。

定理3中关于 带动量的SGD (定理3) 的收敛界, 据我们所知, 是全新的, 并且可能具有独立的兴趣价值。它为基于向量的更新提供了在同一平稳性度量下的清晰参考点。

Theorem 3 (Convergence of SGD with momentum). *Suppose Assumptions 1 and 2 hold, and run SGD with momentum. Choosing $\eta = \min \left\{ \frac{1-\beta}{L}, \frac{(1-\beta)^2}{4L} \right\}$ and $\beta = 1 - \min \left\{ \frac{\sqrt{LD\bar{B}}}{\sigma\sqrt{T}}, 1 \right\}$, the following holds*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(W_{t-1})\|_* \leq \mathcal{O} \left(\sqrt{\frac{rLD}{T}} + \left(\frac{r^2\sigma^2LD}{BT} \right)^{1/4} \right),$$

Consequently, SGD with momentum attains ϵ -stationarity with an iteration complexity of $\mathcal{O} \left(\max \left\{ \frac{rLD}{\epsilon^2}, \frac{r^2\sigma^2LD}{B\epsilon^4} \right\} \right)$.

Note that the rate for MUON with SVD (Theorem 4) is obtained as a special case of MUON with NEWTON-SCHULZ analysis by setting the polar-approximation error to zero (i.e., replacing NEWTON-SCHULZ with an exact polar step). This “oracle” baseline clarifies the gap that NEWTON-SCHULZ needs to close in practice and is useful for interpreting the effect of a finite number of NEWTON-SCHULZ steps.

Theorem 4 (MUON with SVD). *Under Assumptions 1 and 2, setting $\varepsilon_q = 0$ in Theorem 1 yields*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(W_{t-1})\|_* \leq \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2LD}{BT} \right)^{1/4} \right),$$

Consequently, the idealized MUON with SVD attains ϵ -stationarity with iteration complexity of $\mathcal{O} \left(\max \left\{ \frac{LD}{\epsilon^2}, \frac{r^2\sigma^2}{B\epsilon^2}, \frac{r\sigma^2LD}{B\epsilon^4} \right\} \right)$

Discussion of Theorems 3 and 4. Relative to SGD with momentum, the SVD-based polar variant of MUON removes the $\sqrt{r}$ factor from the deterministic (first) term and sharpens rank dependence in the stochastic terms under the same nuclear-norm stationarity metric. Geometrically, the polar step aligns the update with the leading singular structure of the gradient (via spectral-nuclear duality), converting a Frobenius-aligned descent direction into one that is optimally aligned for the nuclear norm, thereby sharpening the r dependence.

Turning to the practical MUON with NEWTON-SCHULZ, Theorem 1 shows that its iteration complexity matches that of the SVD-based polar variant (Theorem 4) up to a multiplicative factor χ_q . By Theorem 2, $\chi_q \rightarrow 1$ doubly exponentially fast in the number of NEWTON-SCHULZ steps q (and improves with the polynomial degree κ). Consequently, a small q already yields rates that are essentially indistinguishable from the ideal SVD-based baseline in iteration count. Since each NEWTON-SCHULZ step uses only matrix multiplications (and avoids SVD), the per-iteration cost is much lower, which explains the superior wall-clock performance of the original SVD-free MUON observed in practice.

5 NUMERICAL EXPERIMENTS

Setup. We conduct a numerical experiment with the CIFAR-10 (50k/10k) dataset and a CNN model, specifically CIFARNET, which has approximately 2M parameters. We compare optimizers: SGD with momentum (baseline), idealized MUON with SVD, and MUON with NEWTON-SCHULZ ($q \in \{1, 2, 3\}$). For the NEWTON-SCHULZ step sweep, we use the NEWTON-SCHULZ polynomial p_κ with degree $\kappa = 2$. We run 50 epochs, and the batch size is $B = 512$. Results are plotted in Fig. 1. Performance is assessed by plotting the training loss (left column) and test loss (right column) over epochs (top row) as well as the cumulative wall-clock time (bottom row). Results represent the average of five runs with different random seeds, including standard deviations. More detailed numerical settings are described in Appendix F.1.

Ablation on NEWTON-SCHULZ step q . As q increases, the learning dynamics per epoch steadily improve: MUON with $q = 1$ already outperforms SGD-M, and $q \in 2, 3$ nearly coincides with the SVD-based MUON in both train loss and test loss, in line with Theorems 1 and 2, which state that the NEWTON-SCHULZ variant matches the SVD iteration complexity up to a factor $\chi_q \rightarrow 1$ that decays doubly exponentially in q . At the same time, the bottom row of Fig. 1 shows that MUON with $q = 2$ or 3 reaches a given test loss substantially faster in wall-clock time than the SVD variant, reflecting the lower per-iteration cost of the NEWTON-SCHULZ update.

定理3 (带动量的SGD收敛). 假设假设1和假设2成立, 并运行带动量的SGD. 选择 $\eta =$ 最小值 $\left\{ \frac{1-\beta}{L}, \frac{(1-\beta)^2}{4L} \right\}$ 和 $\beta = 1 - \min \left\{ \frac{\sqrt{LD\bar{B}}}{\sigma\sqrt{T}}, 1 \right\}$, 则成立

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(W_{t-1})\|_* \leq \mathcal{O} \left(\sqrt{\frac{rLD}{T}} + \left(\frac{r^2\sigma^2LD}{BT} \right)^{1/4} \right),$$

因此, SGD 动量迭代达到 ϵ -平稳性, 迭代复杂度为 $\mathcal{O} \left(\max \left\{ \frac{rLD}{\epsilon^2}, \frac{r^2\sigma^2LD}{B\epsilon^4} \right\} \right)$.

请注意, MUON 与 SVD (定理4) 的速率是通过将极近似误差设置为零 (即, 用精确的极步替换 NEWTON-SCHULZ) 获得的, 它是 NEWTON-SCHULZ 分析的特例。这个“预言者”基线阐明了 NEWTON-SCHULZ 在实际中需要弥补的差距, 并且对于解释有限数量的 NEWTON-SCHULZ 步数的效果很有用。

定理4 (MUON with SVD). 在假设1和2下, 在定理1中设置 $\varepsilon_q = 0$ 得到

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(W_{t-1})\|_* \leq \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2LD}{BT} \right)^{1/4} \right),$$

因此, 理想化的 MUON 与 SVD 通过迭代达到 ϵ -平稳性, 其迭代复杂度为 $\mathcal{O} \left(\max \left\{ \frac{LD}{\epsilon^2}, \frac{r^2\sigma^2}{B\epsilon^2}, \frac{r\sigma^2LD}{B\epsilon^4} \right\} \right)$

关于定理3和定理4的讨论. 相对于带动量的SGD, 基于SVD的MUON极坐标变体去除了确定性 (第一) 项中的 $\sqrt{r}$ 因子, 并在相同的核范数平稳性度量下, 使随机项中的秩依赖性更加显著。从几何角度来看, 极坐标步骤使更新与梯度的主导奇异结构对齐 (通过谱-核对称), 将 Frobenius 对齐的下降方向转换为对核范数最优对齐的方向, 从而使 r 依赖性更加显著。

转向实际的 MUON with NEWTON-SCHULZ, 定理1表明其迭代复杂度与基于SVD的极变体 (定理4) 在乘法因子 χ_q 内相匹配。根据定理2, $\chi_q \rightarrow 1$ 呈双指数速度在 NEWTON-SCHULZ 步数 q (并且随着多项式次数 κ 的提高而改进)。因此, 一个小的 q 就已经产生了在迭代次数上与理想的基于SVD的基线本质上无法区分的速率。由于每个 NEWTON-SCHULZ 步仅使用矩阵乘法 (并避免SVD), 因此每次迭代的成本要低得多, 这解释了在实践中观察到的原始无奇异值分解的 MUON 的优越墙钟性能。

5 数值实验

设置. 我们使用 CIFAR-10 (50k/10k) 数据集和 CNN 模型 (特别是 CifarNet, 参数量约 2M) 进行数值实验。我们比较优化器: 带动量的 SGD (基线)、理想化的带 SVD 的 μ 子 (MUON with SVD) 和带牛顿-舒尔茨的 μ 子 (MUON with NEWTON-SCHULZ $q \in \{1, 2, 3\}$)。对于牛顿-舒尔茨步骤扫描, 我们使用牛顿-舒尔茨多项式 p_κ (度数为 $\kappa = 2$)。运行 50 个周期, 批量大小为 $B = 512$ 。结果绘制在图 1 中。性能评估通过绘制训练损失 (左列) 和测试损失 (右列) 随周期变化 (顶行) 以及累积墙上时钟时间 (底行) 来完成。结果表示五次不同随机种子运行的平均值, 包括标准差。更详细的数值设置在附录 F.1 中描述。**对 N 的消融 NEWTON-SCHULZ 步长 q .** 随着 q 的增加, 每个周期的学习动态稳步提高: MUON 与 $q = 1$ 已经优于 SGD-M, 并且 $q \in 2, 3$ 几乎与基于 SVD 的 MUON 在训练损失和测试损失上重合, 这与定理 1 和 2 一致, 这两个定理指出 NEWTON-SCHULZ 变体与 SVD 迭代复杂度在因子 $\chi_q \rightarrow 1$ 内匹配, 该因子在 q 中双指数衰减。同时, 图 1 的最后一行表明, MUON 与 $q = 2$ 或 3 在墙上时钟时间上达到给定的测试损失速度明显更快, 这反映了 NEWTON-SCHULZ 更新的较低每迭代成本。

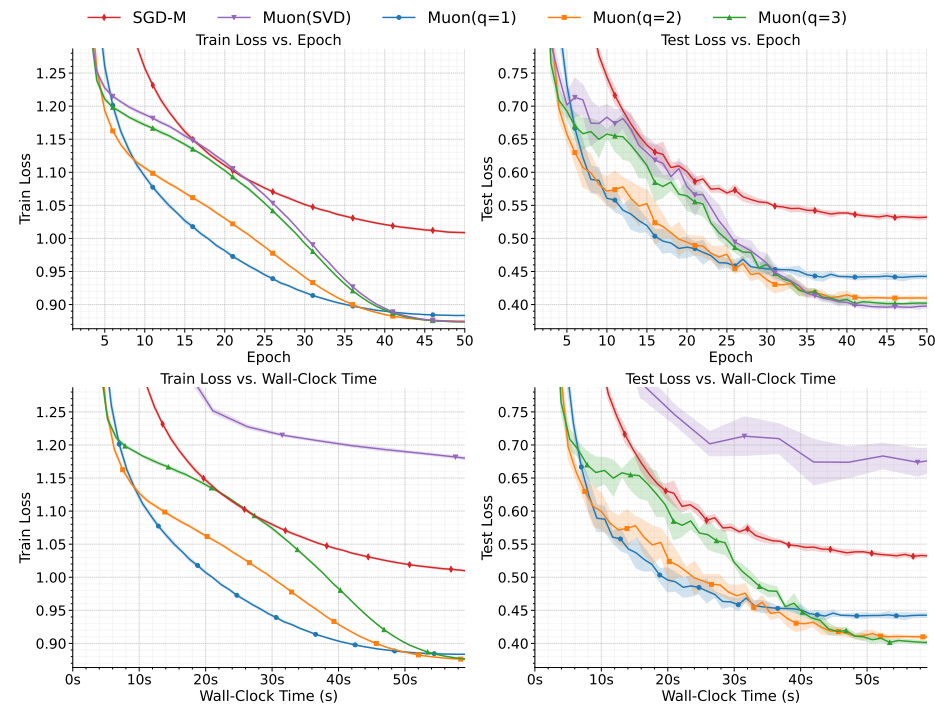


Figure 1: **NEWTON-SCHULZ steps (q) ablation.** MUON with NEWTON-SCHULZ for $q \in \{1, 2, 3\}$ vs. MUON (SVD) and SGD with momentum (SGD-M, baseline).

We additionally performed numerical experiments on various datasets using models at different scales (with the number of parameters indicated in parentheses): a multilayer perceptron (MLP) with 0.5M parameters on MNIST; ResNet-18 (11.2M) on CIFAR-100; WideResNet-28-10 (36.6M) on Tiny-ImageNet; NanoGPT (124M, Transformer) on FineWeb; and a GPT-2-based model (1.3B, Transformer) on FineWeb. All additional experimental results are presented in Appendix G.

Ablation on NEWTON-SCHULZ polynomial degree κ . We perform a controlled ablation that varies the *degree* of the NEWTON-SCHULZ polynomial κ while fixing the number of NEWTON-SCHULZ steps to $q = 3$ for all variants. Increasing the degree $\kappa \in \{1, \dots, 5\}$ improves optimization (the loss drops faster at a fixed epoch) but lengthens each step, yielding a clear accuracy-time trade-off (See Appendix H.1). This mirrors the theory that the residual contracts as $\delta_{j+1} \leq \delta_j^{\kappa+1}$ (Lemma 3), while computation scales with polynomial evaluations.

Rank dependence. We vary the monitored layer’s effective rank $r \in \{16, 32, 64, 128, 216\}$ and plot the epoch-averaged $\|\nabla f(W)\|_*$ on a log-log scale. SGD-M shows a positive slope of approximately 0.3 (grows with r), whereas MUON and its variants are nearly flat. These observations are precisely in line with Theorem 3 and Theorem 4, which state that orthogonalizing momentum removes the deterministic $\sqrt{r}$ penalty and softens the rank dependence of the stochastic terms. The more detailed experimental settings and results are deferred to Appendix H.2.

6 CONCLUSION

We analyzed practical MUON with finite NEWTON-SCHULZ steps and proved nonconvex convergence to ϵ -stationarity with iteration complexity matching the SVD-polar idealization, up to a factor χ_q that shrinks doubly exponentially in q (and improves with κ). Thus, a few NEWTON-SCHULZ steps recover the idealized rate while remaining SVD-free and GPU-friendly. We also provided baselines for SGD with momentum and the SVD-polar variant, showing that MUON weakens rank dependence under the same metric —closing the theory-practice gap and explaining its practical performance.

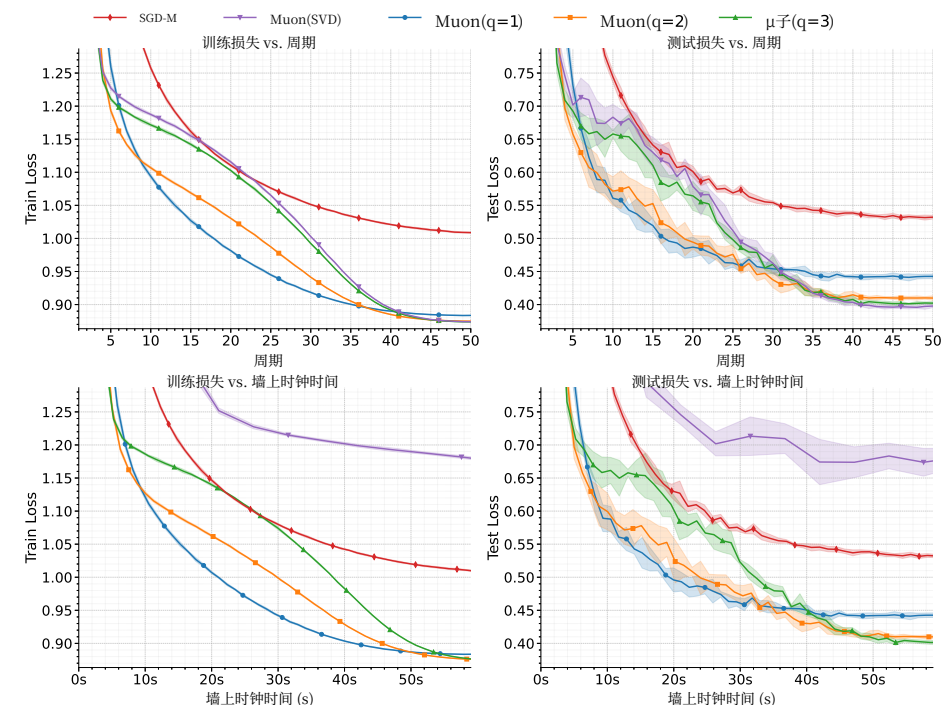


图1: **NE-SCHULZ 步数(q)消融。** μ 子与NEWTON-SCHULZ用于 $q \in \{1, 2, 3\}$ vs. μ 子(SVD)和SGD 带动量(SGD-M, 基线)。

我们此外在各种数据集上使用不同规模的模型进行了数值实验（参数数量括号内所示）：一个具有 0.5M 参数的多层感知器（MLP）在 MNIST 上；ResNet-18 (11.2M) 在 CIFAR-100 上；WideResNet-28-10 (36.6M) 在 Tiny-ImageNet 上；NanoGPT (124M, Transformer) 在 FineWeb 上；以及一个基于 GPT-2 的模型（1.3B, Transformer）在 FineWeb 上。所有附加实验结果在附录 G 中呈现。

NE-SCHULZ 多项式阶数 κ 。 我们进行一项受控消融实验，使 NEWTON-SCHULZ 多项式阶数 κ 变化，同时将 NEWTON-SCHULZ 步数固定为 $q = 3$ ，所有变体均如此。增加阶数 $\kappa \in \{1, \dots, 5\}$ 会提升优化效果（固定周期下损失下降更快），但延长每一步，产生明显的精度-时间权衡（参见附录 H.1）。这反映了理论：残差随 $\delta_{j+1} \leq \delta_j^{\kappa+1}$ 收缩（引理 3），而计算量随多项式求值次数增长。

秩依赖性。 我们改变监控层的有效秩 $r \in \{16, 32, 64, 128, 216\}$ ，并在对数-对数尺度上绘制 epoch 平均的 $\|\nabla f(W)\|_*$ 。SGD-M 显示出约 0.3 的正斜率（随 r 增长），而 MUON 及其变体则几乎平坦。这些观察结果完全符合定理 3 和定理 4，它们指出正交化动量消除了确定性 $\sqrt{r}$ 惩罚，并减弱了随机项的秩依赖性。更详细的实验设置和结果将在附录 H.2 中提供。

6 结论

我们分析了具有有限 NEWTON-SCHULZ 步数的实用 MUON，并证明了非凸收敛到 ϵ -平稳态，其迭代复杂度与 SVD-极化理想化相匹配，误差为一个因子 χ_q ，该因子在 q 中双指数减小（并随着 κ 的提高而改善）。因此，少数 NEWTON-SCHULZ 步数即可恢复理想化速率，同时保持无奇异值分解和 GPU 友好。我们还为带动量的 SGD 和 SVD-极化变体提供了基准，表明 MUON 在相同指标下减弱了秩依赖性——缩小了理论与实际之间的差距，并解释了其实际性能。

USE OF LARGE LANGUAGE MODELS

Large Language Models (LLMs) are used solely as assistive tools for writing. Specifically, we employed an LLM to improve the clarity, grammar, and style of exposition. No part of the research ideation, algorithm design, theoretical analysis, or experimental results involved the use of LLMs. The authors take full responsibility for the content of the paper.

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大型语言模型的运用

大型语言模型（LLMs）仅作为写作辅助工具使用。具体而言，我们使用了一个LLM来提高论述的清晰度、语法和风格。研究的构思、算法设计、理论分析或实验结果均未涉及LLMs的使用。作者对论文内容负全部责任。

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A APPENDIX

A.1 BASIC FACTS FOR MATRIX NORMS

Definition 4 (Schatten Norms). *For each $p \in [1, \infty]$, the Schatten- p norm is defined as $\|X\|_{S_p} := (\sum_i \sigma_i(X)^p)^{1/p}$, where $\sigma_i(X)$ are the singular values of X .*

- *Nuclear/trace norm:* $\|X\|_{S_1} = \|X\|_* = \sum_i \sigma_i(X)$ (Sum of singular values)
- *spectral/operator norm:* $\|X\|_{S_\infty} = \|X\|_{\text{op}} = \max_i \sigma_i(X)$ (largest singular value)
- *Frobenius norm:* $\|X\|_{S_2} = \|X\|_F = \sqrt{\sum_i \sigma_i(X)^2}$

For any conjugate pairs (p, q) , i.e., $\frac{1}{p} + \frac{1}{q} = 1$, the norms $\|\cdot\|_{S_p}$ and $\|\cdot\|_{S_q}$ are dual to each other. In particular, the nuclear norm and the spectral norm are duals.

Lemma 4 (Hölder’s inequality). *For any $A, B \in \mathbb{R}^{m \times n}$,*

$$|\langle A, B \rangle_F| \leq \|A\|_* \|B\|_{\text{op}}.$$

Proof. Let $\text{SVD}(A) = (U, \Sigma, V)$.

$$\langle A, B \rangle_F = \text{tr}(A^\top B) = \text{tr}((U\Sigma V^\top)^\top B) = \text{tr}(V\Sigma U^\top B) = \text{tr}(\Sigma U^\top B V)$$

Let’s define a new matrix $C = U^\top B V$. Since $U^\top$ and V are orthogonal matrices, their operator norm is 1.

$$\|C\|_{\text{op}} = \|U^\top B V\|_{\text{op}} \leq \|U^\top\|_{\text{op}} \|B\|_{\text{op}} \|V\|_{\text{op}} = 1 \cdot \|B\|_{\text{op}} \cdot 1 = \|B\|_{\text{op}}$$

Since Σ is diagonal, $\text{tr}(\Sigma C) = \sum_i \sigma_i(A) C_{ii}$ where $\sigma_i(A)$ is the i -th singular value of A .

$$|\langle A, B \rangle_F| = |\text{tr}(\Sigma C)| = \left| \sum_i \sigma_i(A) C_{ii} \right| \leq \sum_i \sigma_i(A) |C_{ii}| \leq \sum_i \sigma_i(A) \|C\|_{\text{op}} = \|A\|_* \|C\|_{\text{op}}$$

Combining inequalities, we have $|\langle A, B \rangle_F| \leq \|A\|_* \|B\|_{\text{op}}$. $\square$

Lemma 5. *For any $A, B \in \mathbb{R}^{m \times n}$ and any constant $\beta \in (0, 1)$,*

$$\|A + B\|_F^2 \leq \frac{1}{\beta} \|A\|_F^2 + \frac{1}{1-\beta} \|B\|_F^2.$$

Proof. By Young’s inequality (i.e., $\langle x, y \rangle_F \leq \frac{\epsilon}{2} \|x\|_F^2 + \frac{1}{2\epsilon} \|y\|_F^2$ with $\epsilon > 0$), for any positive constant $c > 0$,

$$\begin{aligned} \|A + B\|_F^2 &= \|A\|_F^2 + \|B\|_F^2 + 2\langle A, B \rangle_F \\ &\leq \|A\|^2 + \|B\|_F^2 + c\|A\|_F^2 + \frac{1}{c}\|B\|_F^2 \\ &= (1+c)\|A\|_F^2 + \left(1 + \frac{1}{c}\right)\|B\|_F^2 \end{aligned}$$

If we choose $c = \frac{1-\beta}{\beta}$ where $\beta \in (0, 1)$, we get

$$\|A + B\|_F^2 \leq \frac{1}{\beta} \|A\|_F^2 + \frac{1}{1-\beta} \|B\|_F^2$$

$\square$

Proposition 2. *Let $X \in \mathbb{R}^{m \times n}$ with $r = \min\{m, n\}$. Denote that its singular values are $\{\sigma_i\}_{i=1}^r$. Then the following holds:*

$$(i) \|X\|_F = \sqrt{\text{tr}(X^\top X)} = \sqrt{\sum_{ij} X_{i,j}^2}$$

附录 A

A.1 基本事实 矩阵范数

定义 4 (Schatten范数). 对于每个 $p \in [1, \infty]$, **Schatten- p** 范数定义为 $\|X\|_{S_p} := (\sum_i \sigma_i(X)^p)^{1/p}$, 其中 $\sigma_i(X)$ 是 X 的奇异值。

- **核范数/迹范数:** $\|X\|_{S_1} = \|X\|_* = \sum_i \sigma_i(X)$ (奇异值之和)

- **谱范数/算子范数:** $\|X\|_{S_\infty} = \|X\|_{\text{op}} = \max_i \sigma_i(X)$ (最大奇异值)
- **Frobenius范数:** $\|X\|_{S_2} = \|X\|_F = \sqrt{\sum_i \sigma_i(X)^2}$

对于任何共轭对 (p, q) , 即 $\frac{1}{p} + \frac{1}{q} = 1$, 范数 $\|\cdot\|_{S_p}$ 和 $\|\cdot\|_{S_q}$ 是对偶的。特别是, 核范数和谱范数是对偶的。

引理4 (Hölder不等式). 对于任何 $A, B \in \mathbb{R}^{m \times n}$,

$$|\langle A, B \rangle_F| \leq \|A\|_* \|B\|_{\text{op}}.$$

证明。设 $\text{SVD}(A) = (U, \Sigma, V)$.

$$\langle A, B \rangle_F = \text{tr}(A^\top B) = \text{tr}((U\Sigma V^\top)^\top B) = \text{tr}(V\Sigma U^\top B) = \text{tr}(\Sigma U^\top B V)$$

定义一个新的矩阵 $C = U^\top B V$ 。由于 $U^\top$ 和 V 是正交矩阵, 它们的算子范数是1。

$\|C\|_{\text{op}} = \|U^\top B V\|_{\text{op}} \leq \|U^\top\|_{\text{op}} \|B\|_{\text{op}} \|V\|_{\text{op}} = 1 \cdot \|B\|_{\text{op}} \cdot 1 = \|B\|_{\text{op}}$
由于 Σ 是对角矩阵, $\text{tr}(\Sigma C) = \sum_i \sigma_i(A) C_{ii}$, 其中 $\sigma_i(A)$ 是 A 的第 i 个奇异值。

$$|\langle A, B \rangle_F| = |\text{tr}(\Sigma C)| = \left| \sum_i \sigma_i(A) C_{ii} \right| \leq \sum_i \sigma_i(A) |C_{ii}| \leq \sum_i \sigma_i(A) \|C\|_{\text{op}} = \|A\|_* \|C\|_{\text{op}}$$

结合不等式, 我们得到 $|\langle A, B \rangle_F| \leq \|A\|_* \|B\|_{\text{op}}$. $\square$

引理5. 对于任何 $A, B \in \mathbb{R}^{m \times n}$ 和任何常数 $\beta \in (0, 1)$,

$$\|A + B\|_F^2 \leq \frac{1}{\beta} \|A\|_F^2 + \frac{1}{1-\beta} \|B\|_F^2.$$

证明。根据杨氏不等式 (即 $\langle x, y \rangle_F \leq \frac{\epsilon}{2} \|x\|_F^2 + \frac{1}{2\epsilon} \|y\|_F^2$ 且 $\epsilon > 0$), 对于任何正常数 $c > 0$,

$$\begin{aligned} \|A + B\|_F^2 &= \|A\|_F^2 + \|B\|_F^2 + 2\langle A, B \rangle_F \\ &\leq \|A\|^2 + \|B\|_F^2 + c\|A\|_F^2 + \frac{1}{c}\|B\|_F^2 \\ &= (1+c)\|A\|_F^2 + \left(1 + \frac{1}{c}\right)\|B\|_F^2 \end{aligned}$$

如果我们选择 $c = \frac{1-\beta}{\beta}$ 其中 $\beta \in (0, 1)$, 则得到

$$\|A + B\|_F^2 \leq \frac{1}{\beta} \|A\|_F^2 + \frac{1}{1-\beta} \|B\|_F^2$$

$\square$

命题2. 设 $X \in \mathbb{R}^{m \times n}$ 与 $r = \min\{m, n\}$ 。记其奇异值为 $\{\sigma_i\}_{i=1}^r$ 。则如下成立:

$$(i) \|X\|_F = \sqrt{\text{tr}(X^\top X)} = \sqrt{\sum_{ij} X_{i,j}^2}$$

Proof. Let $\text{SVD}(X) = (U, \Sigma, V)$. Then

$$\begin{aligned}\text{tr}(X^\top X) &= \text{tr}((U\Sigma V^\top)^\top (U\Sigma V^\top)) = \text{tr}(V\Sigma^\top U^\top U\Sigma V^\top) = \text{tr}(V\Sigma^\top \Sigma V^\top) \\ &= \text{tr}(V^\top V\Sigma^\top \Sigma) = \text{tr}(\Sigma^\top \Sigma) = \sum_i \sigma_i(X)^2 = \|X\|_F^2\end{aligned}$$

Since $(X^\top)_{ji} = X_{ij}$, we have

$$(X^\top X)_{jj} = \sum_{i=1}^m (X^\top)_{ji} X_{ij} = \sum_{i=1}^m X_{ij}^2$$

$$\text{Hence, } \text{tr}(X^\top X) = \sum_{j=1}^n (X^\top X)_{jj} = \sum_{j=1}^n \sum_{i=1}^m X_{ij}^2 \quad \square$$

(ii) If $P = \text{Polar}(X)$, then $\langle X, P \rangle_F = \|X\|_*$.

Proof. Let $\text{SVD}(X) = (U, \Sigma, V)$. Then $P = \text{Polar}(X) = UV^\top$.

$$\begin{aligned}\langle X, P \rangle_F &= \langle U\Sigma V^\top, UV^\top \rangle_F = \text{tr}((U\Sigma V^\top)^\top UV^\top) = \text{tr}(V\Sigma U^\top UV^\top) \\ &= \text{tr}(V\Sigma V^\top) = \text{tr}(V^\top V\Sigma) = \text{tr}(\Sigma) = \sum_i \sigma_i(X) = \|X\|_*\end{aligned}$$

□

(iii) $\|X\|_* \leq \sqrt{r} \|X\|_F$.

Proof. By Cauchy-Schwarz inequality,

$$\|X\|_* = \sum_{i=1}^r \sigma_i(X) \leq \sqrt{r \sum_{i=1}^r \sigma_i(X)^2} = \sqrt{r} \|X\|_F$$

□

(iv) $\|X\|_{\text{op}} \leq \|X\|_F \leq \|X\|_*$.

Proof. Since singular values are always non-negative ($\sigma_i(X) \geq 0$),

$$\left(\max_i \sigma_i(X)\right)^2 \leq \sum_i \sigma_i(X)^2 \leq \left(\sum_i \sigma_i(X)\right)^2$$

$$\text{Taking the square root, we get } \|X\|_{\text{op}} \leq \|X\|_F \leq \|X\|_*. \quad \square$$

(v) The polar factor is scale-invariant: $\text{Polar}(cX) = \text{Polar}(X)$ for all $c > 0$.

Proof. Let $\text{SVD}(X) = (U, \Sigma, V)$. Then $P = \text{Polar}(X) = UV^\top$. Also, $\text{SVD}(cX) = (U, c\Sigma, V)$, so $\text{Polar}(cX) = UV^\top$. Thus, $\text{Polar}(cX) = \text{Polar}(X)$ for all $c > 0$. □

A.2 LEMMAS UNDER ASSUMPTIONS

A.2.1 ASSUMPTION 1

Definition 5 (L -smoothness). A differentiable function $f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is L -smooth if its gradient is L -Lipschitz continuous, i.e., for all $X, Y \in \mathbb{R}^{m \times n}$,

$$\|\nabla f(X) - \nabla f(Y)\|_* \leq L\|X - Y\|_{\text{op}}$$

We use smoothness with respect to the operator norm (and the nuclear norm as its dual).

证明。设 $\text{SVD}(X) = (U, \Sigma, V)$ 。则

$$\begin{aligned}\text{tr}(X^\top X) &= \text{tr}((U\Sigma V^\top)^\top (U\Sigma V^\top)) = \text{tr}(V\Sigma^\top U^\top U\Sigma V^\top) = \text{tr}(V\Sigma^\top \Sigma V^\top) \\ &= \text{tr}(V^\top V\Sigma^\top \Sigma) = \text{tr}(\Sigma^\top \Sigma) = \sum_i \sigma_i(X)^2 = \|X\|_F^2\end{aligned}$$

自 $(X^\top)_{ji} = X_{ij}$ 以来，我们就有了

$$(X^\top X)_{jj} = \sum_{i=1}^m (X^\top)_{ji} X_{ij} = \sum_{i=1}^m X_{ij}^2$$

$$\text{因此, } \text{tr}(X^\top X) = \sum_{j=1}^n (X^\top X)_{jj} = \sum_{j=1}^n \sum_{i=1}^m X_{ij}^2 \quad \square$$

(ii) 如果 $P = \text{极}(X)$, 那么 $\langle X, P \rangle_F = \|X\|_*$.

证明。令 $\text{SVD}(X) = (U, \Sigma, V)$ 。那么 $P = \text{极}(X) = UV^\top$ 。

$$\begin{aligned}\langle X, P \rangle_F &= \langle U\Sigma V^\top, UV^\top \rangle_F = \text{tr}((U\Sigma V^\top)^\top UV^\top) = \text{tr}(V\Sigma U^\top UV^\top) \\ &= \text{tr}(V\Sigma V^\top) = \text{tr}(V^\top V\Sigma) = \text{tr}(\Sigma) = \sum_i \sigma_i(X) = \|X\|_*\end{aligned}$$

□

(iii) $\|X\|_* \leq \sqrt{r} \|X\|_F$ 。

证明。根据柯西-施瓦茨不等式,

$$\|X\|_* = \sum_{i=1}^r \sigma_i(X) \leq \sqrt{r \sum_{i=1}^r \sigma_i(X)^2} = \sqrt{r} \|X\|_F$$

□

(iv) $\|X\|_{\text{op}} \leq \|X\|_F \leq \|X\|_*$.

证明。由于奇异值总是非负的 ($\sigma_i(X) \geq 0$),

$$\left(\max_i \sigma_i(X)\right)^2 \leq \sum_i \sigma_i(X)^2 \leq \left(\sum_i \sigma_i(X)\right)^2$$

$$\text{取平方根, 我们得到 } \|X\|_{\text{op}} \leq \|X\|_F \leq \|X\|_*. \quad \square$$

(v) 极因子是尺度不变的: $\text{Polar}(cX) = \text{Polar}(X)$ 对所有 $c > 0$.

Proof. Let $\text{SVD}(X) = (U, \Sigma, V)$. Then $P = \text{Polar}(X) = UV^\top$. Also, $\text{SVD}(cX) = (U, c\Sigma, V)$, so $\text{Polar}(cX) = UV^\top$. Thus, $\text{Polar}(cX) = \text{Polar}(X)$ for all $c > 0$. □

A.2 引理假设

A.2.1 假设1

定义5(L -光滑性)。可微函数 $f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 是 L -光滑的, 如果它的梯度是 L -Lipschitz连续的, 即对于所有 $X, Y \in \mathbb{R}^{m \times n}$,

$$\|\nabla f(X) - \nabla f(Y)\|_* \leq L\|X - Y\|_{\text{op}}$$

我们使用算子范数 (以及作为其 *dual* 的核范数) 的平滑性。

Lemma 6 (Descent Lemma). *Let $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ be L -smooth. Then, for all $X, Y \in \mathbb{R}^{m \times n}$,*

$$f(Y) \leq f(X) + \langle \nabla f(X), Y - X \rangle_F + \frac{L}{2} \|Y - X\|_{\text{op}}^2.$$

Proof. Define an auxiliary scalar function $g : [0, 1] \rightarrow \mathbb{R}$ by considering the value of f along the line segment from X to Y :

$$g(s) := f(X + s(Y - X))$$

By the fundamental theorem of calculus,

$$f(Y) - f(X) = g(1) - g(0) = \int_0^1 g'(s) ds$$

Using the chain rule, $g'(s) = \langle \nabla f(X + s(Y - X)), Y - X \rangle_F$. Substituting this back into the integral equation gives:

$$f(Y) - f(X) = \int_0^1 \langle \nabla f(X + s(Y - X)), Y - X \rangle_F ds$$

Add and subtract $\langle \nabla f(X), Y - X \rangle_F = \int_0^1 \langle \nabla f(X), Y - X \rangle_F ds$, we have:

$$f(Y) - f(X) = \langle \nabla f(X), Y - X \rangle_F + \int_0^1 \langle \nabla f(X + s(Y - X)) - \nabla f(X), Y - X \rangle_F ds$$

We now use the generalized Cauchy-Schwarz inequality or Hölder's inequality (Lemma 4), which states that for any matrices $A, B \in \mathbb{R}^{m \times n}$ and $|\langle A, B \rangle_F| \leq \|A\|_* \|B\|_{\text{op}}$, the nuclear norm ($\|\cdot\|_*$) is the dual norm to the operator norm ($\|\cdot\|_{\text{op}}$). Applying this to the integrand:

$$\langle \nabla f(X + s(Y - X)) - \nabla f(X), Y - X \rangle_F \leq \|\nabla f(X + s(Y - X)) - \nabla f(X)\|_* \|Y - X\|_{\text{op}}$$

By the L -smoothness of f :

$$\|\nabla f(X + s(Y - X)) - \nabla f(X)\|_* \leq L \|s(Y - X)\|_{\text{op}} = Ls \|Y - X\|_{\text{op}}$$

Combining these inequalities, we obtain a bound on the integrand, yielding the final form:

$$f(Y) - f(X) \leq \langle \nabla f(X), Y - X \rangle_F + \int_0^1 Ls \|Y - X\|_{\text{op}}^2 ds = \langle \nabla f(X), Y - X \rangle_F + \frac{L}{2} \|Y - X\|_{\text{op}}^2$$

□

Lemma 7 (Gradient-gap inequality). *Under Assumption 1, for any W , we have*

$$\|\nabla f(W)\|_F^2 \leq 2L(f(W) - f^*)$$

Proof. In the descent lemma (Lemma 6), let $Y = X - \frac{1}{L} \nabla f(X)$. Then, we have

$$\begin{aligned} f\left(X - \frac{1}{L} \nabla f(X)\right) &\leq f(X) + \left\langle \nabla f(X), X - \frac{1}{L} \nabla f(X) - X \right\rangle_F + \frac{L}{2} \left\|X - \frac{1}{L} \nabla f(X) - X\right\|_{\text{op}}^2 \\ &= f(X) - \frac{1}{L} \|\nabla f(X)\|_F + \frac{1}{2L} \|\nabla f(X)\|_{\text{op}}^2 \\ &\leq f(X) - \frac{1}{L} \|\nabla f(X)\|_F + \frac{1}{2L} \|\nabla f(X)\|_F^2 \\ &= f(X) - \frac{1}{2L} \|\nabla f(X)\|_F \end{aligned}$$

where the last inequality is due to Proposition 2(iv). Since $f^* \leq f(X - \frac{1}{L} \nabla f(X))$, we have

$$\|\nabla f(W)\|_F^2 \leq 2L(f(W) - f^*)$$

□

引理 6 (下降引理)。 设 $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ 是 L -平滑的。那么，对于所有 $X, Y \in \mathbb{R}^{m \times n}$,

$$f(Y) \leq f(X) + \langle \nabla f(X), Y - X \rangle_F + \frac{L}{2} \|Y - X\|_{\text{op}}^2.$$

证明。 定义一个辅助标量函数 $g : [0, 1] \rightarrow \mathbb{R}$ ，通过考虑 f 在从 X 到 Y 的线段上的值：

$$g(s) := f(X + s(Y - X))$$

根据微积分基本定理,

$$f(Y) - f(X) = g(1) - g(0) = \int_0^1 g'(s) ds$$

使用链式法则, $g'(s) = \langle \nabla f(X + s(Y - X)), Y - X \rangle_F$ 。 将这个结果代入积分方程得到：

$$f(Y) - f(X) = \int_0^1 \langle \nabla f(X + s(Y - X)), Y - X \rangle_F ds$$

加上并减去 $\langle \nabla f(X), Y - X \rangle_F = \int_0^1 \langle \nabla f(X), Y - X \rangle_F ds$, 我们得到：

$$f(Y) - f(X) = \langle \nabla f(X), Y - X \rangle_F + \int_0^1 \langle \nabla f(X + s(Y - X)) - \nabla f(X), Y - X \rangle_F ds$$

我们现在使用广义Cauchy-Schwarz不等式或Hölder不等式（引理4），该不等式指出，对于任何矩阵 $A, B \in \mathbb{R}^{m \times n}$ 和 $|\langle A, B \rangle_F| \leq \|A\|_* \|B\|_{\text{op}}$ ，核范数($\|\cdot\|_*$)是算子范数($\|\cdot\|_{\text{op}}$)的对偶范数。将其应用于被积函数：

$$\langle \nabla f(X + s(Y - X)) - \nabla f(X), Y - X \rangle_F \leq \|\nabla f(X + s(Y - X)) - \nabla f(X)\|_* \|Y - X\|_{\text{op}}$$

根据 L 的光滑性 f :

$$\|\nabla f(X + s(Y - X)) - \nabla f(X)\|_* \leq L \|s(Y - X)\|_{\text{op}} = Ls \|Y - X\|_{\text{op}}$$

结合这些不等式，我们得到被积函数的界，从而得到最终形式：

$$f(Y) - f(X) \leq \langle \nabla f(X), Y - X \rangle_F + \int_0^1 Ls \|Y - X\|_{\text{op}}^2 ds = \langle \nabla f(X), Y - X \rangle_F + \frac{L}{2} \|Y - X\|_{\text{op}}^2$$

□

引理7 (梯度间隙不等式)。 在假设1下，对于任何 W ，我们有

$$\|\nabla f(W)\|_F^2 \leq 2L(f(W) - f^*)$$

证明。 在下降引理（引理6）中，设 $Y = X - \frac{1}{L} \nabla f(X)$ 。然后，我们有

$$\begin{aligned} f\left(X - \frac{1}{L} \nabla f(X)\right) &\leq f(X) + \left\langle \nabla f(X), X - \frac{1}{L} \nabla f(X) - X \right\rangle_F + \frac{L}{2} \left\|X - \frac{1}{L} \nabla f(X) - X\right\|_{\text{op}}^2 \\ &= f(X) - \frac{1}{L} \|\nabla f(X)\|_F + \frac{1}{2L} \|\nabla f(X)\|_{\text{op}}^2 \\ &\leq f(X) - \frac{1}{L} \|\nabla f(X)\|_F + \frac{1}{2L} \|\nabla f(X)\|_F^2 \\ &= f(X) - \frac{1}{2L} \|\nabla f(X)\|_F \end{aligned}$$

其中最后一个不等式是由于命题2(iv)得出的。由于 $f^* \leq f(X - \frac{1}{L} \nabla f(X))$ ，所以我们有 e

$$\|\nabla f(W)\|_F^2 \leq 2L(f(W) - f^*)$$

□

A.2.2 ASSUMPTION 2

Lemma 8 (Unbiasedness and bounded variance). *If Assumption 2 holds, then*

$$\mathbb{E} [\|G_t - \nabla f(W_t)\|_F^2] \leq \frac{\sigma^2}{B} \quad \text{and} \quad \mathbb{E} [\|G_t - \nabla f(W_t)\|_F] \leq \frac{\sigma}{\sqrt{B}}.$$

where B is the batch size.

Proof. Since $G_t = \frac{1}{B} \sum_{i=1}^B g_i$ with $g_i = \nabla f(W_t; \xi_{t,i})$,

$$\begin{aligned} \mathbb{E} [\|G_t - \nabla f(W_t)\|_F^2] &= \mathbb{E} \left[\left\| \frac{1}{B} \sum_{i=1}^B (\nabla f(W_t; \xi_{t,i}) - \nabla f(W_t)) \right\|_F^2 \right] \\ &= \frac{1}{B^2} \mathbb{E} \left[\left\| \sum_{i=1}^B (\nabla f(W_t; \xi_{t,i}) - \nabla f(W_t)) \right\|_F^2 \right] \\ &= \frac{1}{B^2} \sum_{i=1}^B \mathbb{E} [\|\nabla f(W_t; \xi_{t,i}) - \nabla f(W_t)\|_F^2] \leq \frac{\sigma^2}{B}. \end{aligned}$$

where the last equality is due to i.i.d. uniform sampling with $\mathbb{E}[\nabla f(W_t; \xi_{t,i})] = \nabla f(W_t)$ and independence. The last inequality is due to $\mathbb{E}[\|\nabla f(W; \xi_{t,i}) - \nabla f(W)\|_F^2] \leq \sigma^2$. According to Jensen's inequality, we have

$$\mathbb{E} [\|G_t - \nabla f(W_t)\|_F] \leq \sqrt{\mathbb{E} [\|G_t - \nabla f(W_t)\|_F^2]} = \frac{\sigma}{\sqrt{B}}.$$

□

A.2.2 假设2

引理8 (无偏性和有界方差)。如果假设2成立，那么

$$\mathbb{E} [\|G_t - \nabla f(W_t)\|_F^2] \leq \frac{\sigma^2}{B} \quad \text{and} \quad \mathbb{E} [\|G_t - \nabla f(W_t)\|_F] \leq \frac{\sigma}{\sqrt{B}}.$$

其中 B 是批量大小。

证明。由于 $G_t = \frac{1}{B} \sum_{i=1}^B g_i$ 与 $g_i = \nabla f(W_t; \xi_{t,i})$,

$$\begin{aligned} \mathbb{E} [\|G_t - \nabla f(W_t)\|_F^2] &= \mathbb{E} \left[\left\| \frac{1}{B} \sum_{i=1}^B (\nabla f(W_t; \xi_{t,i}) - \nabla f(W_t)) \right\|_F^2 \right] \\ &= \frac{1}{B^2} \mathbb{E} \left[\left\| \sum_{i=1}^B (\nabla f(W_t; \xi_{t,i}) - \nabla f(W_t)) \right\|_F^2 \right] \\ &= \frac{1}{B^2} \sum_{i=1}^B \mathbb{E} [\|\nabla f(W_t; \xi_{t,i}) - \nabla f(W_t)\|_F^2] \leq \frac{\sigma^2}{B}. \end{aligned}$$

其中最后一个等式是由于独立同分布的均匀采样 $\mathbb{E}[\nabla f(W_t; \xi_{t,i})] = \nabla f(W_t)$ 以及独立性。最后一个不等式是由于 $\mathbb{E}[\|\nabla f(W; \xi_{t,i}) - \nabla f(W)\|_F^2] \leq \sigma^2$ 。根据詹森不等式，我们有

$$\mathbb{E} [\|G_t - \nabla f(W_t)\|_F] \leq \sqrt{\mathbb{E} [\|G_t - \nabla f(W_t)\|_F^2]} = \frac{\sigma}{\sqrt{B}}.$$

□

A.3 NEWTON–SCHULZ POLYNOMIAL

NEWTON–SCHULZ steps orthogonalize a matrix X via NEWTON–SCHULZ steps applied to $A = XX^\top$. Define NEWTON–SCHULZ polynomial p_κ the degree is κ . The following are the properties of p_κ along with their proofs.

NEWTON–SCHULZ polynomial p_κ .

For degree $\kappa \in \mathbb{N}$, the NEWTON–SCHULZ polynomial is the Taylor truncation of $1/\sqrt{\lambda}$ at $\lambda = 1$, i.e.,

$$p^{(s)}(1) = \left. \frac{d^s}{d\lambda^s} \lambda^{-1/2} \right|_{\lambda=1}$$

for $s = 1, \dots, \kappa$. The explicit form of the NEWTON–SCHULZ polynomial for degree κ is

$$p_\kappa(\lambda) = \sum_{s=0}^{\kappa} c_s (1-\lambda)^s, \quad c_s = \frac{(2s)!}{4^s (s!)^2} > 0.$$

Equivalently, with reparametrization $u = 1 - \lambda \in [0, 1]$ and $p_\kappa(1-u) = \sum_{s=0}^{\kappa} c_s u^s$.

Proposition 1 (Properties of p_κ). For $\lambda \in [0, 1]$:

- **Positivity.** $p_\kappa(\lambda) > 0$ and $p_\kappa(\lambda) \geq 1$ with equality if and only if $\lambda = 1$.
- **Monotonicity of τ .** Let $\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$; then we have τ non-decreasing on $[0, 1]$ and $\tau(1) = 1$.

Consequently, for any symmetric $A \succeq 0$ with spectrum in $[0, 1]$, the NEWTON–SCHULZ update $A \mapsto p_\kappa(A)Ap_\kappa(A)$ satisfies $\|p_\kappa(A)Ap_\kappa(A)\|_{\text{op}} \leq 1$: NEWTON–SCHULZ steps preserve the unit spectral ball (see Appendix A.3).

proof.

Positivity. $p_\kappa(\lambda) > 0$ and $p_\kappa(\lambda) \geq 1$ with equality if and only if $\lambda = 1$.

Proof. Separating the first term (for $s = 0$) from the summation that defines $p_\kappa(\lambda)$:

$$p_\kappa(\lambda) = \sum_{s=0}^{\kappa} c_s (1-\lambda)^s = c_0(a-\lambda)^0 + \sum_{s=1}^{\kappa} c_s (1-\lambda)^s$$

The coefficient c_0 is $\frac{(2 \cdot 0)!}{4^0 (0!)^2} = 1$. The rest of the polynomial is the sum $\sum_{s=1}^{\kappa} c_s (1-\lambda)^s$. For any term in this sum and for any $\lambda \in [0, 1]$, the coefficients c_s are strictly positive for all $s \geq 1$. Also, for any exponent $s \geq 1$, $(1-\lambda)^s$ is non-negative in the range $[0, 1]$. Each term $c_s (1-\lambda)^s$ is a product of a positive number and a non-negative number, which means each term is non-negative. Hence,

$$p_\kappa(\lambda) = 1 + \sum_{s=1}^{\kappa} c_s (1-\lambda)^s \geq 1 + 0 = 1$$

This proves that $p_\kappa(\lambda) \geq 1$ for all $\lambda \in [0, 1]$. It is also trivially true that $p_\kappa(\lambda) > 0$. $\square$

Monotonicity of $\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$.

Proof. First note $\tau(1) = 1 \cdot p_\kappa(1)^2 = 1$. For monotonicity, differentiate:

$$\tau'(\lambda) = p_\kappa(\lambda)^2 + 2\lambda p_\kappa(\lambda) p'_\kappa(\lambda) = p_\kappa(\lambda) (p_\kappa(\lambda) + 2\lambda p'_\kappa(\lambda)).$$

Set $u = 1 - \lambda$ and define $q(u) := p_\kappa(1-u) = \sum_{s=0}^{\kappa} c_s u^s$. Then

$$p_\kappa(\lambda) = q(u), \quad p'_\kappa(\lambda) = \frac{d}{d\lambda} q(1-\lambda) = -q'(u), \quad \lambda = 1-u.$$

Hence

$$\tau'(\lambda) = q(u) \underbrace{(q(u) - 2(1-u)q'(u))}_{:=S(u)}.$$

A.3 牛顿-舒尔茨多项式

NEWTON–SCHULZ步数正交化一个矩阵

X $AX = A$ ，通过将NEWTON–SCHULZ步数应用于。定义NEWTON–SCHULZ多项式 p_κ ，其度数为。以下是其性质及其证明。

牛顿-舒尔茨多项式 $p_{\kappa \circ}$

对于度数 $\kappa \in \mathbb{N}$ ，牛顿-舒尔茨多项式是 $1/\sqrt{\lambda}$ 在 $\lambda = 1$ 处的泰勒截断，即，

$$p^{(s)}(1) = \left. \frac{d^s}{d\lambda^s} \lambda^{-1/2} \right|_{\lambda=1}$$

对于 $s = 1, \dots, \kappa$ 。度数为 κ 的牛顿-舒尔茨多项式的显式形式是

$$p_\kappa(\lambda) = \sum_{s=0}^{\kappa} c_s (1-\lambda)^s, \quad c_s = \frac{(2s)!}{4^s (s!)^2} > 0.$$

等价地，通过重新参数化 $u = 1 - \lambda \in [0, 1]$ 和 $p_\kappa(1-u) = \sum_{s=0}^{\kappa} c_s u^s$ 。

命题1 (p_κ 的性质)。 对于 $\lambda \in [0, 1]$:

- **正定性。** $p_\kappa(\lambda) > 0$ 和 $p_\kappa(\lambda) \geq 1$ ，当且仅当 $\lambda = 1$ 时取等号。
- **单调性。** 设 $\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$ ；则我们有 τ 在 $[0, 1]$ 上非递减，且 $\tau(1) = 1$ 。

因此，对于任何谱在 $[0, 1]$ 中的对称 $A \succeq 0$ ，NEWTON–SCHULZ更新 $A \mapsto p_\kappa(A)Ap_\kappa(A)$ 满足 $\|p_\kappa(A)Ap_\kappa(A)\|_{\text{op}} \leq 1$ ：NEWTON–SCHULZ步数保持单位谱球（参见附录A.3）。

证明。

正定性。 $p_\kappa(\lambda) > 0$ 和 $p_\kappa(\lambda) \geq 1$ ，当且仅当 $\lambda = 1$ 时取等号。

证明。将第一项（对于 $s = 0$ ）从定义 $p_\kappa(\lambda)$ 的求和中分离出来：

$$p_\kappa(\lambda) = \sum_{s=0}^{\kappa} c_s (1-\lambda)^s = c_0(a-\lambda)^0 + \sum_{s=1}^{\kappa} c_s (1-\lambda)^s$$

系数 c_0 是 $\frac{(2 \cdot 0)!}{4^0 (0!)^2} = 1$ 。多项式的其余部分是求和 $\sum_{s=1}^{\kappa} c_s (1-\lambda)^s$ 。对于这个求和中的任何一项和任何 $\lambda \in [0, 1]$ ，系数 c_s 对于所有 $s \geq 1$ 都严格为正。此外，对于任何指数 $s \geq 1$ ， $(1-\lambda)^s$ 在范围 $[0, 1]$ 内非负。每一项 $c_s (1-\lambda)^s$ 都是一个正数和一个非负数的乘积，这意味着每一项都是非负的。因此，

$$p_\kappa(\lambda) = 1 + \sum_{s=1}^{\kappa} c_s (1-\lambda)^s \geq 1 + 0 = 1$$

这证明了 $p_\kappa(\lambda) \geq 1$ 对于所有 $\lambda \in [0, 1]$ 成立。它也显然成立，即 $p_\kappa(\lambda) > 0$ 。 $\square$

$\tau(\lambda) := \lambda[p_\kappa(\lambda)]^2$ 的单调性。

证明。首先注意 $\tau(1) = 1 \cdot p_\kappa(1)^2 = 1$ 。为证明单调性，求导数：

$$\tau'(\lambda) = p_\kappa(\lambda)^2 + 2\lambda p_\kappa(\lambda) p'_\kappa(\lambda) = p_\kappa(\lambda) (p_\kappa(\lambda) + 2\lambda p'_\kappa(\lambda)).$$

设 $u = 1 - \lambda$ 并定义 $q(u) := p_\kappa(1-u) = \sum_{s=0}^{\kappa} c_s u^s$ 。则

$$p_\kappa(\lambda) = q(u), \quad p'_\kappa(\lambda) = \frac{d}{d\lambda} q(1-\lambda) = -q'(u), \quad \lambda = 1-u.$$

因此

$$\tau'(\lambda) = q(u) \underbrace{(q(u) - 2(1-u)q'(u))}_{:=S(u)}.$$

Now, we compute $S(u)$ in closed form. Since

$$q(u) = \sum_{s=0}^{\kappa} c_s u^s, \quad q'(u) = \sum_{s=1}^{\kappa} s c_s u^{s-1},$$

we obtain

$$S(u) = \sum_{s=0}^{\kappa} c_s u^s - 2(1-u) \sum_{s=1}^{\kappa} s c_s u^{s-1}.$$

Expanding and collecting coefficients of u^t gives

$$S(u) = (c_0 - 2c_1) + \sum_{t=1}^{\kappa-1} ((2t+1)c_t - 2(t+1)c_{t+1}) u^t + (2\kappa+1)c_{\kappa} u^{\kappa}.$$

Using the ratio identity

$$\frac{c_{s+1}}{c_s} = \frac{(2s+2)!(s!)^2 4^s}{(2s)!((s+1)!)^2 4^{s+1}} = \frac{(2s+2)(2s+1)}{4(s+1)^2} = \frac{2s+1}{2(s+1)},$$

for $t = 0, \dots, \kappa-1$,

$$(2t+1)c_t - 2(t+1)c_{t+1} = (2t+1)c_t - 2(t+1) \left(\frac{2t+1}{2(t+1)} c_t \right) = 0,$$

and also $c_0 - 2c_1 = 1 - 2 \cdot \frac{1}{2} = 0$. Therefore, all coefficients up to degree $\kappa-1$ vanish, and we are left with

$$S(u) = (2\kappa+1)c_{\kappa} u^{\kappa} \geq 0 \quad \text{for } u \in [0, 1],$$

with strict positivity for $u > 0$ when $\kappa \geq 1$. Since $q(u) > 0$ on $[0, 1]$, we conclude

$$\tau'(\lambda) = q(u)S(u) \geq 0 \quad \text{for all } \lambda \in [0, 1],$$

i.e., τ is non-decreasing on $[0, 1]$, and $\tau'(\lambda) > 0$ for $\lambda \in [0, 1)$ when $\kappa \geq 1$. Together with $\tau(1) = 1$, this proves the proposition. $\square$

NS step preserves the unit spectral ball. Let $A \succeq 0$ be symmetric with spectrum $\sigma(A) \subset [0, 1]$. By spectral calculus,

$$p_{\kappa}(A) A p_{\kappa}(A) = U \operatorname{diag} (\lambda_i p_{\kappa}(\lambda_i)^2) U^{\top},$$

where $A = U \operatorname{diag}(\lambda_i) U^{\top}$. Thus

$$\|p_{\kappa}(A) A p_{\kappa}(A)\|_{\text{op}} = \max_i (\lambda_i p_{\kappa}(\lambda_i)^2) = \max_i \tau(\lambda_i) \leq \tau(1) = 1,$$

because τ is non-decreasing on $[0, 1]$. Hence, the NEWTON–SCHULZ update maps the unit spectral ball into itself.

现在, 我们计算 $S(u)$ 的闭式解。由于

$$q(u) = \sum_{s=0}^{\kappa} c_s u^s, \quad q'(u) = \sum_{s=1}^{\kappa} s c_s u^{s-1},$$

我们得到

$$S(u) = \sum_{s=0}^{\kappa} c_s u^s - 2(1-u) \sum_{s=1}^{\kappa} s c_s u^{s-1}.$$

展开并收集 u^t 的系数给出

$$S(u) = (c_0 - 2c_1) + \sum_{t=1}^{\kappa-1} ((2t+1)c_t - 2(t+1)c_{t+1}) u^t + (2\kappa+1)c_{\kappa} u^{\kappa}.$$

使用比率恒等式

$$\frac{c_{s+1}}{c_s} = \frac{(2s+2)!(s!)^2 4^s}{(2s)!((s+1)!)^2 4^{s+1}} = \frac{(2s+2)(2s+1)}{4(s+1)^2} = \frac{2s+1}{2(s+1)},$$

对于 $t = 0, \dots, \kappa-1$,

$$(2t+1)c_t - 2(t+1)c_{t+1} = (2t+1)c_t - 2(t+1) \left(\frac{2t+1}{2(t+1)} c_t \right) = 0,$$

并且还有 $c_0 - 2c_1 = 1 - 2 \cdot \frac{1}{2} = 0$ 。因此, 所有最高次数为 $\kappa-1$ 的系数都消失, 我们得到

$$S(u) = (2\kappa+1)c_{\kappa} u^{\kappa} \geq 0 \quad \text{for } u \in [0, 1],$$

当 $u > 0$ 严格为正时 $\kappa \geq 1$ 。由于 $q(u) > 0$ 在 $[0, 1]$ 上, 我们得出结论

$$\tau'(\lambda) = q(u)S(u) \geq 0 \quad \text{for all } \lambda \in [0, 1],$$

即, τ 在 $[0, 1]$ 上非递减, 且 $\tau'(\lambda) > 0$ 对于 $\lambda \in [0, 1)$ 在 $\kappa \geq 1$ 时。结合 $\tau(1) = 1$, 这证明了命题。 $\square$

NS步骤保持单位谱球。 设 $A \succeq 0$ 是对称的, 其谱为 $\sigma(A) \subset [0, 1]$ 。通过谱计算,

$$p_{\kappa}(A) A p_{\kappa}(A) = U \operatorname{diag} (\lambda_i p_{\kappa}(\lambda_i)^2) U^{\top},$$

其中 $A = U \operatorname{diag}(\lambda_i) U^{\top}$ 。因此

$$\|p_{\kappa}(A) A p_{\kappa}(A)\|_{\text{op}} = \max_i (\lambda_i p_{\kappa}(\lambda_i)^2) = \max_i \tau(\lambda_i) \leq \tau(1) = 1,$$

因为 τ 在 $[0, 1]$ 上非递减。因此, NEWTON–SCHULZ更新将单位谱球映射到自身。

B MUON WITH FINITE NEWTON–SCHULZ ITERATION

Algorithm 1 MUON with NEWTON–SCHULZ Orthogonalization

Require: learning rate $\eta > 0$, momentum $\beta \in [0, 1)$, NEWTON-SCHULZ steps $q \in \mathbb{N}$,
NEWTON-SCHULZ polynomial p_κ with degree κ , batch size B , total iteration T .

$$\begin{aligned}
1: & \textbf{Initialize: } M_0 \leftarrow 0, W_0 \in \mathbb{R}^{m \times n} \\
2: & \textbf{for } t = 1 \text{ to } T \textbf{ do} \\
3: & \quad G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_{t-1}; \xi_{t,i}) \\
4: & \quad M_t \leftarrow \beta M_{t-1} + G_t \\
5: & \quad X_{t,0} \leftarrow M_t / \alpha_t \text{ with } \alpha_t = \max\{1, \|M_t\|_F\} \quad (\text{scaling ensures } \|X_{t,0}\|_{\text{op}} \leq 1) \\
6: & \quad \textbf{for } j = 1 \text{ to } q \textbf{ do} \\
7: & \quad \quad X_{t,j} \leftarrow p_\kappa(X_{t,j-1} X_{t,j-1}^\top) X_{t,j-1} \\
8: & \quad \textbf{end for} \\
9: & \quad O_t \leftarrow X_{t,q} \\
10: & \quad W_t \leftarrow W_{t-1} - \eta O_t \\
11: & \textbf{end for}
\end{aligned}$$

Proof of Theorem 1. First, we introduce the *scaled* EMA momentum: $N_t := (1 - \beta)M_t$. Then, we get $N_t = \beta N_{t-1} + (1 - \beta)G_t$. Note that the polar factor is scale-invariant (Proposition 2(v)). Let P_t be the polar factor of M_t , i.e., $P_t = \text{Polar}(M_t)$. Hence, $P_t := \text{Polar}(N_t) = \text{Polar}(M_t)$.

By Assumption 1, we start from *descent lemma* (Lemma 6). Since $W_t = W_{t-1} - \eta O_t$, we have

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) + \langle \nabla f(W_{t-1}), W_t - W_{t-1} \rangle_F + \frac{L}{2} \|W_t - W_{t-1}\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), O_t \rangle_F + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle N_t, O_t \rangle_F + \eta \langle N_t - \nabla f(W_{t-1}), O_t \rangle_F + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&\leq f(W_{t-1}) - \eta \langle N_t, O_t \rangle_F + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle N_t, P_t \rangle_F + \eta \langle N_t, P_t - O_t \rangle_F + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \|N_t\|_* + \eta \langle N_t, P_t - O_t \rangle_F + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&\leq f(W_{t-1}) - \eta \|N_t\|_* + \eta \|N_t\|_* \|P_t - O_t\|_{\text{op}} + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2
\end{aligned}$$

where the second inequality and the third inequality are due to Hölder's inequality (Lemma 4), and the last equality is due to $\langle N_t, P_t \rangle_F = \|N_t\|_*$ (Proposition 2(ii)).

Now, we define the *polar approximation error*, which is the error between the exact polar factor $P_t = \text{Polar}(N_t) = \text{Polar}(M_t)$ and the actual step O_t generated by q -step NEWTON-SCHULZ steps, i.e., $O_t = \text{NEWTON-SCHULZ}(M_t, q)$, measured in the operator norm.

$$\varepsilon_{t,q} := \|O_t - P_t\|_{\text{op}} \quad \text{and} \quad \varepsilon_q := \sup_t \varepsilon_{t,q}$$

Since $\|P_t\|_{\text{op}} \leq 1$, $\|O_t\|_{\text{op}} \leq 1 + \varepsilon_{t,q} \leq 1 + \varepsilon_q$, we have

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) - \eta(1 - \|P_t - O_t\|_{\text{op}}) \|N_t\|_* + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&\leq f(W_{t-1}) - \eta(1 - \varepsilon_{t,q}) \|N_t\|_* + \eta(1 + \varepsilon_{t,q}) \|N_t - \nabla f(W_{t-1})\|_* + \frac{L}{2} \eta^2 (1 + \varepsilon_{t,q})^2 \\
&\leq f(W_{t-1}) - \eta(1 - \varepsilon_{t,q}) \|\nabla f(W_{t-1})\|_* + \eta(1 - \varepsilon_{t,q}) \|\nabla f(W_{t-1}) - N_t\|_* \\
&\quad + \eta(1 + \varepsilon_{t,q}) \|N_t - \nabla f(W_{t-1})\|_* + \frac{L}{2} \eta^2 (1 + \varepsilon_{t,q})^2 \\
&= f(W_{t-1}) - \eta(1 - \varepsilon_{t,q}) \|\nabla f(W_{t-1})\|_* + 2\eta \|N_t - \nabla f(W_t)\|_* + \frac{L}{2} \eta^2 (1 + \varepsilon_{t,q})^2
\end{aligned}$$

where the last inequality is due to $- \|A\|_* \leq - \|B\|_* + \|A - B\|_*$ and the last equality holds because $(1 - \varepsilon_{t,q}) + (1 + \varepsilon_{t,q}) = 2$. Since $\varepsilon_{t,q} \leq \varepsilon_q$, we arrive at the clean one-step inequality as

$$f(W_t) \leq f(W_{t-1}) - \eta(1 - \varepsilon_q) \|\nabla f(W_{t-1})\|_* + 2\eta \|\nabla f(W_{t-1}) - N_t\|_* + \frac{L}{2} \eta^2 (1 + \varepsilon_q)^2. \quad (1)$$

批量大小 MUON WITH FINITE NEWTON-SCHULZ 迭代

算法1 MUON with NEWTON-SCHULZ 正交化

要求: 学习率 $\eta > 0$, 动量 $\beta \in [0, 1)$, NEWTON-SCHULZ步数 $q \in \mathbb{N}$,
NEWTON-SCHULZ多项式 p_κ with degree κ , 批量大小 B , 总迭代次数 T .

```

1: 初始化:  $M_0 \leftarrow 0, \quad W_0 \in \mathbb{R}^{m \times n}$ 
2: 期望  $t = 1$  到  $T$  do
3:    $G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_{t-1}; \xi_{t,i})$ 
4:    $M_t \leftarrow \beta M_{t-1} + G_t$ 
5:    $X_{t,0} \leftarrow M_t / \alpha_t$  与  $\alpha_t = \text{最大}\{1, \|M_t\|_F\}$  (缩放确保  $\|X_{t,0}\|_0 \leq 1$ )
6:   for  $j = 1$  到  $q$  do
7:      $X_{t,j} \leftarrow p_\kappa(X_{t,j-1} X_{t,j-1}^\top) X_{t,j-1}$ 
8:   结束for循环
9:    $O_t \leftarrow X_{t,q}$ 
10:   $W_t \leftarrow W_{t-1} - \eta O_t$ 
11: 结束for循环

```

命题1的证明。 首先, 我们引入 缩放 EMA动量: $N_t := (1 - \beta)M_t$ 。然后, 我们得到 $N_t = \beta N_{t-1} + (1 - \beta)G_t$ 。注意极因子是尺度不变的 (命题2(v))。令 P_t 为 M_t 的极因子, 即 $P_t = \text{Polar}(M_t)$ 。因此, $P_t := \text{Polar}(N_t) = \text{Polar}(M_t)$ 。

根据假设1, 我们从下降引理(引理6)开始。由于 $W_t = W_{t-1} - \eta O_t$, 我们有

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) + \langle \nabla f(W_{t-1}), W_t - W_{t-1} \rangle_F + \frac{L}{2} \|W_t - W_{t-1}\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), O_t \rangle_F + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle N_t, O_t \rangle_F + \eta \langle N_t - \nabla f(W_{t-1}), O_t \rangle_F + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&\leq f(W_{t-1}) - \eta \langle N_t, O_t \rangle_F + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle N_t, P_t \rangle_F + \eta \langle N_t, P_t - O_t \rangle_F + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \|N_t\|_* + \eta \langle N_t, P_t - O_t \rangle_F + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&\leq f(W_{t-1}) - \eta \|N_t\|_* + \eta \|N_t\|_* \|P_t - O_t\|_{\text{op}} + \eta \|N_t - \nabla f(W_{t-1})\|_* \|O_t\|_{\text{op}} + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2
\end{aligned}$$

其中第二个不等式和第三个不等式归因于Hölder不等式(引理4), 最后一个等式归因于 $\langle N_t, \text{样式id}='2' \rangle, \langle P_t, \text{样式id}='2' \rangle_F = \|N_t\|_*$ (命题2(ii))。

现在, 我们定义 $\langle \text{样式id}='1' \rangle$ 极化近似误差 $\langle \text{样式} \rangle$, 这是精确极因子 $P_t = \text{Polar}(N_t) = \text{Polar}(M_t)$ 与实际生成的步骤 O_t 之间的误差, 该步骤由 q -步NEWTON-SCHULZ步骤生成, 即 $O_t = \text{NEWTON-SCHULZ}(M_t \langle \text{样式id}='11' \rangle, \langle \text{样式} \rangle q)$, 在算子范数中衡量。

$$\varepsilon_{t,q} := \|O_t - P_t\|_{\text{op}} \quad \text{and} \quad \varepsilon_q := \sup_t \varepsilon_{t,q}$$

由于 $\|P_t\|_{\text{op}} \leq 1$, $\|O_t\|_{\text{op}} \leq 1 + \varepsilon_t$, $q \leq 1 + \varepsilon_q$, 所以我们有

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) - \eta(1 - \|P_t - O_t\|_{\text{op}})\|N_t\|_* + \eta\|N_t - \nabla f(W_{t-1})\|_*\|O_t\|_{\text{op}} + \frac{L}{2}\eta^2\|O_t\|_{\text{op}}^2 \\
&\leq f(W_{t-1}) - \eta(1 - \varepsilon_{t,q})\|N_t\|_* + \eta(1 + \varepsilon_{t,q})\|N_t - \nabla f(W_{t-1})\|_* + \frac{L}{2}\eta^2(1 + \varepsilon_{t,q})^2 \\
&\leq f(W_{t-1}) - \eta(1 - \varepsilon_{t,q})\|\nabla f(W_{t-1})\|_* + \eta(1 - \varepsilon_{t,q})\|\nabla f(W_{t-1}) - N_t\|_* \\
&\quad + \eta(1 + \varepsilon_{t,q})\|N_t - \nabla f(W_{t-1})\|_* + \frac{L}{2}\eta^2(1 + \varepsilon_{t,q})^2 \\
&= f(W_{t-1}) - \eta(1 - \varepsilon_{t,q})\|\nabla f(W_{t-1})\|_* + 2\eta\|N_t - \nabla f(W_t)\|_* + \frac{L}{2}\eta^2(1 + \varepsilon_{t,q})^2
\end{aligned}$$

其中最后一个不等式是由于 $\|A\|_* \leq -\|B\|_* + \|A - B\|_*$ 1- 导致的, 而最

后一个等式成立是因为 $\epsilon_{t,q} + 1 + \epsilon_{t,q} = 2$ $\epsilon_{t,q} \leq \epsilon_q$ 由于 $\epsilon_{t,q} \leq \epsilon_q$, 我们得到干净的单一步骤不等式为

$$f(W_t) \leq f(W_{t-1}) - \eta(1 - \varepsilon_q) \|\nabla f(W_{t-1})\|_* + 2\eta \|\nabla f(W_{t-1}) - N_t\|_* + \frac{L}{2} \eta^2 (1 + \varepsilon_q)^2. \quad (1)$$

Rearranging and taking the expectation yields

$$\mathbb{E}\|\nabla f(W_{t-1})\|_* \leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta(1 - \varepsilon_q)} + \frac{2}{1 - \varepsilon_q} \mathbb{E}\|\nabla f(W_{t-1}) - N_t\|_* + \frac{L\eta(1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}. \quad (2)$$

Bounding $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$.

In order to bound $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$, we introduce the true scaled momentum $\bar{N}_t$ defined by the true (full-batch) gradient $\nabla f(W_t)$ instead of G_t for each step t :

- $\bar{N}_t = \beta\bar{N}_{t-1} + (1 - \beta)\nabla f(W_{t-1})$ for $t > 0$ and $\bar{N}_0 = 0$.
- Note that $N_t = \beta N_{t-1} + (1 - \beta)G_t$ for $t > 0$ and $N_0 = 0$.

Then we can decompose $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$ as

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \underbrace{\mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_*]}_{\text{Term (A)}} + \underbrace{\mathbb{E}[\|\bar{N}_t - N_t\|_*]}_{\text{Term (B)}} \quad (3)$$

Bounding the Term (A). Let $D_t = \|\nabla f(W_{t-1}) - \bar{N}_t\|_*$. From the recursion,

$$\nabla f(W_{t-1}) - \bar{N}_t = \beta(\nabla f(W_{t-1}) - \bar{N}_{t-1})$$

Hence, we have

$$\begin{aligned} D_t &= \beta\|\nabla f(W_{t-1}) - \bar{N}_{t-1}\|_* \\ &\leq \beta\|\nabla f(W_{t-2}) - \bar{N}_{t-1}\| + \beta\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \\ &= \beta D_{t-1} + \beta\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_*. \end{aligned}$$

Applying Assumption 1 and using the fact that $\|O_{t-1}\|_{\text{op}} \leq 1 + \varepsilon_{t-1,q} \leq 1 + \varepsilon_q$, we have

$$\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \leq L\|W_{t-1} - W_{t-2}\|_{\text{op}} = L\eta\|O_{t-1}\|_{\text{op}} \leq L\eta(1 + \varepsilon_q)$$

Hence, we have the following recursion,

$$D_t \leq \beta D_{t-1} + \beta L\eta(1 + \varepsilon_q)$$

Since $\bar{N}_0 = 0$, we have

$$D_1 = \|\nabla f(W_0) - \bar{N}_1\|_* = \|\nabla f(W_0) - (\beta\bar{N}_0 + (1 - \beta)\nabla f(W_0))\|_* = \beta\|\nabla f(W_0)\|_*$$

By unrolling the recursion and taking the expectation, we get

$$\mathbb{E}[D_t] \leq \beta^t \mathbb{E}[\|\nabla f(W_0)\|_*] + \sum_{i=1}^t \beta^i L\eta(1 + \varepsilon_q) \leq \beta^t \|\nabla f(W_0)\|_* + \frac{\beta L\eta(1 + \varepsilon_q)}{1 - \beta} \quad (4)$$

Bounding the Term (B). When we unroll the EMA recursion of both N_t and $\bar{N}_t$,

$$\begin{aligned} N_t &= \beta N_{t-1} + (1 - \beta)G_t = \beta^t N_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} G_i \\ \bar{N}_t &= \beta \bar{N}_{t-1} + (1 - \beta)\nabla f(W_t) = \beta^t \bar{N}_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} \nabla f(W_i) \end{aligned}$$

Then, we compute the expectation of the Frobenius norm bias caused by M_t and $\bar{M}_t$,

$$\mathbb{E}[\|N_t - \bar{N}_t\|_F] \leq (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] \quad (5)$$

Applying Jensen's inequality and the linearity of expectation gives

$$\begin{aligned} (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] &\leq \sqrt{(1 - \beta)^2 \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\mathbb{E} \left[\sum_{i=1}^t \beta^{2(t-i)} \|G_i - \nabla f(W_i)\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \end{aligned}$$

重新排列并取期望得到

$$\mathbb{E}\|\nabla f(W_{t-1})\|_* \leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta(1 - \varepsilon_q)} + \frac{2}{1 - \varepsilon_q} \mathbb{E}\|\nabla f(W_{t-1}) - N_t\|_* + \frac{L\eta(1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}. \quad (2)$$

界限项 $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$.

为了界限项 $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$, 我们引入了由真实 (全批次) 梯度 $\nabla f(W_t)$ 定义的真实缩放动量 N_t , 而不是每个步骤 t 的 G_t :

- $\bar{N}_t = \beta\bar{N}_{t-1} + (1 - \beta)\nabla f(W_{t-1})$ for $t > 0$ and $\bar{N}_0 = 0$.
- Note that $N_t = \beta N_{t-1} + (1 - \beta)G_t$ for $t > 0$ and $N_0 = 0$.

然后我们可以将 $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$ 分解为

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \underbrace{\mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_*]}_{\text{Term (A)}} + \underbrace{\mathbb{E}[\|\bar{N}_t - N_t\|_*]}_{\text{Term (B)}} \quad (3)$$

界限项(A). 设 $D_t = \|\nabla f(W_{t-1}) - \bar{N}_t\|_*$. 根据递归关系,

$$\nabla f(W_{t-1}) - \bar{N}_t = \beta(\nabla f(W_{t-1}) - \bar{N}_{t-1})$$

因此, 我们有

$$\begin{aligned} D_t &= \beta\|\nabla f(W_{t-1}) - \bar{N}_{t-1}\|_* \\ &\leq \beta\|\nabla f(W_{t-2}) - \bar{N}_{t-1}\| + \beta\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \\ &= \beta D_{t-1} + \beta\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_*. \end{aligned}$$

应用假设1, 并利用 $\|O_{t-1}\|_{\text{op}} \leq 1 + \varepsilon_{t-1,q} \leq 1 + \varepsilon_q$ 这一事实, 我们得到

$$\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \leq L\|W_{t-1} - W_{t-2}\|_{\text{op}} = L\eta\|O_{t-1}\|_{\text{op}} \leq L\eta(1 + \varepsilon_q)$$

因此, 我们有以下递推关系

$$D_t \leq \beta D_{t-1} + \beta L\eta(1 + \varepsilon_q)$$

由于 $N_0 = 0$, 我们得到

$$D_1 = \|\nabla f(W_0) - \bar{N}_1\|_* = \|\nabla f(W_0) - (\beta\bar{N}_0 + (1 - \beta)\nabla f(W_0))\|_* = \beta\|\nabla f(W_0)\|_*$$

通过展开递推关系并取期望, 我们得到

$$\mathbb{E}[D_t] \leq \beta^t \mathbb{E}[\|\nabla f(W_0)\|_*] + \sum_{i=1}^t \beta^i L\eta(1 + \varepsilon_q) \leq \beta^t \|\nabla f(W_0)\|_* + \frac{\beta L\eta(1 + \varepsilon_q)}{1 - \beta} \quad (4)$$

界限项 (B). 当我们展开 N_t 和 N_t 的EMA递推关系时,

$$N_t = \beta N_{t-1} + (1 - \beta)G_t = \beta^t N_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} G_i$$

$$\bar{N}_t = \beta \bar{N}_{t-1} + (1 - \beta)\nabla f(W_t) = \beta^t \bar{N}_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} \nabla f(W_i)$$

然后, 我们计算由 M_t 和 M_t 引起的Frobenius范数偏差的期望,

$$\mathbb{E}[\|N_t - \bar{N}_t\|_F] \leq (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] \quad (5)$$

应用詹森不等式和期望的线性, 可以得到

$$\begin{aligned} (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] &\leq \sqrt{(1 - \beta)^2 \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\mathbb{E} \left[\sum_{i=1}^t \beta^{2(t-i)} \|G_i - \nabla f(W_i)\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \end{aligned}$$

By Lemma 8, Eq.(5) becomes

$$\begin{aligned}\mathbb{E}[\|N_t - \bar{N}_t\|_F] &\leq (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \\ &\leq (1 - \beta) \sqrt{\frac{1 - \beta^{2t}}{1 - \beta^2} \frac{\sigma^2}{B}} \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sigma}{\sqrt{B}}\end{aligned}$$

Using the fact that $\|X\|_* \leq \sqrt{r}\|X\|_F$ for $X \in \mathbb{R}^{m \times n}$ with $r = \min\{m, n\}$ (Proposition 2(iii)), we have

$$\mathbb{E}[\|\bar{N}_t - N_t\|_*] \leq \sqrt{r} \mathbb{E}[\|\bar{N}_t - N_t\|_F] \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (6)$$

Plugging Eq.(4) and Eq.(6) into Eq.(3), we obtain

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (7)$$

Averaging and tuning. Plugging Eq.(7) into Eq.(2), we get

$$\begin{aligned}\mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta(1 - \varepsilon_q)} \\ &\quad + \frac{2}{1 - \varepsilon_q} \left(\frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{L \eta (1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}\end{aligned}$$

Averaging over $t = 1, \dots, T$, we obtain

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{D}{T \eta (1 - \varepsilon_q)} + \frac{2}{1 - \varepsilon_q} \left(\frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) \\ &\quad + \frac{2}{1 - \varepsilon_q} \left(\frac{1}{T} \sum_{t=1}^T \beta^t \|\nabla f(W_0)\|_* \right) + \frac{L \eta (1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}\end{aligned}$$

Using Proposition 2(iii) and Lemma 7, $\|\nabla f(W_0)\|_* \leq \sqrt{r} \|\nabla f(W_0)\|_F \leq \sqrt{2rLD}$, where $D = f(W_0) - f^*$. Then, we have

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{D}{T \eta (1 - \varepsilon_q)} + \frac{2}{1 - \varepsilon_q} \left(\frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) \\ &\quad + \frac{2\beta\sqrt{2rLD}}{(1 - \varepsilon_q)T(1 - \beta)} + \frac{L \eta (1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}\end{aligned}$$

where $D = f(W_0) - f^*$. By setting $\eta = \sqrt{\frac{(1 - \beta)D}{TL}}$, we obtain

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{1}{1 - \varepsilon_q} \left(\frac{D}{T \eta} + \frac{L \eta}{2} (1 + \varepsilon_q)^2 + \frac{2\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \right) \\ &= \frac{1}{1 - \varepsilon_q} \left(\sqrt{\frac{LD}{T}} \left(\frac{1 + 2\beta(1 + \varepsilon_q)}{\sqrt{1 - \beta}} + \frac{(1 + \varepsilon_q)^2 \sqrt{1 - \beta}}{2} \right) + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \right)\end{aligned}$$

Setting $\beta = 1 - \min\left\{\frac{\sqrt{LD\beta}}{\sigma\sqrt{rT}}, 1\right\}$, we get

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{1}{1 - \varepsilon_q} \left(\frac{(1 + \varepsilon_q)^2 \sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \frac{1 + 2\beta(1 + \varepsilon_q)}{\sqrt{1 - \beta}} \sqrt{\frac{LD}{T}} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \right) \\ &= \frac{1}{1 - \varepsilon_q} \left(\frac{(1 + \varepsilon_q)^2 \sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \left(1 + 2\beta(1 + \varepsilon_q) + \frac{2}{\sqrt{1 + \beta}} \right) \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} + 2\sqrt{2}\beta \frac{\sigma r}{\sqrt{BT}} \right) \\ &= \frac{1}{1 - \varepsilon_q} \cdot \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} \right) \quad (8)\end{aligned}$$

根据引理8, 公式 (5) 变为

$$\begin{aligned}\mathbb{E}[\|N_t - \bar{N}_t\|_F] &\leq (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \\ &\leq (1 - \beta) \sqrt{\frac{1 - \beta^{2t}}{1 - \beta^2} \frac{\sigma^2}{B}} \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sigma}{\sqrt{B}}\end{aligned}$$

利用事实 $\|X\|_* \leq \sqrt{r}\|X\|_F$ 对于 $X \in \mathbb{R}^{m \times n}$ 使用 $r = \min\{m, n\}$ (命题 2(iii)) , 我们得到

$$\mathbb{E}[\|\bar{N}_t - N_t\|_*] \leq \sqrt{r} \mathbb{E}[\|\bar{N}_t - N_t\|_F] \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (6)$$

将公式 (4) 和公式 (6) 代入公式 (3), 我们得到

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (7)$$

平均和调整. 将公式 (7) 代入公式 (2), 我们得到

$$\begin{aligned}\mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta(1 - \varepsilon_q)} \\ &\quad + \frac{2}{1 - \varepsilon_q} \left(\frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{L \eta (1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}\end{aligned}$$

对 $t = 1, \dots, T$ 进行平均, 我们得到

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{D}{T \eta (1 - \varepsilon_q)} + \frac{2}{1 - \varepsilon_q} \left(\frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) \\ &\quad + \frac{2}{1 - \varepsilon_q} \left(\frac{1}{T} \sum_{t=1}^T \beta^t \|\nabla f(W_0)\|_* \right) + \frac{L \eta (1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}\end{aligned}$$

使用命题 2(iii) 和引理7, $\|\nabla f(W_0)\|_* \leq \sqrt{r} \|\nabla f(W_0)\|_F \leq \sqrt{2rLD}$, 其中 $D = f(W_0) - f^*$ 。然后, 我们得到

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{D}{T \eta (1 - \varepsilon_q)} + \frac{2}{1 - \varepsilon_q} \left(\frac{\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) \\ &\quad + \frac{2\beta\sqrt{2rLD}}{(1 - \varepsilon_q)T(1 - \beta)} + \frac{L \eta (1 + \varepsilon_q)^2}{2(1 - \varepsilon_q)}\end{aligned}$$

在 $D = f(W_0) - f^*$ 处。通过设置 $\eta = \sqrt{\frac{(1 - \beta)D}{TL}}$, 我们得到

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{1}{1 - \varepsilon_q} \left(\frac{D}{T \eta} + \frac{L \eta}{2} (1 + \varepsilon_q)^2 + \frac{2\beta L \eta (1 + \varepsilon_q)}{1 - \beta} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \right) \\ &= \frac{1}{1 - \varepsilon_q} \left(\sqrt{\frac{LD}{T}} \left(\frac{1 + 2\beta(1 + \varepsilon_q)}{\sqrt{1 - \beta}} + \frac{(1 + \varepsilon_q)^2 \sqrt{1 - \beta}}{2} \right) + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \right)\end{aligned}$$

设置 $\beta = 1 - \min\left\{\frac{\sqrt{LD\beta}}{\sigma\sqrt{rT}}, 1\right\}$, 我

们得到

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{1}{1 - \varepsilon_q} \left(\frac{(1 + \varepsilon_q)^2 \sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \frac{1 + 2\beta(1 + \varepsilon_q)}{\sqrt{1 - \beta}} \sqrt{\frac{LD}{T}} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \right) \\ &= \frac{1}{1 - \varepsilon_q} \left(\frac{(1 + \varepsilon_q)^2 \sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \left(1 + 2\beta(1 + \varepsilon_q) + \frac{2}{\sqrt{1 + \beta}} \right) \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} + 2\sqrt{2}\beta \frac{\sigma r}{\sqrt{BT}} \right) \\ &= \frac{1}{1 - \varepsilon_q} \cdot \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} \right) \quad (8)\end{aligned}$$

From NEWTON–SCHULZ residual to factor χ_q . By the NEWTON–SCHULZ residual–error link (Lemma 1) and the residual contraction (Lemma 3) as assembled in Corollary 1, we have

$$\varepsilon_q = \sup_t \varepsilon_{t,q} = \sup_t \left(1 - \sqrt{1 - \delta_{t,q}}\right) \leq 1 - \sqrt{1 - \delta_0^{(\kappa+1)^q}}.$$

Hence, we bound the factor χ_q as

$$\chi_q = (1 - \varepsilon_q)^{-1} \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}$$

with $\delta_0 = \sup_t \delta_{t,0} < 1$ (Remark 1). Finally, we conclude from Eq.(8) that

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(W_{t-1})\|_* \leq \chi_q \cdot \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} \right), \quad \chi_q = \frac{1}{1 - \varepsilon_q},$$

with the constant factor bounded by

$$\chi_q \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}.$$

Finally, we can find an ϵ -nuclear norm stationary point of f with a complexity of

$$\mathcal{O} \left(\max \left\{ \frac{\chi_q^2 LD}{\epsilon^2}, \frac{\chi_q^2 r^2 \sigma^2}{B\epsilon^2}, \frac{\chi_q^4 r \sigma^2 LD}{B\epsilon^4} \right\} \right)$$

□

从 NEWTON–SCHULZ 残差到因式分解 χ_q . 根据 NEWTON–SCHULZ 残差-误差关系 (引理1) 以及 Corollary 1 中组装的残差收缩 (引理3), 我们有

$$\varepsilon_q = \sup_t \varepsilon_{t,q} = \sup_t \left(1 - \sqrt{1 - \delta_{t,q}}\right) \leq 1 - \sqrt{1 - \delta_0^{(\kappa+1)^q}}.$$

因此, 我们将因子 χ_q 界定为

$$\chi_q = (1 - \varepsilon_q)^{-1} \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}$$

其中 $\delta_0 = \sup_t \delta_{t,0} < 1$ (注释1)。最后, 我们从公式(8)得出

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(W_{t-1})\|_* \leq \chi_q \cdot \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} \right), \quad \chi_q = \frac{1}{1 - \varepsilon_q},$$

其中常数因子被界定为

$$\chi_q \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}.$$

最后, 我们可以找到一个 ϵ -核范数驻点, 其复杂度为

$$\mathcal{O} \left(\max \left\{ \frac{\chi_q^2 LD}{\epsilon^2}, \frac{\chi_q^2 r^2 \sigma^2}{B\epsilon^2}, \frac{\chi_q^4 r \sigma^2 LD}{B\epsilon^4} \right\} \right)$$

□

C MUON WITH SVD AND SGD WITH MOMENTUM

C.1 THEOREM 4 (MUON WITH SVD)

Algorithm 2 MUON with SVD

Require: learning rate $\eta > 0$, momentum $\beta \in [0, 1]$, NEWTON–SCHULZ steps $q \in \mathbb{N}$, batch size B , Total iteration T .

```

1: Initialize:  $M_0 \leftarrow 0, W_0 \in \mathbb{R}^{m \times n}$ 
2: for  $t = 1$  to  $T$  do
3:    $G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_{t-1}; \xi_{t,i})$ 
4:    $M_t \leftarrow \beta M_{t-1} + G_t$ 
5:    $O_t \leftarrow \text{Polar}(M_t)$  (SVD( $M_t$ ) = ( $U_t, \Sigma_t, V_t$ ), then  $\text{Polar}(M_t) = U_t V_t^\top$ )
6:    $W_t \leftarrow W_{t-1} - \eta O_t$ 
7: end for
8: return  $W_T$ 

```

Proof of Theorem 4. First, we introduce the *scaled* EMA momentum: $N_t := (1 - \beta)M_t$. Then, we get $N_t = \beta N_{t-1} + (1 - \beta)G_t$. Note that the polar factor is scale-invariant (Proposition 2(v)). Let P_t be the polar factor of M_t , i.e., $P_t = \text{Polar}(M_t)$. Hence, $P_t := \text{Polar}(N_t) = \text{Polar}(M_t)$.

By Assumption 1, we start from *descent lemma* (Lemma 6). Since $W_t = W_{t-1} - \eta O_t$, we have

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) + \langle \nabla f(W_{t-1}), W_t - W_{t-1} \rangle_F + \frac{L}{2} \|W_t - W_{t-1}\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), O_t \rangle_F + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), P_t \rangle_F + \frac{L}{2} \eta^2 \|P_t\|_{\text{op}}^2
\end{aligned}$$

where the last equality is due to $O_t = P_t = \text{Polar}(M_t)$. Let $\text{SVD}(M_t) = (U_t, \Sigma_t, V_t)$. Then $P_t = \text{Polar}(M_t) = U_t V_t^\top$. Since $P_t = \text{Polar}(M_t)$ is a partial isometry, $\|P_t\|_{\text{op}} \leq 1$ (and $\|P_t\|_{\text{op}} = 1$ if $\text{rank}(M_t) > 0$). Then, we have

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), P_t \rangle_F + \frac{L}{2} \eta^2 \\
&= f(W_{t-1}) - \eta \langle N_t, P_t \rangle_F + \eta \langle N_t - \nabla f(W_{t-1}), P_t \rangle_F + \frac{L\eta^2}{2} \\
&= f(W_{t-1}) - \eta \|N_t\|_* + \eta \langle N_t - \nabla f(W_{t-1}), P_t \rangle_F + \frac{L\eta^2}{2}
\end{aligned}$$

where the last equality is due to $\langle N_t, P_t \rangle_F = \|N_t\|_*$ (Proposition 2(ii)). By Hölder’s inequality (Lemma 4) and the triangle inequality, we have

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) - \eta \|N_t\|_* + \eta \|N_t - \nabla f(W_{t-1})\|_* \|P_t\|_{\text{op}} + \frac{L\eta^2}{2} \\
&\leq f(W_{t-1}) - \eta \|\nabla f(W_{t-1})\|_* + \eta \|\nabla f(W_{t-1}) - N_t\|_* + \eta \|N_t - \nabla f(W_{t-1})\|_* + \frac{L\eta^2}{2} \\
&= f(W_{t-1}) - \eta \|\nabla f(W_{t-1})\|_* + 2\eta \|\nabla f(W_{t-1}) - N_t\|_* + \frac{L\eta^2}{2}.
\end{aligned}$$

Rearranging and taking expectation gives

$$\mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta} + 2\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] + \frac{L\eta}{2} \quad (9)$$

Bounding $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$.

In order to bound $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$, we introduce the true scaled momentum $\bar{N}_t$ defined by the true (full-batch) gradient $\nabla f(W_t)$ instead of G_t for each step t :

C 期望 MUON WITHSVD AND 带动量的SGD

C.1 定理4（带SVD的μ子）

算法2 MUON with SVD

要求: 学习率 $\eta > 0$, 动量 $\beta \in [0, 1]$, NEWTON–SCHULZ步数 $q \in \mathbb{N}$, 批量大小 B , 总迭代次数 T . **1: 初始化:** $M_0 \leftarrow 0, W_0 \in \mathbb{R}^{m \times n}$ **2: for** $t = 1$ 到 T **do** **3:** $G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_{t-1}; \xi_{t,i})$ **4:** $M_t \leftarrow \beta M_{t-1} + G_t$ **5:** $O_t \leftarrow \text{Polar}(M_t)$ (SVD(M_t) = (U_t, Σ_t, V_t), 然后 $\text{Polar}(M_t) = U_t V_t^\top$) **6:** $W_t \leftarrow W_{t-1} - \eta O_t$ **7: end for** **8: return** W_T

定理4的证明。首先，我们引入<em id='3'>缩放EMA动量: $N_t := (1 - \beta)M_{t_0}$ 。然后，我们得到 $N_t = \beta N_{t-1} + (1 - \beta)G_{t_0}$ 。注意极因子是尺度不变的（命题2(v)）。令 P_t 为 M_t 的极因子，即 $P_t = \text{Polar}(M_t)$ 。因此， $P_t := \text{Polar}(N_t) = \text{Polar}(M_t)$ 。

根据假设1，我们从<em id='1'>下降引理(引理6)开始。由于 $W_t = W_{t-1} - \eta O_t$ ，我们有

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) + \langle \nabla f(W_{t-1}), W_t - W_{t-1} \rangle_F + \frac{L}{2} \|W_t - W_{t-1}\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), O_t \rangle_F + \frac{L}{2} \eta^2 \|O_t\|_{\text{op}}^2 \\
&= f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), P_t \rangle_F + \frac{L}{2} \eta^2 \|P_t\|_{\text{op}}^2
\end{aligned}$$

其中最后一个等式是由于 $O_t = P_t = \text{Polar}(M_t)$ 。令 $\text{SVD}(M_t) = (U_t, \Sigma_t, V_t)$ 。然后 $P_t = \text{Polar}(M_t) = U_t V_t^\top$ 。由于 $P_t = \text{Polar}(M_t)$ 是一个部分等距映射， $\|P_t\|_{\text{op}} \leq 1$ (并且 $\|P_t\|_{\text{op}} = 1$ 如果 $\text{rank}(M_t) > 0$)。然后，我们有

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) - \eta \langle \nabla f(W_{t-1}), P_t \rangle_F + \frac{L}{2} \eta^2 \\
&= f(W_{t-1}) - \eta \langle N_t, P_t \rangle_F + \eta \langle N_t - \nabla f(W_{t-1}), P_t \rangle_F + \frac{L\eta^2}{2} \\
&= f(W_{t-1}) - \eta \|N_t\|_* + \eta \langle N_t - \nabla f(W_{t-1}), P_t \rangle_F + \frac{L\eta^2}{2}
\end{aligned}$$

其中最后一个等式是由于 $\langle N_t, P_t \rangle_F = \|N_t\|_*$ (命题2(ii))。根据 Hölder不等式(引理4)和三角不等式，我们有

$$\begin{aligned}
f(W_t) &\leq f(W_{t-1}) - \eta \|N_t\|_* + \eta \|N_t - \nabla f(W_{t-1})\|_* \|P_t\|_{\text{op}} + \frac{L\eta^2}{2} \\
&\leq f(W_{t-1}) - \eta \|\nabla f(W_{t-1})\|_* + \eta \|\nabla f(W_{t-1}) - N_t\|_* + \eta \|N_t - \nabla f(W_{t-1})\|_* + \frac{L\eta^2}{2} \\
&= f(W_{t-1}) - \eta \|\nabla f(W_{t-1})\|_* + 2\eta \|\nabla f(W_{t-1}) - N_t\|_* + \frac{L\eta^2}{2}.
\end{aligned}$$

重新排列并取期望得到

$$\mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta} + 2\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] + \frac{L\eta}{2} \quad (9)$$

界定 $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$ 。

为了界定 $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$ ，我们引入了由真实（全批次）梯度 $\nabla f(W_t)$ 定义的真实缩放动量 N_t ，而不是为每一步 t 使用 G_t ：

- $\bar{N}_t = \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_{t-1})$ for $t > 0$ and $\bar{N}_0 = 0$.
- Note that $N_t = \beta N_{t-1} + (1 - \beta) G_t$ for $t > 0$ and $N_0 = 0$.

Then we can decompose $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$ as

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \underbrace{\mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_*]}_{\text{Term (A)}} + \underbrace{\mathbb{E}[\|\bar{N}_t - N_t\|_*]}_{\text{Term (B)}} \quad (10)$$

Bounding the Term (A). Let $D_t = \|\nabla f(W_{t-1}) - \bar{N}_t\|_*$. From the recursion,

$$\nabla f(W_{t-1}) - \bar{N}_t = \beta(\nabla f(W_{t-1}) - \bar{N}_{t-1})$$

Hence, we have

$$\begin{aligned} D_t &= \beta \|\nabla f(W_{t-1}) - \bar{N}_{t-1}\|_* \\ &\leq \beta \|\nabla f(W_{t-2}) - \bar{N}_{t-1}\| + \beta \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \\ &= \beta D_{t-1} + \beta \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_*. \end{aligned}$$

Applying Assumption 1 and using the fact that $\|O_{t-1}\|_{\text{op}} = \|P_{t-1}\|_{\text{op}} \leq 1$, we have

$$\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \leq L \|W_{t-1} - W_{t-2}\|_{\text{op}} = L\eta \|O_{t-1}\|_{\text{op}} \leq L\eta$$

Hence, we have the following recursion,

$$D_t \leq \beta D_{t-1} + \beta L\eta$$

Since $\bar{N}_0 = 0$, we have

$$D_1 = \|\nabla f(W_0) - \bar{N}_1\|_* = \|\nabla f(W_0) - (\beta \bar{N}_0 + (1 - \beta) \nabla f(W_0))\|_* = \beta \|\nabla f(W_0)\|_*$$

By unrolling the recursion and taking the expectation, we get

$$\mathbb{E}[D_t] \leq \beta^t \mathbb{E}[\|\nabla f(W_0)\|_*] + \sum_{i=1}^t \beta^i L\eta \leq \beta^t \|\nabla f(W_0)\|_* + \frac{\beta L\eta}{1 - \beta} \quad (11)$$

Bounding the Term (B). When we unroll the EMA recursion of both N_t and $\bar{N}_t$,

$$\begin{aligned} N_t &= \beta N_{t-1} + (1 - \beta) G_t = \beta^t N_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} G_i \\ \bar{N}_t &= \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t) = \beta^t \bar{N}_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} \nabla f(W_i) \end{aligned}$$

Then, we compute the expectation of the Frobenius norm bias caused by M_t and $\bar{M}_t$,

$$\mathbb{E}[\|N_t - \bar{N}_t\|_F] \leq (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] \quad (12)$$

Applying Jensen's inequality and the linearity of expectation gives

$$\begin{aligned} (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] &\leq \sqrt{(1 - \beta)^2 \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\mathbb{E} \left[\sum_{i=1}^t \beta^{2(t-i)} \|G_i - \nabla f(W_i)\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \end{aligned}$$

- $\bar{N}_t = \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_{t-1})$ for $t > 0$ and $\bar{N}_0 = 0$.
- Note that $N_t = \beta N_{t-1} + (1 - \beta) G_t$ for $t > 0$ and $N_0 = 0$.

然后我们可以将 $\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*]$ 分解为

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \underbrace{\mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_*]}_{\text{Term (A)}} + \underbrace{\mathbb{E}[\|\bar{N}_t - N_t\|_*]}_{\text{Term (B)}} \quad (10)$$

界限项 (A). 令 $D_t = \|\nabla f(W_{t-1}) - \bar{N}_t\|_*$. 根据递归关系,

$$\nabla f(W_{t-1}) - \bar{N}_t = \beta(\nabla f(W_{t-1}) - \bar{N}_{t-1})$$

因此, 我们有

$$\begin{aligned} D_t &= \beta \|\nabla f(W_{t-1}) - \bar{N}_{t-1}\|_* \\ &\leq \beta \|\nabla f(W_{t-2}) - \bar{N}_{t-1}\| + \beta \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \\ &= \beta D_{t-1} + \beta \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_*. \end{aligned}$$

应用假设1并利用 $\|O_{t-1}\|_{\text{op}} = \|P_{t-1}\|_{\text{op}} \leq 1$ 这一事实, 我们有

$$\|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_* \leq L \|W_{t-1} - W_{t-2}\|_{\text{op}} = L\eta \|O_{t-1}\|_{\text{op}} \leq L\eta$$

因此, 我们有以下递归关系,

$$D_t \leq \beta D_{t-1} + \beta L\eta$$

由于 $N_0 = 0$, 我们有

$$D_1 = \|\nabla f(W_0) - \bar{N}_1\|_* = \|\nabla f(W_0) - (\beta \bar{N}_0 + (1 - \beta) \nabla f(W_0))\|_* = \beta \|\nabla f(W_0)\|_*$$

通过展开递归并取期望, 我们得到

$$\mathbb{E}[D_t] \leq \beta^t \mathbb{E}[\|\nabla f(W_0)\|_*] + \sum_{i=1}^t \beta^i L\eta \leq \beta^t \|\nabla f(W_0)\|_* + \frac{\beta L\eta}{1 - \beta} \quad (11)$$

界限项 (B). 当我们展开 N_t 和 $\bar{N}_t$ 的 EMA 递归时,

$$\begin{aligned} N_t &= \beta N_{t-1} + (1 - \beta) G_t = \beta^t N_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} G_i \\ \bar{N}_t &= \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t) = \beta^t \bar{N}_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} \nabla f(W_i) \end{aligned}$$

然后, 我们计算由 M_t 和 $\bar{M}_t$ 引起的 Frobenius 范数偏差的期望,

$$\mathbb{E}[\|N_t - \bar{N}_t\|_F] \leq (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] \quad (12)$$

应用詹森不等式和期望的线性, 可以得到

$$\begin{aligned} (1 - \beta) \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F \right] &\leq \sqrt{(1 - \beta)^2 \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\mathbb{E} \left[\sum_{i=1}^t \beta^{2(t-i)} \|G_i - \nabla f(W_i)\|_F^2 \right]} \\ &= (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \end{aligned}$$

By Lemma 8, Eq.(12) becomes

$$\begin{aligned}\mathbb{E}[\|N_t - \bar{N}_t\|_F] &\leq (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \\ &\leq (1 - \beta) \sqrt{\frac{1 - \beta^{2t}}{1 - \beta^2} \frac{\sigma^2}{B}} \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sigma}{\sqrt{B}}\end{aligned}$$

Using the fact that $\|X\|_* \leq \sqrt{r}\|X\|_F$ for $X \in \mathbb{R}^{m \times n}$ with $r = \min\{m, n\}$ (Proposition 2(iii)), we have

$$\mathbb{E}[\|\bar{N}_t - N_t\|_*] \leq \sqrt{r} \mathbb{E}[\|\bar{N}_t - N_t\|_F] \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (13)$$

Plugging Eq.(11) and Eq.(13) into Eq.(10), we obtain

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \frac{\beta L \eta}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (14)$$

Averaging and tuning. Plugging Eq.(14) into Eq.(9), we get

$$\mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta} + 2 \left(\frac{\beta L \eta}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{L \eta}{2}$$

Averaging over $t = 1, \dots, T$, we obtain

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{D}{T \eta} + 2 \left(\frac{\beta L \eta}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{2}{T} \sum_{t=1}^T \beta^t \|\nabla f(W_0)\|_* + \frac{L \eta}{2}$$

Using Proposition 2(iii) and Lemma 7, $\|\nabla f(W_0)\|_* \leq \sqrt{r} \|\nabla f(W_0)\|_F \leq \sqrt{2rLD}$, where $D = f(W_0) - f^*$. Then, we have

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{D}{T \eta} + 2 \left(\frac{\beta L \eta}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} + \frac{L \eta}{2}$$

where $D = f(W_0) - f^*$. By setting $\eta = \sqrt{\frac{(1 - \beta)D}{TL}}$, we obtain

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{D}{T \eta} + \frac{L \eta}{2} + \frac{2\beta L \eta}{1 - \beta} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \\ &= \sqrt{\frac{LD}{T}} \left(\frac{1 + 2\beta}{\sqrt{1 - \beta}} + \frac{\sqrt{1 - \beta}}{2} \right) + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)}\end{aligned}$$

Setting $\beta = 1 - \min \left\{ \frac{\sqrt{LD\bar{B}}}{\sigma\sqrt{rT}}, 1 \right\}$, we get

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{\sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \frac{1 + 2\beta}{\sqrt{1 - \beta}} \sqrt{\frac{LD}{T}} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \\ &= \frac{\sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \left(1 + 2\beta + \frac{2}{\sqrt{1 + \beta}} \right) \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} + 2\sqrt{2}\beta \frac{\sigma r}{\sqrt{BT}} \\ &= \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} \right) \quad (15)\end{aligned}$$

Thus, we can find an ϵ -nuclear norm stationary point of f with a complexity of

$$\mathcal{O} \left(\max \left\{ \frac{LD}{\epsilon^2}, \frac{r^2 \sigma^2}{B \epsilon^2}, \frac{r \sigma^2 LD}{B \epsilon^4} \right\} \right)$$

□

根据引理8, 公式(12)变为

$$\begin{aligned}\mathbb{E}[\|N_t - \bar{N}_t\|_F] &\leq (1 - \beta) \sqrt{\sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2]} \\ &\leq (1 - \beta) \sqrt{\frac{1 - \beta^{2t}}{1 - \beta^2} \frac{\sigma^2}{B}} \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sigma}{\sqrt{B}}\end{aligned}$$

利用事实 $\|X\|_* \leq \sqrt{r}\|X\|_F$ 对于 $X \in \mathbb{R}^{m \times n}$ 使用 $r = \min\{m, n\}$ (命题 2(iii)), 我们得到

$$\mathbb{E}[\|\bar{N}_t - N_t\|_*] \leq \sqrt{r} \mathbb{E}[\|\bar{N}_t - N_t\|_F] \leq \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (13)$$

将公式(11)和公式(13)代入公式(10), 我们得到

$$\mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_*] \leq \frac{\beta L \eta}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \quad (14)$$

平均和调整. 将公式(14)代入公式(9), 我们得到

$$\mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\eta} + 2 \left(\frac{\beta L \eta}{1 - \beta} + \beta^t \|\nabla f(W_0)\|_* + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{L \eta}{2}$$

对 $t = 1, \dots, T$ 进行平均, 我们得到

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{D}{T \eta} + 2 \left(\frac{\beta L \eta}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{2}{T} \sum_{t=1}^T \beta^t \|\nabla f(W_0)\|_* + \frac{L \eta}{2}$$

使用命题 2(iii) 和引理7, $\|\nabla f(W_0)\|_* \leq \sqrt{r} \|\nabla f(W_0)\|_F \leq \sqrt{2rLD}$, 其中 $D = f(W_0) - f^*$. 然后, 我们得到

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] \leq \frac{D}{T \eta} + 2 \left(\frac{\beta L \eta}{1 - \beta} + \sqrt{\frac{1 - \beta}{1 + \beta}} \frac{\sqrt{r}\sigma}{\sqrt{B}} \right) + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} + \frac{L \eta}{2}$$

在 $D = f(W_0) - f^*$ 处。通过设置 $\eta = \sqrt{\frac{(1 - \beta)D}{TL}}$, 我们

得到

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{D}{T \eta} + \frac{L \eta}{2} + \frac{2\beta L \eta}{1 - \beta} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \\ &= \sqrt{\frac{LD}{T}} \left(\frac{1 + 2\beta}{\sqrt{1 - \beta}} + \frac{\sqrt{1 - \beta}}{2} \right) + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)}\end{aligned}$$

设置 $\beta = 1 - \min \left\{ \frac{\sqrt{LD\bar{B}}}{\sigma\sqrt{rT}}, 1 \right\}$, 我们得

到

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_*] &\leq \frac{\sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \frac{1 + 2\beta}{\sqrt{1 - \beta}} \sqrt{\frac{LD}{T}} + \frac{2\sigma\sqrt{r}}{\sqrt{B}} \sqrt{\frac{1 - \beta}{1 + \beta}} + \frac{2\beta\sqrt{2rLD}}{T(1 - \beta)} \\ &= \frac{\sqrt{1 - \beta}}{2} \sqrt{\frac{LD}{T}} + \left(1 + 2\beta + \frac{2}{\sqrt{1 + \beta}} \right) \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} + 2\sqrt{2}\beta \frac{\sigma r}{\sqrt{BT}} \\ &= \mathcal{O} \left(\sqrt{\frac{LD}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 LD}{BT} \right)^{1/4} \right) \quad (15)\end{aligned}$$

因此, 我们可以找到一个具有 ϵ -核范数驻点, 其复杂度为

$$\mathcal{O} \left(\max \left\{ \frac{LD}{\epsilon^2}, \frac{r^2 \sigma^2}{B \epsilon^2}, \frac{r \sigma^2 LD}{B \epsilon^4} \right\} \right)$$

□

Algorithm 3 SGD with momentum (SGD-M)**Require:** learning rate $\eta > 0$, momentum $\beta \in [0, 1)$, batch size B

```

1: Initialize:  $M_0 \leftarrow 0$ ,  $W_0 \in \mathbb{R}^{m \times n}$ 
2: for  $t = 1$  to  $T$  do
3:    $G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_t; \xi_{t,i})$ 
4:    $M_t \leftarrow \beta M_{t-1} + G_t$ 
5:    $W_t \leftarrow W_{t-1} - \eta M_t$ 
6: end for
7: return  $W_T$ 

```

C.2 THEOREM 3 (SGD WITH MOMENTUM)

Proof of Theorem 3. First, we introduce the *scaled* EMA momentum: $N_t := (1 - \beta)M_t$. Then, we get $N_t = \beta N_{t-1} + (1 - \beta)G_t$ with $N_0 = 0$ and the scaled learning rate $\tilde{\eta} := \frac{\eta}{1-\beta}$, yielding $W_t = W_{t-1} - \tilde{\eta}N_t$.

By Assumption 1, we start from *the descent lemma* (Lemma 6). Since $W_t = W_{t-1} - \tilde{\eta}N_t$, we have

$$\begin{aligned} f(W_t) &\leq f(W_{t-1}) + \langle \nabla f(W_{t-1}), W_t - W_{t-1} \rangle_F + \frac{L}{2} \|W_t - W_{t-1}\|_{\text{op}}^2 \\ &= f(W_{t-1}) - \tilde{\eta} \langle \nabla f(W_{t-1}), N_t \rangle_F + \frac{L}{2} \tilde{\eta}^2 \|N_t\|_{\text{op}}^2 \end{aligned}$$

By using $\langle a, b \rangle_F = \frac{1}{2} (\|a\|_F^2 + \|b\|_F^2 - \|a - b\|_F^2)$, we obtain

$$\begin{aligned} f(W_t) &\leq f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 - \frac{\tilde{\eta}}{2} \|N_t\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 + \frac{L}{2} \tilde{\eta}^2 \|N_t\|_{\text{op}}^2 \\ &\leq f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 - \frac{\tilde{\eta}}{2} \|N_t\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 + \frac{L}{2} \tilde{\eta}^2 \|N_t\|_F^2 \\ &= f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 - \frac{\tilde{\eta}(1 - L\tilde{\eta})}{2} \|N_t\|_F^2 \\ &\leq f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 \end{aligned}$$

where the second inequality is due to $\|N_t\|_{\text{op}} \leq \|N_t\|_F$ (Proposition 2(iv)). The last inequality holds by choosing $\tilde{\eta} \leq 1/L$ so that we can drop the last term.

Rearranging and taking expectation gives

$$\mathbb{E} [\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\tilde{\eta}} + \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2]$$

Averaging over $t = 1$ to T , we obtain

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2D}{T\tilde{\eta}} + \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2] \quad (16)$$

where $D = f(W_0) - f^*$. Denote

$$S_A := \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2D}{T\tilde{\eta}} + S_B, \quad S_B := \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2] \quad (17)$$

Bounding $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2]$.

We introduce the true scaled momentum $\bar{N}_t$ defined by the true (full-batch) gradient $\nabla f(W_t)$ instead of G_t for each step t :

- $\bar{N}_t = \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t)$ for $t > 0$ and $\bar{N}_0 = 0$.

算法3 带动量的SGD (SGD-M)**要求:** 学习率 $\eta > 0$, 动量 $\beta \in [0, 1)$, 批量大小 B

```

1: 初始化:  $M_0 \leftarrow 0$ ,  $W_0 \in \mathbb{R}^{m \times n}$ 
2: 对于  $t = 1$  到  $T$  执行
3:    $G_t \leftarrow \frac{1}{B} \sum_{i=1}^B \nabla f(W_t; \xi_{t,i})$ 
4:    $M_t \leftarrow \beta M_{t-1} + G_t$ 
5:    $W_t \leftarrow W_{t-1} - \eta M_t$ 
6: 结束for循环
7: 返回  $W_T$ 

```

C.2 定理3 (SGDWITH动量)

定理3的证明。首先，我们引入缩放EMA动量： $N_t := (1 - \beta)M_t$ 。然后，我们得到 $N_t = \beta N_{t-1} + (1 - \beta)G_t$ ，其中 $N_0 = 0$ ，以及缩放学习率 $\tilde{\eta} := \frac{\eta}{1-\beta}$ ，从而得到 $W_t = W_{t-1} - \tilde{\eta}N_t$ 。

根据假设1，我们从下降引理 (引理6)开始。由于 $W_t = W_{t-1} - \tilde{\eta}N_t$ ，我们有

$$\begin{aligned} f(W_t) &\leq f(W_{t-1}) + \langle \nabla f(W_{t-1}), W_t - W_{t-1} \rangle_F + \frac{L}{2} \|W_t - W_{t-1}\|_{\text{op}}^2 \\ &= f(W_{t-1}) - \tilde{\eta} \langle \nabla f(W_{t-1}), N_t \rangle_F + \frac{L}{2} \tilde{\eta}^2 \|N_t\|_{\text{op}}^2 \end{aligned}$$

通过使用 $\langle a, b \rangle_F = \frac{1}{2} (\|a\|_F^2 + \|b\|_F^2 - \|a - b\|_F^2)$ ，我们得到

$$\begin{aligned} f(W_t) &\leq f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 - \frac{\tilde{\eta}}{2} \|N_t\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 + \frac{L}{2} \tilde{\eta}^2 \|N_t\|_{\text{op}}^2 \\ &\leq f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 - \frac{\tilde{\eta}}{2} \|N_t\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 + \frac{L}{2} \tilde{\eta}^2 \|N_t\|_F^2 \\ &= f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 - \frac{\tilde{\eta}(1 - L\tilde{\eta})}{2} \|N_t\|_F^2 \\ &\leq f(W_{t-1}) - \frac{\tilde{\eta}}{2} \|\nabla f(W_{t-1})\|_F^2 + \frac{\tilde{\eta}}{2} \|N_t - \nabla f(W_{t-1})\|_F^2 \end{aligned}$$

其中第二个不等式是由于 $\|N_t\|_{\text{op}} \leq \|N_t\|_F$ (命题2(iv))。最后一个不等式通过选择 $\tilde{\eta} \leq 1/L$ 成立，这样我们可以去掉最后一项。

重新排列并取期望得到

$$\mathbb{E} [\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2\mathbb{E}[f(W_{t-1}) - f(W_t)]}{\tilde{\eta}} + \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2]$$

对 $t = 1$ 到 T 求平均，我们得到

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2D}{T\tilde{\eta}} + \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2] \quad (16)$$

其中 $D = f(W_0) - f^*$ 。记

$$S_A := \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2D}{T\tilde{\eta}} + S_B, \quad S_B := \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2] \quad (17)$$

界定 $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2]$ 。

我们引入由真实（全批次）梯度 $\nabla f(W_t)$ 定义的真正缩放动量 N_t ，而不是 G_t 每一步 t ：

- $\bar{N}_t = \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t)$ for $t > 0$ and $\bar{N}_0 = 0$.

- Note that $N_t = \beta N_{t-1} + (1 - \beta)G_t$ for $t > 0$ and $N_0 = 0$.

Then, we can decompose $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_F^2]$ using $\|a + b\|_F^2 \leq 2\|a\|_F^2 + 2\|b\|_F^2$,

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_F^2] \leq 2 \underbrace{\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2]}_{\text{Term (A)}} + 2 \underbrace{\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\bar{N}_t - N_t\|_F^2]}_{\text{Term (B)}} \quad (18)$$

Bounding the Term (A). Let $e_t := \nabla f(W_{t-1}) - \bar{N}_t$. Using the recursion for $\bar{N}_t$, we have

$$\begin{aligned} e_t &= \nabla f(W_{t-1}) - (\beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t)) \\ &= \beta (\nabla f(W_{t-1}) - \bar{N}_{t-1}) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t)) \\ &= \beta e_{t-1} + \beta (\nabla f(W_{t-1}) - \nabla f(W_{t-2})) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t)) \end{aligned}$$

Apply the variation of Young's inequality (Lemma 5) with $c = \frac{1-\beta}{\beta}$:

$$\begin{aligned} \|e_t\|_F^2 &= \|\beta e_{t-1} + \beta (\nabla f(W_{t-1}) - \nabla f(W_{t-2})) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t))\|_F^2 \\ &= (1 + c) \beta^2 \|e_{t-1}\|_F^2 + (1 + 1/c) \|\beta (\nabla f(W_{t-1}) - \nabla f(W_{t-2})) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t))\|_F^2 \end{aligned}$$

Since $1 + c = \frac{1}{\beta}$ and $1 + 1/c = 1/(1 - \beta)$, and $\|x + y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$, we get

$$\|e_t\|_F^2 \leq \beta \|e_{t-1}\|_F^2 + \frac{2\beta^2}{1 - \beta} \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_F^2 + 2(1 - \beta) \|\nabla f(W_{t-1}) - \nabla f(W_t)\|_F^2.$$

By Assumption 1 and the norm monotonicity $\|\cdot\|_{\text{op}} \leq \|\cdot\|_F \leq \|\cdot\|_*$ (Proposition 2(iv)), we have

$$\begin{aligned} \|\nabla f(W_t) - \nabla f(W_{t-1})\|_F^2 &\leq \|\nabla f(W_t) - \nabla f(W_{t-1})\|_*^2 \leq L^2 \|W_t - W_{t-1}\|_{\text{op}}^2 \\ &= L^2 \tilde{\eta}^2 \|N_t\|_{\text{op}}^2 \leq L^2 \tilde{\eta}^2 \|N_t\|_F^2 \end{aligned}$$

and

$$\begin{aligned} \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_F^2 &\leq \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_*^2 \leq L^2 \|W_{t-1} - W_{t-2}\|_{\text{op}}^2 \\ &= L^2 \tilde{\eta}^2 \|N_{t-1}\|_{\text{op}}^2 \leq L^2 \tilde{\eta}^2 \|N_{t-1}\|_F^2 \end{aligned}$$

Hence, we have the following recursion,

$$\|e_t\|_F^2 \leq \beta \|e_{t-1}\|_F^2 + \frac{2\beta^2 L^2 \tilde{\eta}^2}{1 - \beta} \|N_{t-1}\|_F^2 + 2(1 - \beta) L^2 \tilde{\eta} \|N_t\|_F^2$$

Taking expectation and averaging over $t = 1$ to T , we obtain

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2] &\leq \beta \frac{1}{T} \sum_{t=1}^T \|\nabla f(W_{t-2}) - \bar{N}_{t-1}\|_F^2 \\ &\quad + \frac{2\beta^2 L^2 \tilde{\eta}^2}{1 - \beta} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_{t-1}\|_F^2] + 2(1 - \beta) L^2 \tilde{\eta}^2 \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t\|_F^2] \end{aligned}$$

Since $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-2}) - \bar{N}_{t-1}\|_F^2] \leq \frac{1}{T} \mathbb{E}\|e_0\|_F^2 + \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2]$ and $\frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_{t-1}\|_F^2 \leq \frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_t\|_F^2$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2] \leq \frac{\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} + \frac{2L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) \left(\frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_t\|_F^2 \right). \quad (19)$$

- Note that $N_t = \beta N_{t-1} + (1 - \beta)G_t$ for $t > 0$ and $N_0 = 0$.

然后, 我们可以使用 $\|a + b\|_F^2 \leq 2\|a\|_F^2 + 2\|b\|_F^2$ 对 $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_F^2]$ 进行分解,

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_F^2] \leq 2 \underbrace{\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2]}_{\text{Term (A)}} + 2 \underbrace{\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\bar{N}_t - N_t\|_F^2]}_{\text{Term (B)}} \quad (18)$$

界限项 (A). 令 $e_t := \nabla f(W_{t-1}) - N_t$. 使用 N_t 的递归公式, 我们有

$$\begin{aligned} e_t &= \nabla f(W_{t-1}) - (\beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t)) \\ &= \beta (\nabla f(W_{t-1}) - \bar{N}_{t-1}) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t)) \\ &= \beta e_{t-1} + \beta (\nabla f(W_{t-1}) - \nabla f(W_{t-2})) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t)) \end{aligned}$$

应用杨氏不等式 (引理5) 的变体, 参数为 $c = \frac{1-\beta}{\beta}$:

$$\begin{aligned} \|e_t\|_F^2 &= \|\beta e_{t-1} + \beta (\nabla f(W_{t-1}) - \nabla f(W_{t-2})) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t))\|_F^2 \\ &= (1 + c) \beta^2 \|e_{t-1}\|_F^2 + (1 + 1/c) \|\beta (\nabla f(W_{t-1}) - \nabla f(W_{t-2})) + (1 - \beta) (\nabla f(W_{t-1}) - \nabla f(W_t))\|_F^2 \end{aligned}$$

由于 $1 + c = \frac{1}{\beta}$ 和 $1 + 1/c = 1/(1 - \beta)$, 以及 $\|x + y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$, 我们得到

$$\|e_t\|_F^2 \leq \beta \|e_{t-1}\|_F^2 + \frac{2\beta^2}{1 - \beta} \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_F^2 + 2(1 - \beta) \|\nabla f(W_{t-1}) - \nabla f(W_t)\|_F^2.$$

By 假设1和范数单调性 $\|\cdot\|_{\text{op}} \leq \|\cdot\|_F \leq \|\cdot\|_*$ (命题2(iv)), 我们有

$$\begin{aligned} \|\nabla f(W_t) - \nabla f(W_{t-1})\|_F^2 &\leq \|\nabla f(W_t) - \nabla f(W_{t-1})\|_*^2 \leq L^2 \|W_t - W_{t-1}\|_{\text{op}}^2 \\ &= L^2 \tilde{\eta}^2 \|N_t\|_{\text{op}}^2 \leq L^2 \tilde{\eta}^2 \|N_t\|_F^2 \end{aligned}$$

and

$$\begin{aligned} \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_F^2 &\leq \|\nabla f(W_{t-1}) - \nabla f(W_{t-2})\|_*^2 \leq L^2 \|W_{t-1} - W_{t-2}\|_{\text{op}}^2 \\ &= L^2 \tilde{\eta}^2 \|N_{t-1}\|_{\text{op}}^2 \leq L^2 \tilde{\eta}^2 \|N_{t-1}\|_F^2 \end{aligned}$$

因此, 我们有以下递归公式:

$$\|e_t\|_F^2 \leq \beta \|e_{t-1}\|_F^2 + \frac{2\beta^2 L^2 \tilde{\eta}^2}{1 - \beta} \|N_{t-1}\|_F^2 + 2(1 - \beta) L^2 \tilde{\eta} \|N_t\|_F^2$$

对 $t = 1$ 到 T 进行期望和平均, 我们得到

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2] &\leq \beta \frac{1}{T} \sum_{t=1}^T \|\nabla f(W_{t-2}) - \bar{N}_{t-1}\|_F^2 \\ &\quad + \frac{2\beta^2 L^2 \tilde{\eta}^2}{1 - \beta} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_{t-1}\|_F^2] + 2(1 - \beta) L^2 \tilde{\eta}^2 \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t\|_F^2] \end{aligned}$$

由于 $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-2}) - \bar{N}_{t-1}\|_F^2] \leq \frac{1}{T} \mathbb{E}\|e_0\|_F^2 + \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2]$ 和 $\frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_{t-1}\|_F^2 \leq \frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_t\|_F^2$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_{t-1}) - \bar{N}_t\|_F^2] \leq \frac{\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} + \frac{2L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) \left(\frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_t\|_F^2 \right). \quad (19)$$

Bounding the Term (B). When we unroll the EMA recursion of both N_t and $\bar{N}_t$,

$$N_t = \beta N_{t-1} + (1 - \beta)G_t = \beta^t N_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} G_i$$

$$\bar{N}_t = \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t) = \beta^t \bar{N}_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} \nabla f(W_i)$$

Then, we compute the expectation of the Frobenius squared norm bias caused by N_t and $\bar{N}_t$,

$$\begin{aligned} \mathbb{E}[\|N_t - \bar{N}_t\|_F^2] &\leq \mathbb{E} \left[\left\| \beta^t (N_0 - \bar{N}_0) + (1 - \beta) \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right] \\ &= \beta^{2t} \mathbb{E}[\|N_0 - \bar{N}_0\|_F^2] + (1 - \beta)^2 \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right] \\ &= \beta^{2t} \mathbb{E}[\|N_0 - \bar{N}_0\|_F^2] + (1 - \beta)^2 \sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2] \end{aligned}$$

where equalities are due to Assumption 2, which states $\mathbb{E}[G_t] = \nabla f(W_t)$ so that the cross-term vanishes. Note that $\mathbb{E}[\|N_0 - \bar{N}_0\|_F^2] = 0$. By Lemma 8, we get

$$\begin{aligned} \mathbb{E}[\|N_t - \bar{N}_t\|_F^2] &\leq (1 - \beta)^2 \sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2] \\ &\leq (1 - \beta)^2 \sum_{i=1}^t \beta^{2(t-i)} \frac{\sigma^2}{B} \leq \frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \end{aligned}$$

By averaging over $t = 1, \dots, T$, we have

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \bar{N}_t\|_F^2] \leq \frac{1}{T} \sum_{t=1}^T \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) \leq \frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \quad (20)$$

Plugging Eq.(19) and Eq.(20) to Eq.(18), we obtain

$$\begin{aligned} &\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_F^2] \\ &\leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + 2 \left(\frac{\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} + \frac{2L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) \left(\frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_t\|_F^2 \right) \right) \\ &\leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + \frac{2\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} \\ &\quad + \frac{8L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) \left(\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_F^2] + \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2] \right) \end{aligned} \quad (21)$$

where the last inequality is due to $\|a + b\|_F^2 \leq 2\|a\|_F^2 + 2\|b\|_F^2$. Eq.(21) can be expressed as

$$S_B \leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + \frac{2\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} + \frac{8L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) (S_A + S_B)$$

By Lemma 7 and Proposition 2(iv), we have $\mathbb{E}\|e_0\|_F^2 = \mathbb{E}\|\nabla f(W_0)\|_F^2 \leq \mathbb{E}\|\nabla f(W_0)\|_*^2 \leq 2LD$. Therefore,

$$S_B \leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + \frac{4\beta LD}{T(1 - \beta)} + \frac{8L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) (S_A + S_B)$$

界限项 (B). 当我们展开 N_t 和 $\bar{N}_t$ 的 EMA递归时,

$$N_t = \beta N_{t-1} + (1 - \beta)G_t = \beta^t N_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} G_i$$

$$\bar{N}_t = \beta \bar{N}_{t-1} + (1 - \beta) \nabla f(W_t) = \beta^t \bar{N}_0 + (1 - \beta) \sum_{i=1}^t \beta^{t-i} \nabla f(W_i)$$

接着, 我们计算由 N_t 和 $\bar{N}_t$ 引起的Frobenius平方范数偏差的期望,

$$\begin{aligned} \mathbb{E}[\|N_t - \bar{N}_t\|_F^2] &\leq \mathbb{E} \left[\left\| \beta^t (N_0 - \bar{N}_0) + (1 - \beta) \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right] \\ &= \beta^{2t} \mathbb{E}[\|N_0 - \bar{N}_0\|_F^2] + (1 - \beta)^2 \mathbb{E} \left[\left\| \sum_{i=1}^t \beta^{t-i} (G_i - \nabla f(W_i)) \right\|_F^2 \right] \\ &= \beta^{2t} \mathbb{E}[\|N_0 - \bar{N}_0\|_F^2] + (1 - \beta)^2 \sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2] \end{aligned}$$

其中等式源于假设2, 该假设指出 $\mathbb{E}[G_t] = \nabla f(W_t)$, 因此交叉项消失。注意 $\mathbb{E}[\|N_0 - \bar{N}_0\|_F^2] = 0$ 。根据引理8, 我们得到

$$\begin{aligned} \mathbb{E}[\|N_t - \bar{N}_t\|_F^2] &\leq (1 - \beta)^2 \sum_{i=1}^t \beta^{2(t-i)} \mathbb{E}[\|G_i - \nabla f(W_i)\|_F^2] \\ &\leq (1 - \beta)^2 \sum_{i=1}^t \beta^{2(t-i)} \frac{\sigma^2}{B} \leq \frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \end{aligned}$$

通过对 $t = 1, \dots, T$ 进行平均, 我们有

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \bar{N}_t\|_F^2] \leq \frac{1}{T} \sum_{t=1}^T \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) \leq \frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \quad (20)$$

将公式(19)和公式(20)代入公式(18), 我们得到

$$\begin{aligned} &\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1}) - N_t\|_F^2] \\ &\leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + 2 \left(\frac{\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} + \frac{2L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) \left(\frac{1}{T} \sum_{t=1}^T \mathbb{E}\|N_t\|_F^2 \right) \right) \\ &\leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + \frac{2\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} \\ &\quad + \frac{8L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) \left(\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_F^2] + \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|N_t - \nabla f(W_{t-1})\|_F^2] \right) \end{aligned} \quad (21)$$

其中最后一个不等式源于 $\|a + b\|_F^2 \leq 2\|a\|_F^2 + 2\|b\|_F^2$ 。公式(21)可以表示为

$$S_B \leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + \frac{2\beta \mathbb{E}\|e_0\|_F^2}{T(1 - \beta)} + \frac{8L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) (S_A + S_B)$$

根据引理7和命题2(iv), 我们有 $\mathbb{E}\|e_0\|_F^2 = \mathbb{E}\|\nabla f(W_0)\|_F^2 \leq \mathbb{E}\|\nabla f(W_0)\|_*^2 \leq 2LD$ 。因此,

$$S_B \leq 2 \left(\frac{1 - \beta}{1 + \beta} \frac{\sigma^2}{B} \right) + \frac{4\beta LD}{T(1 - \beta)} + \frac{8L^2 \tilde{\eta}^2}{1 - \beta} \left(\frac{\beta^2}{1 - \beta} + 1 - \beta \right) (S_A + S_B)$$

Let $\theta := \frac{4L^2\tilde{\eta}^2}{1-\beta} \left(\frac{\beta^2}{1-\beta} + 1 - \beta \right) = \frac{4L^2\tilde{\eta}^2}{(1-\beta)^2} K$ and $K := \beta^2 + (1-\beta)^2$. Then, we have

$$S_B \leq 2\theta(S_A + S_B) + \frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right)$$

Hence

$$(1-2\theta)S_B \leq 2\theta S_A + \frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right). \quad (22)$$

Insert Eq.(22) into Eq.(17), we obtain

$$S_A \leq \frac{2D}{T\tilde{\eta}} + S_B \leq \frac{2D}{T\tilde{\eta}} + \frac{2\theta}{1-2\theta} S_A + \frac{1}{1-2\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right)$$

Therefore, provided $\theta < \frac{1}{4}$,

$$S_A \leq \frac{2(1-2\theta)}{1-4\theta} \cdot \frac{D}{\tilde{\eta}T} + \frac{1}{1-4\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right). \quad (23)$$

Finally, we have

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2(1-2\theta)}{1-4\theta} \cdot \frac{(1-\beta)D}{\eta T} + \frac{1}{1-4\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right). \quad (24)$$

Tuning η and β . We need $\tilde{\eta} \leq 1/L$ and $\theta < \frac{1}{4}$. Since $K = \beta^2 + (1-\beta)^2 \in [1/2, 1]$, a convenient sufficient choice is

$$\eta \leq \min \left\{ \frac{1-\beta}{L}, \frac{(1-\beta)^2}{4L\sqrt{K}} \right\}$$

Applying Jensen's inequality and using the fact that $\|X\|_* \leq \sqrt{r}\|X\|_F$ for $X \in \mathbb{R}^{m \times n}$ with $r = \min\{m, n\}$ (Proposition 2(iii)), we get

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_*] &\leq \sqrt{r} \left(\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_F] \right) \leq \sqrt{r} \sqrt{\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_F^2]} \\ &\leq \sqrt{r} \sqrt{\frac{2(1-2\theta)}{1-4\theta} \cdot \frac{(1-\beta)D}{\eta T} + \frac{1}{1-4\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right)} \\ &\leq \sqrt{\frac{2r(1-2\theta)}{1-4\theta} \cdot \frac{(1-\beta)D}{\eta T}} + \sqrt{\frac{r}{1-4\theta} \cdot \frac{4\beta LD}{(1-\beta)T}} + \sqrt{\frac{r}{1-4\theta} \cdot \frac{2(1-\beta)}{1+\beta} \cdot \frac{\sigma^2}{B}} \end{aligned}$$

where the last inequality is due to $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$. By setting $\eta = \min \left\{ \frac{1-\beta}{L}, \frac{(1-\beta)^2}{4L\sqrt{K}} \right\}$, which guarantees $\tilde{\eta} \leq 1/L$ and $\theta < \frac{1}{4}$, we obtain,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_*] \leq \underbrace{\mathcal{O} \left(\sqrt{\frac{rLD}{T}} \right)}_{\text{Deterministic Term}} + \underbrace{\mathcal{O} \left(\sqrt{\frac{r\beta LD}{(1-\beta)T}} \right) + \mathcal{O} \left(\sqrt{\frac{r\sigma^2(1-\beta)}{(1+\beta)B}} \right)}_{\text{Stochastic Term}}$$

By setting $\beta = 1 - \min \left\{ \frac{\sqrt{LDB}}{\sigma\sqrt{T}}, 1 \right\}$, we get

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_*] &\leq \mathcal{O} \left(\sqrt{\frac{rLD}{T}} \right) + \mathcal{O} \left(\sqrt{\frac{r\beta LD}{(1-\beta)T}} \right) + \mathcal{O} \left(\sqrt{\frac{r\sigma^2(1-\beta)}{(1+\beta)B}} \right) \\ &\leq \mathcal{O} \left(\sqrt{\frac{rLD}{T}} + \left(\frac{r^2\sigma^2 LD}{BT} \right)^{1/4} \right) \end{aligned}$$

Thus, we can find an ϵ -nuclear norm stationary point of f with a complexity of

$$\mathcal{O} \left(\max \left\{ \frac{rLD}{\epsilon^2}, \frac{r^2\sigma^2 LD}{B\epsilon^4} \right\} \right)$$

□

令 $\theta := \frac{4L^2\tilde{\eta}^2}{1-\beta} \left(\frac{\beta^2}{1-\beta} + 1 - \beta \right) = \frac{4L^2\tilde{\eta}^2}{(1-\beta)^2} K$ 且 $K := \beta^2 + (1-\beta)^2$ 。然后，我们有

$$S_B \leq 2\theta(S_A + S_B) + \frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right)$$

因此

$$(1-2\theta)S_B \leq 2\theta S_A + \frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right). \quad (22)$$

将公式(22)代入公式(17)，我们得到

$$S_A \leq \frac{2D}{T\tilde{\eta}} + S_B \leq \frac{2D}{T\tilde{\eta}} + \frac{2\theta}{1-2\theta} S_A + \frac{1}{1-2\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right)$$

因此，假设 $\theta < \frac{1}{4}$,

$$S_A \leq \frac{2(1-2\theta)}{1-4\theta} \cdot \frac{D}{\tilde{\eta}T} + \frac{1}{1-4\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right). \quad (23)$$

最后，我们有

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla f(W_{t-1})\|_F^2] \leq \frac{2(1-2\theta)}{1-4\theta} \cdot \frac{(1-\beta)D}{\eta T} + \frac{1}{1-4\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right). \quad (24)$$

调整 η 和 β 。 我们需要 $\tilde{\eta} \leq 1/L$ 和 $\theta < \frac{1}{4}$ 。由于 $K = \beta^2 + (1-\beta)^2 \in [1/2, 1]$ ，一个方便的充分选择是

$$\eta \leq \min \left\{ \frac{1-\beta}{L}, \frac{(1-\beta)^2}{4L\sqrt{K}} \right\}$$

应用詹森不等式，并利用 $\|X\|_* \leq \sqrt{r}\|X\|_F$ 对于 $X \in \mathbb{R}^{m \times n}$ 具有最小值 $\{m, n\}$ （命题 2(iii)）这一事实，我们得到

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_*] &\leq \sqrt{r} \left(\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_F] \right) \leq \sqrt{r} \sqrt{\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_F^2]} \\ &\leq \sqrt{r} \sqrt{\frac{2(1-2\theta)}{1-4\theta} \cdot \frac{(1-\beta)D}{\eta T} + \frac{1}{1-4\theta} \left(\frac{4\beta LD}{(1-\beta)T} + 2 \left(\frac{1-\beta}{1+\beta} \frac{\sigma^2}{B} \right) \right)} \\ &\leq \sqrt{\frac{2r(1-2\theta)}{1-4\theta} \cdot \frac{(1-\beta)D}{\eta T}} + \sqrt{\frac{r}{1-4\theta} \cdot \frac{4\beta LD}{(1-\beta)T}} + \sqrt{\frac{r}{1-4\theta} \cdot \frac{2(1-\beta)}{1+\beta} \cdot \frac{\sigma^2}{B}} \end{aligned}$$

其中最后一个不等式归因于 $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ 。通过设置 $\eta = \min \left\{ \frac{1-\beta}{L}, \frac{(1-\beta)^2}{4L\sqrt{K}} \right\}$ ，

这保证了 $\tilde{\eta} \leq 1/L$ 和 $\theta < \frac{1}{4}$ ，我们得到，

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_*] \leq \underbrace{\mathcal{O} \left(\sqrt{\frac{rLD}{T}} \right)}_{\text{Deterministic Term}} + \underbrace{\mathcal{O} \left(\sqrt{\frac{r\beta LD}{(1-\beta)T}} \right) + \mathcal{O} \left(\sqrt{\frac{r\sigma^2(1-\beta)}{(1+\beta)B}} \right)}_{\text{Stochastic Term}}$$

通过设置 $\beta = 1 - \min \left\{ \frac{\sqrt{LDB}}{\sigma\sqrt{T}}, 1 \right\}$ ，我们得到

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(W_t)\|_*] &\leq \mathcal{O} \left(\sqrt{\frac{rLD}{T}} \right) + \mathcal{O} \left(\sqrt{\frac{r\beta LD}{(1-\beta)T}} \right) + \mathcal{O} \left(\sqrt{\frac{r\sigma^2(1-\beta)}{(1+\beta)B}} \right) \\ &\leq \mathcal{O} \left(\sqrt{\frac{rLD}{T}} + \left(\frac{r^2\sigma^2 LD}{BT} \right)^{1/4} \right) \end{aligned}$$

因此，我们可以找到一个具有 ϵ -核范数复杂度的 f 的稳定点

$$\mathcal{O} \left(\max \left\{ \frac{rLD}{\epsilon^2}, \frac{r^2\sigma^2 LD}{B\epsilon^4} \right\} \right)$$

□

D NEWTON–SCHULZ LEMMAS: PROOFS

Remark 1 (Initial residual below one).

With $X_{t,0} = M_t/\alpha_t$ and $\alpha_t = \max\{1, \|M_t\|_F\}$, we have $\delta_{t,0} \in [0, 1)$ for every t ; moreover $\delta_{t,0} = 0$ when $M_t = 0$.

Proof. Let $M_t = U\Sigma V^\top$ (rank r_t). Then $X_{t,0}X_{t,0}^\top = U(\Sigma^2/\alpha_t^2)U^\top$, so on $\text{range}(M_t)$ the eigenvalues are $\sigma_i^2/\alpha_t^2 \in (0, 1]$. If $r_t = 0$, set $\delta_{t,0} = 0$. Otherwise, the minimal positive eigenvalue $\lambda_{\min}^+ = \min_{i \leq r_t} \sigma_i^2/\alpha_t^2 > 0$, hence by Lemma 1, $\delta_{t,0} = 1 - \lambda_{\min}^+ < 1$. $\square$

Lemma 9. (*Polar factor invariance under NEWTON–SCHULZ*).

As NEWTON–SCHULZ iterates by the polynomial p_κ , i.e., $X_{t,0} = M_t/\alpha_t$, $X_{t,j+1} = p_\kappa(X_{t,j}X_{t,j}^\top)X_{t,j}$, the polar factor is invariant: $\text{Polar}(M_t) = \text{Polar}(X_{t,j})$ for all $j \geq 0$.

Proof. First, the polar factor is invariant under scalar multiplication. ($\because$ Let $\text{SVD}(X) = (U, \Sigma, V)$. Then $\text{Polar}(X) = UV^\top$. Now, let $Y = cX$ for $c > 0$. Then $\text{SVD}(Y) = \text{SVD}(cX) = (U, c\Sigma, V)$. Thus, $\text{Polar}(Y) = UV^\top$. Hence, $P_t = \text{Polar}(M_t) = \text{Polar}(M_t/\alpha_t) = \text{Polar}(X_{t,0})$ holds.

Now, to prove by induction on $j \geq 0$, we assume that $P_t = \text{Polar}(X_{t,j})$. Note that one step of NEWTON–SCHULZ for polynomial p is defined as

$$X_{t,j+1} = p(X_{t,j}X_{t,j}^\top)X_{t,j}$$

Let $\text{SVD}(X_{t,j}) = (U, \Sigma, V)$ with $\Sigma \succ 0$ (on $\text{range}(X_{t,j})$), so that $\text{Polar}(X_{t,j}) = UV^\top$. Then

$$p(X_{t,j}X_{t,j}^\top)X_{t,j} = p(U\Sigma V^\top(U\Sigma V^\top)^\top)U\Sigma V^\top = p(U\Sigma^2U^\top)U\Sigma V^\top = Up(\Sigma^2)\Sigma V^\top$$

Thus, $p(X_{t,j}X_{t,j}^\top)X_{t,j}$ has the same left/right singular vectors U, V as $X_{t,j}$. Hence,

$$\text{Polar}(X_{t,j+1}) = \text{Polar}(p(X_{t,j}X_{t,j}^\top)X_{t,j}) = \text{Polar}(X_{t,j}) = UV^\top$$

By induction, we have

$$P_t = \text{Polar}(M_t) = \text{Polar}(X_{t,0}) = \text{Polar}(X_{t,1}) = \dots = \text{Polar}(X_{t,q})$$

which implies the polar factor invariance under NEWTON–SCHULZ updates. $\square$

Lemma 10. (*Support invariance under NEWTON–SCHULZ*).

As NEWTON–SCHULZ iterates by the polynomial p_κ , i.e., $X_{t,0} = M_t/\alpha_t$, $X_{t,j+1} = p_\kappa(X_{t,j}X_{t,j}^\top)X_{t,j}$, the support is invariant: $\text{range}(M_t) = \text{range}(X_{t,j})$ for all $j \geq 0$, and $p_\kappa(X_{t,j}X_{t,j}^\top)$ is positive definite on $\text{range}(X_{t,j})$.

Proof. On $\text{range}(X_{t,j})$ the spectrum of $A := X_{t,j}X_{t,j}^\top$ lies in $(0, 1]$. p_κ is strictly positive on $(0, 1]$ (Proposition 1), so $p_\kappa(A)|_{\text{range}(X_{t,j})}$ is invertible. Hence

$$\text{range}(X_{t,j+1}) = \text{range}(p_\kappa(A)X_{t,j}) = \text{range}(X_{t,j}),$$

and by induction $\text{range}(X_{t,j}) = \text{range}(M_t)$ for all j . $\square$

Remark 2 (Support-aware details). As in the main text, all NEWTON–SCHULZ quantities are restricted to $\text{range}(M_t)$ (no standing full-rank assumption). Lemma 10 preserves the column space; P_t , $X_{t,j}X_{t,j}^\top$, and Π_t vanish on $\text{range}(M_t)^\perp$.

Lemma 1 (Orthogonality residual vs. Polar approximation error).

Let $\lambda_{\min}^+$ be the smallest positive eigenvalue of $X_{t,q}X_{t,q}^\top$ restricted to $\text{range}(M_t)$ (set $\lambda_{\min}^+ = 1$ if $\text{rank}(M_t) = 0$). Then

$$\delta_{t,q} = 1 - \lambda_{\min}^+, \quad \varepsilon_{t,q} = 1 - \sqrt{\lambda_{\min}^+} = 1 - \sqrt{1 - \delta_{t,q}}.$$

D NEWTON–SCHULZLEMMAS: 证明

注释1 (初始残差小于1)。

当 $X_{t,0} = M_t/\alpha_t$ 和 $\alpha_t = \max\{1, \|M_t\|_F\}$ 时, 我们对于每个 t 都有 $\delta_{t,0} \in [0, 1)$; 此外 $\delta_{t,0} = 0$, 当 $M_t = 0$

证明。设 $M_t = U\Sigma V^\top$ (秩 r_t)。然后 $X_{t,0}X_{t,0}^\top = U(\Sigma^2/\alpha_t^2)U^\top$, 所以在范围 (M_t) 内特征值是 $\sigma_i^2/\alpha_t^2 \in (0, 1]$ 。如果 $r_t = 0$, 设置 $\delta_{t,0} = 0$ 。否则, 最小正特征值 $\lambda_{\min}^+ = \min_{i \leq r_t} \sigma_i^2/\alpha_t^2 > 0$, 因此根据引理1, $\delta_{t,0} = 1 - \lambda_{\min}^+ < 1$ 。 $\square$

引理9。(极因子在 $\langle \text{样式id}=3 \rangle N \langle \text{样式}\rangle \langle \text{样式id}=5 \rangle \text{EWTON} \langle \text{样式}\rangle \langle \text{样式id}=7 \rangle \langle \text{样式}\rangle \langle \text{样式id}=9 \rangle \text{CHULZ} \langle \text{样式}\rangle$ 下的不变性)。

因为 $\langle \text{样式id}=1 \rangle N \langle \text{样式}\rangle \langle \text{样式id}=3 \rangle \text{EWTON} \langle \text{样式}\rangle \langle \text{样式id}=5 \rangle \langle \text{样式}\rangle \langle \text{样式id}=7 \rangle \text{CHULZ} \langle \text{样式}\rangle$ 通过多项式 p_κ 迭代, 即 $X_t \langle \text{样式id}=11 \rangle, 0 = M_t/\alpha_t, X_t \langle \text{样式id}=15 \rangle, j+1 = p_\kappa \langle \text{样式id}=19 \rangle \langle \text{样式}\rangle X_t \langle \text{样式id}=22 \rangle, j X_t^\top \langle \text{样式id}=25 \rangle, j \langle \text{样式id}=28 \rangle \rangle \langle \text{样式}\rangle X_t \langle \text{样式id}=31 \rangle, j$, 极因子是不变的: $\langle \text{样式id}=34 \rangle \text{Polar} \langle \text{样式}\rangle M_t \langle \text{样式id}=37 \rangle \rangle \langle \text{样式}\rangle = \langle \text{样式id}=40 \rangle \text{Polar} \langle \text{样式}\rangle X_t \langle \text{样式id}=43 \rangle, \langle \text{样式}\rangle \langle \text{样式id}=46 \rangle \rangle \langle \text{样式}\rangle$ 对所有 $j \geq 0$ 。

证明。首先, 极因子在标量乘法下是不变的。($\because$ 设 $\text{SVD}(X) = (U, \Sigma, V)$ 。则 $\text{Polar}(X) = UV^\top$ 。现在, 设 $Y = cX$ 对 $c > 0$ 。则 $\text{SVD}(Y) = \text{SVD}(cX) = (U, c\Sigma, V)$ 。因此, $\text{Polar}(Y) = UV^\top$ 。所以, $P_t = \text{Polar}(M_t) = \text{Polar}(M_t/\alpha_t) = \text{Polar}(X_{t,0})$ 成立。

现在, 我们要对 $j \geq 0$ 归纳地证明, 假设 $P_t = \text{Polar}(X_{t,j})$ 。注意NEWTON–SCHULZ的一个步骤对于多项式 p 定义为

$$X_{t,j+1} = p(X_{t,j}X_{t,j}^\top)X_{t,j}$$

设 $\text{SVD}(X_{t,j}) = (U, \Sigma, V)$, 其中 $\Sigma \succ 0$ 在范围 $(X_{t,j})$ 内, 使得 $\text{Polar}(X_{t,j}) = UV^\top$ 。则

$$p(X_{t,j}X_{t,j}^\top)X_{t,j} = p(U\Sigma V^\top(U\Sigma V^\top)^\top)U\Sigma V^\top = p(U\Sigma^2U^\top)U\Sigma V^\top = Up(\Sigma^2)\Sigma V^\top$$

因此, $p(X_{t,j}X_{t,j}^\top)X_{t,j}$ 具有与 $X_{t,j}$ 相同的左/右奇异向量 U, V 。所以,

$$\text{Polar}(X_{t,j+1}) = \text{Polar}(p(X_{t,j}X_{t,j}^\top)X_{t,j}) = \text{Polar}(X_{t,j}) = UV^\top$$

归纳地, 我们有

$$P_t = \text{Polar}(M_t) = \text{Polar}(X_{t,0}) = \text{Polar}(X_{t,1}) = \dots = \text{Polar}(X_{t,q})$$

这表明极因子在NEWTON–SCHULZ更新下保持不变。 $\square$

引理10. (在NEWTON–SCHULZ下支持保持不变)。

随着NEWTON–SCHULZ通过多项式 p_κ 迭代, 即 $X_{t,0} = M_t/\alpha_t, X_{t,j+1} = p_\kappa(X_{t,j}X_{t,j}^\top)X_{t,j}$, 支持保持不变: $\text{range}(M_t) = \text{range}(X_{t,j})$ 对所有 $j \geq 0$, 并且 $p_\kappa(X_{t,j}X_{t,j}^\top)$ 在 $\text{range}(X_{t,j})$ 上是正定的。

证明。在 $\text{range}(X_{t,j})$ 上, A 的谱:= $X_{t,j}X_{t,j}^\top$ 位于 $(0, 1]$ 。 p_κ 在 $(0, 1]$ (命题1)上严格为正, 所以 $p_\kappa(A)|_{\text{range}(X_{t,j})}$ 是可逆的。因此

$$\text{range}(X_{t,j+1}) = \text{range}(p_\kappa(A)X_{t,j}) = \text{range}(X_{t,j}),$$

并且归纳地, $\text{range}(X_{t,j}) = \text{range}(M_t)$ 对所有 j 。 $\square$

备注2 (支持感知细节)。与正文一样, 所有NEWTON–SCHULZ量都限制在范围 (M_t) 内 (没有全秩假设)。引理10保留了列空间; $P_t, X_{t,j}X_{t,j}^\top$ 和 Π_t 在范围 $(M_t)^\perp$ 上消失。

引理1 (正交残差与极化近似误差)。

令 $\lambda_{\min}^+$ 最小值为 $X_{t,q}X_{t,q}^\top$ 在范围 (M_t) 内的最小正特征值 (若秩 $(M_t) = 0$, 则设 $\lambda_{\min}^+ = 1$)。然后

$$\delta_{t,q} = 1 - \lambda_{\min}^+, \quad \varepsilon_{t,q} = 1 - \sqrt{\lambda_{\min}^+} = 1 - \sqrt{1 - \delta_{t,q}}.$$

Proof. Let $X_{t,q} = U\Sigma V^\top$ and $\Pi_t = UU^\top$ with $\Sigma = \text{diag}(\sigma_i)$. Then,

$$\Pi_t - X_{t,q}X_{t,q}^\top = U(I - \Sigma^2)U^\top$$

on $\text{range}(M_t)$ and vanishes on $\text{range}(M_t)^\perp$. Thus, the orthogonality residual $\delta_{t,q}$ is computed as

$$\delta_{t,q} = \|\Pi_t - X_{t,q}X_{t,q}^\top\|_{\text{op}} = \max_i |1 - \sigma_i^2| = 1 - \lambda_{\min}^+ \quad (25)$$

where $\lambda_{\min}^+$ is the smallest positive eigenvalue of $X_{t,q}X_{t,q}^\top$ on $\text{range}(M_t)$. By the polar factor invariance under NEWTON–SCHULZ updates (Lemma 9), we have

$$\varepsilon_{t,q} = \|X_{t,q} - \text{Polar}(M_t)\|_{\text{op}} = \|U\Sigma V^\top - UV^\top\|_{\text{op}} = \|\Sigma - I\|_{\text{op}} = 1 - \min_i \sigma_i = 1 - \sqrt{\lambda_{\min}^+} \quad (26)$$

Combining Eq.(25) and Eq.(26), we conclude

$$\varepsilon_{t,q} = 1 - \sqrt{\lambda_{\min}^+} = 1 - \sqrt{1 - \delta_{t,j}}$$

□

Lemma 2 (Residual update).

For NEWTON–SCHULZ polynomial p_κ the orthogonality residual $\delta_{t,j}$ is updated by NEWTON–SCHULZ per step as

$$\delta_{t,j+1} = \phi(\delta_{t,j}),$$

where $\phi(u) := 1 - (1 - u)[p_\kappa(1 - u)]^2$.

Proof. Recall that NEWTON–SCHULZ update is defined as

$$X_{t,j+1} = p(X_{t,j}X_{t,j}^\top)X_{t,j}, \quad j = 0, 1, \dots, q-1$$

where $\|X_{t,0}\|_{\text{op}} \leq 1$. Let $A_{t,j}$ be $X_{t,j}X_{t,j}^\top$. Then $A_{t,j}$ is symmetric, because

$$A_{t,j}^\top = (X_{t,j}X_{t,j}^\top)^\top = (X_{t,j}^\top)^\top X_{t,j}^\top = X_{t,j}X_{t,j}^\top = A_{t,j}$$

Applying NEWTON–SCHULZ update, we get

$$\begin{aligned} A_{t,j+1} &= X_{t,j+1}X_{t,j+1}^\top = (p(A_{t,j})X_{t,j})(p(A_{t,j})X_{t,j})^\top \\ &= p(A_{t,j})X_{t,j}X_{t,j}^\top p(A_{t,j})^\top = p(A_{t,j})A_{t,j}p(A_{t,j})^\top \end{aligned}$$

Since $A_{t,j}$ is symmetric, any polynomial $p(A_{t,j})$ is also symmetric. Thus, $p(A_{t,j})^\top = p(A_{t,j})$. Therefore, we have

$$A_{t,j+1} = p(A_{t,j})A_{t,j}p(A_{t,j}) \quad (27)$$

Since $A_{t,j}$ is symmetric, $A_{t,j}$ is orthogonally diagonalizable. Let $\{\lambda_0, \lambda_1, \dots\}$ be the eigenvalues of $A_{t,j}$. Hence, we can write its spectral decomposition as $A_{t,j} = U\Lambda U^\top = U \text{diag}(\lambda_l)U^\top$. By the key property of matrix polynomials, $p(A_{t,j}) = Up(\Lambda)U^\top = U \text{diag}(p(\lambda_l))U^\top$. Now, we substitute the spectral decompositions of $A_{t,j}$ and $p(A_{t,j})$ into Eq.(27),

$$A_{t,j+1} = (Up(\Lambda)U^\top)(U\Lambda U^\top)(Up(\Lambda)U^\top) = U(p(\Lambda)\Lambda p(\Lambda))U^\top$$

Let the new diagonal matrix be $\Lambda' := p(\Lambda)\Lambda p(\Lambda)$. Then, the diagonal entries of Λ' are $\lambda_\ell[p(\lambda_\ell)]^2$. The expression for $A_{t,j+1} = U\Lambda'U^\top$, which is the spectral decomposition of $A_{t,j+1}$. This form explicitly shows that the eigenvalues of $A_{t,j+1}$ are the diagonal entries of Λ' , which are $\lambda_\ell[p(\lambda_\ell)]^2$.

We conclude that $A_{t,j+1} = p(A_{t,j})A_{t,j}p(A_{t,j}) = U \text{diag}(\lambda_i[p(\lambda_\ell)]^2)U^\top$ on $\text{range}(M_t)$ and both Π_t and $A_{t,j}$ vanish on $\text{range}(M_t)^\perp$.

Claim: $\|X_{t,j}\|_{\text{op}} \leq 1$ for all $j \geq 0$.

证明。令 $X_{t,q} = U\Sigma V^\top$ 和 $\Pi_t = UU^\top$, 其中 Σ 为 σ_i 的对角矩阵。然后,

$$\Pi_t - X_{t,q}X_{t,q}^\top = U(I - \Sigma^2)U^\top$$

on M_t 的范围内, 并在 M_t 的范围内消失[⊥]. 因此, 计算正交残差 $\delta_{t,q}$ as

$$\delta_{t,q} = \|\Pi_t - X_{t,q}X_{t,q}^\top\|_{\text{op}} = \max_i |1 - \sigma_i^2| = 1 - \lambda_{\min}^+ \quad (25)$$

其中 λ^+ 最小值是 $X_{t,q}X_{t,q}^\top$ 在 M_t 范围内的最小正特征值。根据NEWTON–SCHULZ更新下的极因子不变性（引理9），我们有

$$\varepsilon_{t,q} = \|X_{t,q} - \text{Polar}(M_t)\|_{\text{op}} = \|U\Sigma V^\top - UV^\top\|_{\text{op}} = \|\Sigma - I\|_{\text{op}} = 1 - \min_i \sigma_i = 1 - \sqrt{\lambda_{\min}^+} \quad (26)$$

结合公式(25)和公式(26), 我们得出

$$\varepsilon_{t,q} = 1 - \sqrt{\lambda_{\min}^+} = 1 - \sqrt{1 - \delta_{t,j}}$$

□

引理2（残差更新）。

对于NEWTON–SCHULZ多项式 p_κ , 正交残差 $\delta_{t,j}$ 通过NEWTON–SCHULZ每步更新为

$$\delta_{t,j+1} = \phi(\delta_{t,j}),$$

在 $\phi(u) := 1 - (1 - u)[p_\kappa(1 - u)]^2$ 处。

证明。回想 NEWTON–SCHULZ 更新是定义为

$$X_{t,j+1} = p(X_{t,j}X_{t,j}^\top)X_{t,j}, \quad j = 0, 1, \dots, q-1$$

在 $\|X_{t,0}\|_{\text{op}} \leq 1$ 处。令 $A_{t,j}$ 为 $X_{t,j}X_{t,j}^\top$ 。那么 $A_{t,j}$ 是对称的, 因为

$$A_{t,j}^\top = (X_{t,j}X_{t,j}^\top)^\top = (X_{t,j}^\top)^\top X_{t,j}^\top = X_{t,j}X_{t,j}^\top = A_{t,j}$$

应用 NEWTON–SCHULZ 更新, 我们得到

$$\begin{aligned} A_{t,j+1} &= X_{t,j+1}X_{t,j+1}^\top = (p(A_{t,j})X_{t,j})(p(A_{t,j})X_{t,j})^\top \\ &= p(A_{t,j})X_{t,j}X_{t,j}^\top p(A_{t,j})^\top = p(A_{t,j})A_{t,j}p(A_{t,j})^\top \end{aligned}$$

由于 $A_{t,j}$ 是对称的, 任何多项式 $p(A_{t,j})$ 也是对称的。因此, $p(A_{t,j})^\top = p(A_{t,j})$ 。因此, 我们有

$$A_{t,j+1} = p(A_{t,j})A_{t,j}p(A_{t,j}) \quad (27)$$

由于 $A_{t,j}$ 是对称的, $A_{t,j}$ 是正交对角化的。令 $\{\lambda_0, \lambda_1, \dots\}$ 为 $A_{t,j}$ 的特征值。因此, 我们可以将其谱分解写为 $A_{t,j} = U\Lambda U^\top = U \text{diag}(\lambda_l)U^\top$ 。根据矩阵多项式的关键性质, $p(A_{t,j}) = Up(\Lambda)U^\top = U \text{diag}(p(\lambda_l))U^\top$ 。现在, 我们将 $A_{t,j}$ 和 $p(A_{t,j})$ 的谱分解代入式(27),

$$A_{t,j+1} = (Up(\Lambda)U^\top)(U\Lambda U^\top)(Up(\Lambda)U^\top) = U(p(\Lambda)\Lambda p(\Lambda))U^\top$$

设新的对角矩阵为 $\Lambda' := p(\Lambda)\Lambda p(\Lambda)$ 。然后, Λ' 的对角元为 $\lambda_\ell[p(\lambda_\ell)]^2$ 。 $A_{t,j+1} = U\Lambda'U^\top$ 的表达式, 即 $A_{t,j+1}$ 的谱分解。这种形式明确地表明 $A_{t,j+1}$ 的特征值是 Λ' 的对角元, 即 $\lambda_\ell[p(\lambda_\ell)]^2$ 。

我们得出结论, $A_{t,j+1} = p(A_{t,j})A_{t,j}p(A_{t,j}) = U \text{diag}(\lambda_i[p(\lambda_\ell)]^2)U^\top$ 在 M_t 的范围内成立, 并且 Π_t 和 $A_{t,j}$ 在 $M_t^\perp$ 的范围内都消失。

声明: $\|X_{t,j}\|_{\text{op}} \leq 1$ 对所有 $j \geq 0$ 适用。

For $j = 0$, it is trivial. Assume that $\|X_{t,j}\|_{\text{op}} \leq 1$ holds. Then, $\|A_{t,j}\|_{\text{op}} = \|X_{t,j}X_{t,j}^\top\|_{\text{op}} = \|X_{t,j}\|_{\text{op}}^2 \leq 1$. Hence, the largest singular value of $A_{t,j}$ is in $[0, 1]$, which implies all eigenvalues of symmetric $A_{t,j}$ are in $[0, 1]$, i.e., $\max_l \lambda_l = \lambda_{\max} \leq 1$.

$$\|X_{t,j+1}\|_{\text{op}}^2 = \|X_{t,j+1}X_{t,j+1}^\top\|_{\text{op}} = \|A_{t,j+1}\|_{\text{op}} = \max_{\ell} (\lambda_{\ell}[p(\lambda_{\ell})]^2)$$

Since $\tau(\lambda) = \lambda[p(\lambda)]^2$ is non-decreasing on $[0, 1]$ with $\tau(1) \leq 1$ (Proposition 1), we have

$$\|X_{t,j+1}\|_{\text{op}}^2 = \max_{\ell} (\tau(\lambda_l)) = \tau(\max_{\ell} \lambda_l) = \tau(\lambda_{\max}) \leq \tau(1) \leq 1$$

By induction, we get $\|X_{t,j}\|_{\text{op}} \leq 1$ for all $j \geq 0$. The claim also implies $\|A_{t,j}\|_{\text{op}} \leq 1$ for all $j \geq 0$.

Now, recall that the orthogonality residual at step j is defined as

$$\delta_{t,j} = \|\Pi_t - X_{t,j}X_{t,j}^\top\|_{\text{op}} = \|\Pi_t - A_{t,j}\|_{\text{op}}$$

Let $\lambda_{\min}^+$ be the minimum eigenvalue of $A_{t,j}$, i.e., $\min_l \lambda_l = \lambda_{\min}^+ \leq 1$. Then, by the claim, we have

$$\delta_{t,j} = \|\Pi_t - A_{t,j}\|_{\text{op}} = \max_l |1 - \lambda_l| = 1 - \lambda_{\min}^+ \quad (28)$$

Now, we consider the next step residual $\delta_{t,j+1}$,

$$\delta_{t,j+1} = \|\Pi_t - A_{t,j+1}\|_{\text{op}} = \max_l |1 - \lambda_{\ell}[p(\lambda_{\ell})]^2| = \max_l (1 - \lambda_{\ell}[p(\lambda_{\ell})]^2)$$

where the last equality is due to $\|A_{t,j+1}\|_{\text{op}} \leq 1$. Since $\tau(\lambda) = \lambda[p(\lambda)]^2$ is non-decreasing on $[0, 1]$ with $\tau(1) \leq 1$, we have

$$\begin{aligned} \delta_{t,j+1} &= 1 - \lambda_{\min}^+[p(\lambda_{\min}^+)]^2 \\ &= 1 - (1 - \delta_{t,j})[p(1 - \delta_{t,j})]^2 \end{aligned}$$

where the last equality is due to Eq.(28). Therefore, we conclude that the orthogonality residual is updated by NEWTON-SCHULZ as

$$\delta_{t,j+1} = \phi(\delta_{t,j})$$

where $\phi(u) = 1 - (1 - u)[p(1 - u)]^2$ □

Lemma 3 (Residual Decay by NEWTON-SCHULZ Polynomial). For NEWTON-SCHULZ polynomial p_{κ} , $\phi(u) \leq u^{\kappa+1}$ on $[0, 1]$ where ϕ is a function defined in Lemma 2. Hence, for every t and all $j \geq 0$,

$$\delta_{t,j+1} \leq \delta_{t,j}^{\kappa+1}, \quad \delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}.$$

Proof. The NEWTON-SCHULZ polynomial $p(\lambda)$ of degree κ is defined as the truncation of the Taylor series of $g(\lambda) = \lambda^{-1/2}$ expanded around $\lambda = 1$. This means that the first κ derivatives of $p(\lambda)$ and $g(\lambda)$ are identical at $\lambda = 1$:

$$p^{(s)}(1) = g^{(s)}(1) \quad \text{for } s = 0, 1, \dots, \kappa$$

Let's consider the variable substitution $\lambda = 1 - u$. The function becomes $f(u) = g(1 - u) = (1 - u)^{-1/2}$. The polynomial $p(1 - u)$ is then the Taylor expansion of $f(u)$ around $u = 0$ up to the term u^{κ} :

$$p(1 - u) = \sum_{s=0}^{\kappa} \frac{f^{(s)}(0)}{s!} u^s = \sum_{s=0}^{\kappa} c_s u^s \quad (29)$$

The derivatives of $f(u) = (1 - u)^{-1/2}$ at $u = 0$ are $f^{(s)}(0) = \frac{(2s)!}{s!4^s}$, so that the coefficients are $c_s = \frac{(2s)!}{(s!)^2 4^s}$. Moreover, since $p(1 - u)$ and $f(u)$ share the same first κ derivatives at $u = 0$, we get

$$f^{(s)}(0) = (-1)^s p^{(s)}(1), \quad \text{for } s = 1, \dots, \kappa \quad (30)$$

对于 $j = 0$, 这是显而易见的。假设 $\|X_{t,j}\|_{\text{op}} \leq 1$ 成立。那么, $\|A_{t,j}\|_{\text{op}} = \|X_{t,j}X_{t,j}^\top\|_{\text{op}} = \|X_{t,j}\|_{\text{op}}^2 \leq 1$ 。因此, A_t 的最大奇异值位于 $[0, 1]$, , 这意味着对称 A_t 的所有特征值都在 $[0, 1]$, 内, 即 $\max_l \lambda_l = \lambda_{\max} \leq 1$ 。

$$\|X_{t,j+1}\|_{\text{op}}^2 = \|X_{t,j+1}X_{t,j+1}^\top\|_{\text{op}} = \|A_{t,j+1}\|_{\text{op}} = \max_{\ell} (\lambda_{\ell}[p(\lambda_{\ell})]^2)$$

由于 $\tau(\lambda) = \lambda[p(\lambda)]^2$ 在 $[0, 1]$ 上非递减, 且 $\tau(1) \leq 1$ (命题1), 所以我们有

$$\|X_{t,j+1}\|_{\text{op}}^2 = \max_{\ell} (\tau(\lambda_l)) = \tau(\max_{\ell} \lambda_l) = \tau(\lambda_{\max}) \leq \tau(1) \leq 1$$

通过归纳, 我们得到 $\|X_{t,j}\|_{\text{op}} \leq 1$ 对所有 $j \geq 0$ 成立。该命题还意味着 $\|A_{t,j}\|_{\text{op}} \leq 1$ 对所有 $j \geq 0$ 成立。现在, 回想一下步骤 j 处的正交残差定义为

$$\delta_{t,j} = \|\Pi_t - X_{t,j}X_{t,j}^\top\|_{\text{op}} = \|\Pi_t - A_{t,j}\|_{\text{op}}$$

令 λ^+ 最小值是 A_t 的最小特征值, 即 $\min_l \lambda_l = \lambda^+$ 最小值 ≤ 1 。那么, 根据该命题, 我们有

$$\delta_{t,j} = \|\Pi_t - A_{t,j}\|_{\text{op}} = \max_l |1 - \lambda_l| = 1 - \lambda_{\min}^+ \quad (28)$$

现在, 我们考虑下一步残差 $\delta_{t,j+1}$,

$$\delta_{t,j+1} = \|\Pi_t - A_{t,j+1}\|_{\text{op}} = \max_l |1 - \lambda_{\ell}[p(\lambda_{\ell})]^2| = \max_l (1 - \lambda_{\ell}[p(\lambda_{\ell})]^2)$$

其中最后一个等式是由于 $\|A_{t,j+1}\|_{\text{op}} \leq 1$ 。由于 $\tau(\lambda) = \lambda[p(\lambda)]^2$ 在 $[0, 1]$ 上非递减, 且 $\tau(1) \leq 1$, 所以我们有

$$\begin{aligned} \delta_{t,j+1} &= 1 - \lambda_{\min}^+[p(\lambda_{\min}^+)]^2 \\ &= 1 - (1 - \delta_{t,j})[p(1 - \delta_{t,j})]^2 \end{aligned}$$

其中最后一个等式源于公式(28)。因此, 我们得出正交残差由牛顿-舒尔茨更新的结论

$$\delta_{t,j+1} = \phi(\delta_{t,j})$$

其中 $\phi(u) = 1 - (1 - u)[p(1 - u)]^2$ □

引理3 (残差通过牛顿-舒尔茨多项式衰减). 对于牛顿-舒尔茨多项式 p_{κ} , $\phi(u) \leq u^{\kappa+1}$ 在 $[0, 1]$ 上, 其中 ϕ 是引理2中定义的函数。因此, 对于每一个 t 和所有 $j \geq 0$,

$$\delta_{t,j+1} \leq \delta_{t,j}^{\kappa+1}, \quad \delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}.$$

证明。牛顿-舒尔茨多项式 $p(\lambda)$ 是一个 κ 次多项式, 定义为 $g(\lambda) = \lambda^{-1/2}$ 在 $\lambda = 1$ 处展开的泰勒级数的截断。这意味着 $p(\lambda)$ 和 $g(\lambda)$ 在 $\lambda = 1$ 处的第一个 κ 个导数是相同的:

$$p^{(s)}(1) = g^{(s)}(1) \quad \text{for } s = 0, 1, \dots, \kappa$$

让我们考虑变量替换 $\lambda = 1 - u$ 。函数变为 $f(u) = g(1 - u) = (1 - u)^{-1/2}$ 。多项式 $p(1 - u)$ 然后是 $f(u)$ 在 $u = 0$ 处的泰勒展开, 直到项 u^{κ} :

$$p(1 - u) = \sum_{s=0}^{\kappa} \frac{f^{(s)}(0)}{s!} u^s = \sum_{s=0}^{\kappa} c_s u^s \quad (29)$$

$f(u)$ 的导数 $= (1 - u)^{-1/2}$ 在 $u = 0$ 处的值是 $f^{(s)}(0) = \frac{(2s)!}{s!4^s}$, 因此系数为 $c_s = \frac{(2s)!}{(s!)^2 4^s}$ 。此外, 由于 $p(1 - u)$ 和 $f(u)$ 在 $u = 0$ 处的第一个 κ 导数相同, 我们得到

$$f^{(s)}(0) = (-1)^s p^{(s)}(1), \quad \text{for } s = 1, \dots, \kappa \quad (30)$$

Consider the function $\tau(\lambda) := \lambda[p(\lambda)]^2$.

Claim: $\tau(\lambda)$ is non-decreasing on $[0, 1]$ with $\tau(1) \leq 1$.

$\tau(1) = 1 \cdot [p(1)]^2 = [g(1)]^2 = 1$ holds. Now, the derivative of $\tau(\lambda)$ is

$$\tau'(\lambda) = [p(\lambda)]^2 + 2\lambda p(\lambda)p'(\lambda) = p(\lambda) (p(\lambda) + 2\lambda p'(\lambda))$$

With $\lambda = 1 - u$, we have $p'(\lambda) = -p'(1 - u)$. Then, we obtain

$$\tau'(\lambda) = p(1 - u) \underbrace{(p(1 - u) - 2(1 - u)p'(1 - u))}_{:=S(u)} \quad (31)$$

Since all coefficients of $p(1 - u)$ are positive, i.e., $c_s = \frac{(2s)!}{(s!)^2 4^s} > 0$, we have

$$p(1 - u) > 0 \text{ for } u \in [0, 1]$$

Now, it suffices to prove $S(u) \geq 0$ for $u \in [0, 1]$. From Eq.(29), we can compute $S(u)$:

$$\begin{aligned} S(u) &= p(1 - u) - 2(1 - u)p'(1 - u) \\ &= \sum_{s=0}^{\kappa} c_s u^s - 2(1 - u) \sum_{s=1}^{\kappa} s c_s u^{s-1} \\ &= (c_0 - 2c_1) + \sum_{s=1}^{\kappa-1} ((2s+1)c_s - 2(s+1)c_{s+1}) u^s + (2\kappa+1)c_{\kappa} u^{\kappa} \end{aligned} \quad (32)$$

Note that the ratio for the coefficients is,

$$\frac{c_{s+1}}{c_s} = \frac{\frac{(2s+2)!}{((s+1)!)^2 4^{s+1}}}{\frac{(2s)!}{(s!)^2 4^s}} = \frac{2s+1}{2(s+1)}$$

Therefore,

$$(2s+1)c_s - 2(s+1)c_{s+1} = (2s+1)c_s - 2(s+1) \left(\frac{2s+1}{2(s+1)} c_s \right) = 0,$$

and also $c_0 - 2c_1 = 1 - 2 \cdot \frac{1}{2} = 0$. Hence, all coefficients up to degree $\kappa - 1$ cancel in Eq.(32), leaving the exact factorization

$$S(u) = (2\kappa+1)c_{\kappa} u^{\kappa}, \quad c_{\kappa} = \frac{(2\kappa)!}{4^{\kappa}(\kappa!)^2} > 0. \quad (33)$$

Putting Eq.(33) into Eq.(31), which is $\tau'(\lambda) = p(1 - u)S(u)$, gives

$$\tau'(\lambda) = p(1 - u)(2\kappa+1)c_{\kappa} u^{\kappa}, \quad \text{with } u = 1 - \lambda \in [0, 1].$$

Since $p(1 - u) > 0$, $c_{\kappa} > 0$, and $u^{\kappa} \geq 0$ on $[0, 1]$, we have $\tau'(\lambda) \geq 0$ for $\lambda \in [0, 1]$. Moreover, $\tau'(\lambda) > 0$ for $\lambda \in [0, 1)$ (because then $u > 0$), and $\tau'(1) = 0$. Therefore, τ is non-decreasing on $[0, 1]$ (in fact, it is strictly increasing on $[0, 1)$), as claimed.

Now, we can apply Lemma 2, since $\tau(\lambda)$ is non-decreasing on $[0, 1]$ with $\tau(1) \leq 1$. From Lemma 2, we know the orthogonality residual is updated according to the rule:

$$\delta_{t,j+1} = \phi(\delta_{t,j}) \quad (34)$$

where the function $\phi(u)$ is defined as:

$$\phi(u) = 1 - (1 - u)[p(1 - u)]^2 \quad (35)$$

Claim: $\phi^{(s)}(0) = 0$ for all $s = 1, \dots, \kappa$. The function $\phi(u)$ can be rewritten using $f(u) = (1 - u)^{-1/2}$ and $p(1 - u)$:

$$\phi(u) = 1 - \frac{1}{f(u)^2} [p(1 - u)]^2 \quad (36)$$

考虑函数 $\tau(\lambda) := \lambda[p(\lambda)]^2$ 。

命题: $\tau(\lambda)$ 在 $[0, 1]$ 上非递减, 且满足 $\tau(1) \leq 1$ 。

$\tau(1) = 1 \cdot [p(1)]^2 = [g(1)]^2 = 1$ 成立。现在, $\tau(\lambda)$ 的导数是

$$\tau'(\lambda) = [p(\lambda)]^2 + 2\lambda p(\lambda)p'(\lambda) = p(\lambda) (p(\lambda) + 2\lambda p'(\lambda))$$

给定 $\lambda = 1 - u$, 我们有 $p'(\lambda) = -p'(1 - u)$ 。然后, 我们得到

$$\tau'(\lambda) = p(1 - u) \underbrace{(p(1 - u) - 2(1 - u)p'(1 - u))}_{:=S(u)} \quad (31)$$

由于 $p(1 - u)$ 的所有系数都是正的, 即 $c_s = \frac{(2s)!}{(s!)^2 4^s} \geq 0$, 因此我们有

$$p(1 - u) > 0 \text{ for } u \in [0, 1]$$

现在, 只需证明 $S(u) \geq 0$ 对于 $u \in [0, 1]$ 成立。根据式(29), 我们可以计算 $S(u)$:

$$\begin{aligned} S(u) &= p(1 - u) - 2(1 - u)p'(1 - u) \\ &= \sum_{s=0}^{\kappa} c_s u^s - 2(1 - u) \sum_{s=1}^{\kappa} s c_s u^{s-1} \\ &= (c_0 - 2c_1) + \sum_{s=1}^{\kappa-1} ((2s+1)c_s - 2(s+1)c_{s+1}) u^s + (2\kappa+1)c_{\kappa} u^{\kappa} \end{aligned} \quad (32)$$

注意系数的比率是,

$$\frac{c_{s+1}}{c_s} = \frac{\frac{(2s+2)!}{((s+1)!)^2 4^{s+1}}}{\frac{(2s)!}{(s!)^2 4^s}} = \frac{2s+1}{2(s+1)}$$

因此,

$$(2s+1)c_s - 2(s+1)c_{s+1} = (2s+1)c_s - 2(s+1) \left(\frac{2s+1}{2(s+1)} c_s \right) = 0,$$

以及 $c_0 - 2c_1 = 1 - 2 \cdot \frac{1}{2} = 0$ 。因此, 所有最高次数为 $\kappa - 1$ 的系数在等式(32)中抵消, 留下精确的分解

$$S(u) = (2\kappa+1)c_{\kappa} u^{\kappa}, \quad c_{\kappa} = \frac{(2\kappa)!}{4^{\kappa}(\kappa!)^2} > 0. \quad (33)$$

将等式(33)代入等式(31), 即 $\tau'(\lambda) = p(1 - u)S(u)$, 得到

$$\tau'(\lambda) = p(1 - u)(2\kappa+1)c_{\kappa} u^{\kappa}, \quad \text{with } u = 1 - \lambda \in [0, 1].$$

由于 $p(1 - u) > 0$, $c_{\kappa} > 0$ 和 $u^{\kappa} \geq 0$ 在 $[0, 1]$ 上, 我们有 $\tau'(\lambda) \geq 0$ 对于 $\lambda \in [0, 1]$ 。此外, $\tau'(\lambda) > 0$ 对于 $\lambda \in [0, 1)$ (因为那时 $u > 0$), 并且 $\tau'(1) = 0$ 。因此, τ 在 $[0, 1]$ 上是非递减的 (实际上, 它在 $[0, 1]$ 上是严格递增的), 正如所声称的那样。

现在, 我们可以应用引理2, 因为 $\tau(\lambda)$ 在 $[0, 1]$ 上是单调递增的, 并且满足 $\tau(1) \leq 1$ 。根据引理2, 我们知道正交残差是按照以下规则进行更新的:

$$\delta_{t,j+1} = \phi(\delta_{t,j}) \quad (34)$$

其中函数 $\phi(u)$ 定义为:

$$\phi(u) = 1 - (1 - u)[p(1 - u)]^2 \quad (35)$$

声明: $\phi^{(s)}(0) = 0$ 对所有 $s = 1, \dots, \kappa$ 。函数 $\phi(u)$ 可以使用 $f(u) = (1 - u)^{-1/2}$ 和 $p(1 - u)$ 重新表示:

$$\phi(u) = 1 - \frac{1}{f(u)^2} [p(1 - u)]^2 \quad (36)$$

Let's define a remainder function $R_\kappa(u)$, which represents the difference between $f(u)$ and its Taylor approximation $p(1-u)$:

$$R_\kappa(u) = f(u) - p(1-u) \quad (37)$$

Since $p(1-u)$ matches the derivatives of $f(u)$ up to order κ at $u=0$, which implies Eq.(30), the remainder function and its first κ derivatives are all zero at $u=0$:

$$R_\kappa^{(s)}(0) = f^{(s)}(0) - p^{(s)}(1) \cdot (-1)^s = 0, \quad \text{for } s = 1, \dots, \kappa \quad (38)$$

Substituting $p(1-u) = f(u) - R_\kappa(u)$ from Eq.(37) to Eq.(36),

$$\begin{aligned} \phi(u) &= 1 - \frac{[f(u) - R_\kappa(u)]^2}{f(u)^2} \\ &= 1 - \frac{f(u)^2 - 2f(u)R_\kappa(u) + R_\kappa(u)^2}{f(u)^2} \\ &= 1 - \left(1 - \frac{2R_\kappa(u)}{f(u)} + \frac{R_\kappa(u)^2}{f(u)^2}\right) \\ &= \frac{2R_\kappa(u)}{f(u)} - \frac{R_\kappa(u)^2}{f(u)^2} = R_\kappa(u) \cdot \left(\frac{2f(u) - R_\kappa(u)}{f(u)^2}\right) \end{aligned}$$

Let's define the term in the brackets as a new function, $H(u)$:

$$H(u) = \frac{2f(u) - R_\kappa(u)}{f(u)^2}$$

So, we have a simple product form: $\phi(u) = R_\kappa(u)H(u)$. Note that $H(u)$ is well-behaved around $u=0$ since $f(0)=1$ and $R_\kappa(0)=0$, making the denominator non-zero. To find the derivatives of $\phi(u)$, we use the Leibniz rule (the generalized product rule) for the s -th derivative of a product:

$$\phi^{(s)}(u) = \frac{d^s}{du^s} [R_\kappa(u)H(u)] = \sum_{j=0}^s \binom{s}{j} R_\kappa^{(j)}(u) H^{(s-j)}(u)$$

Now, we evaluate this s -th derivative at $u=0$:

$$\phi^{(s)}(0) = \sum_{j=0}^s \binom{s}{j} R_\kappa^{(j)}(0) H^{(s-j)}(0)$$

Let's consider any integer s in the range $0 \leq s \leq \kappa$. In the summation above, the index j runs from 0 to s . For every term in this sum, the condition $j \leq s \leq \kappa$ holds. From Eq.(38), we know that $R_\kappa^{(j)}(0) = 0$ for all $j \leq \kappa$. Therefore, every single term in the summation contains a factor of $R_\kappa^{(j)}(0)$ which is equal to zero.

$$\phi^{(s)}(0) = \sum_{j=0}^s \binom{s}{j} (0) \cdot H^{(s-j)}(0) = 0$$

This result holds for all $s = 1, \dots, \kappa$. Thus, we have proven that the first κ derivatives of $\phi(u)$ at $u=0$ are all zero:

$$\phi(0) = \phi'(0) = \dots = \phi^{(\kappa)}(0) = 0$$

This implies that the Taylor series for $\phi(u)$ starts with a term of order $u^{\kappa+1}$:

$$\phi(u) = \frac{\phi^{(\kappa+1)}(0)}{(\kappa+1)!} u^{\kappa+1} + \mathcal{O}(u^{\kappa+2})$$

Find the leading term of $\phi(u)$.

Recall that we can express $p(1-u)$ using the Taylor remainder theorem:

$$p(1-u) = f(u) - R_\kappa(u) \quad (39)$$

让我们定义一个余数函数 $R_\kappa(u)$, 它表示 $f(u)$ 与其泰勒近似 $p(1-u)$ 之间的差值:

$$R_\kappa(u) = f(u) - p(1-u) \quad (37)$$

由于 $p(1-u)$ 与 $f(u)$ 的导数在 $u=0$ 处匹配, 最高阶为 κ , 这表明等式(30), 因此余函数及其前 κ 阶导数在 $u=0$ 处均为零:

$$R_\kappa^{(s)}(0) = f^{(s)}(0) - p^{(s)}(1) \cdot (-1)^s = 0, \quad \text{for } s = 1, \dots, \kappa \quad (38)$$

将等式(37)中的 $p(1-u)$ 、 $f(u)$ 和 $-R_\kappa(u)$ 代入等式(36),

$$\begin{aligned} \phi(u) &= 1 - \frac{[f(u) - R_\kappa(u)]^2}{f(u)^2} \\ &= 1 - \frac{f(u)^2 - 2f(u)R_\kappa(u) + R_\kappa(u)^2}{f(u)^2} \\ &= 1 - \left(1 - \frac{2R_\kappa(u)}{f(u)} + \frac{R_\kappa(u)^2}{f(u)^2}\right) \\ &= \frac{2R_\kappa(u)}{f(u)} - \frac{R_\kappa(u)^2}{f(u)^2} = R_\kappa(u) \cdot \left(\frac{2f(u) - R_\kappa(u)}{f(u)^2}\right) \end{aligned}$$

让我们将括号中的项定义为一个新函数, $H(u)$:

$$H(u) = \frac{2f(u) - R_\kappa(u)}{f(u)^2}$$

因此, 我们得到一个简单的乘积形式: $\phi(u) = R_\kappa(u)H(u)$ 。请注意 $H(u)$ 在 $u=0$ 附近表现良好, 因为 $f(0)=1$ 且 $R_\kappa(0)=0$, 使得分母不为零。为了找到 $\phi(u)$ 的导数, 我们使用Leibniz规则 (乘积的广义乘积规则) 来求第 s 阶导数:

$$\phi^{(s)}(u) = \frac{d^s}{du^s} [R_\kappa(u)H(u)] = \sum_{j=0}^s \binom{s}{j} R_\kappa^{(j)}(u) H^{(s-j)}(u)$$

现在, 我们在 $u=0$ 处计算这个 s 阶导数:

$$\phi^{(s)}(0) = \sum_{j=0}^s \binom{s}{j} R_\kappa^{(j)}(0) H^{(s-j)}(0)$$

让我们考虑任意整数 s 在范围 $0 \leq s \leq \kappa$ 内。在上述求和中, 索引 j 从 0 运行到 s 。对于这个求和中的每一项, 条件 $j \leq s \leq \kappa$ 都成立。根据公式(38), 我们知道 $R_\kappa^{(j)}(0) = 0$ 对所有 $j \leq \kappa$ 成立。因此, 求和中的每一项都包含一个因子 $R_\kappa^{(j)}(0)$, 它等于零。

$$\phi^{(s)}(0) = \sum_{j=0}^s \binom{s}{j} (0) \cdot H^{(s-j)}(0) = 0$$

这个结果对所有 $s = 1, \dots, \kappa$ 都成立。因此, 我们已经证明了 $\phi(u)$ 在 $u=0$ 处的所有前 κ 阶导数都为零:

$$\phi(0) = \phi'(0) = \dots = \phi^{(\kappa)}(0) = 0$$

这意味着 $\phi(u)$ 的泰勒级数以一个阶数为 $u^{\kappa+1}$ 的项开始:

$$\phi(u) = \frac{\phi^{(\kappa+1)}(0)}{(\kappa+1)!} u^{\kappa+1} + \mathcal{O}(u^{\kappa+2})$$

找到 $\phi(u)$ 的主导项。

回想我们可以用泰勒余项定理来表示 $p(1-u)$:

$$p(1-u) = f(u) - R_\kappa(u) \quad (39)$$

where $R_\kappa(u)$ is the remainder term, which is of order $\mathcal{O}(u^{\kappa+1})$. Substituting Eq.(39) into the expression for $\phi(u)$ in Eq.(35):

$$\begin{aligned}\phi(u) &= 1 - (1-u)[f(u) - R_\kappa(u)]^2 \\ &= 1 - (1-u)[f(u)^2 - 2f(u)R_\kappa(u) + R_\kappa(u)^2] \\ &= 1 - (1-u) \left[\frac{1}{1-u} - \frac{2R_\kappa(u)}{\sqrt{1-u}} + R_\kappa(u)^2 \right] \\ &= 1 - [1 - 2\sqrt{1-u}R_\kappa(u) + (1-u)R_\kappa(u)^2] \\ &= 2\sqrt{1-u}R_\kappa(u) - (1-u)R_\kappa(u)^2\end{aligned}$$

The remainder $R_\kappa(u)$ is given by $R_\kappa(u) = \frac{f^{(\kappa+1)}(c)}{(\kappa+1)!} u^{\kappa+1}$ for some $c \in (0, u)$. As $u \rightarrow 0$, the leading term of $\phi(u)$ is determined by $2\sqrt{1-u} \cdot \frac{f^{(\kappa+1)}(0)}{(\kappa+1)!} u^{\kappa+1}$. The derivatives of $f(u) = (1-u)^{-1/2}$ at $u = 0$ are $f^{(s)}(0) = \frac{(2s)!}{s!4^s}$. Thus, the leading coefficient of the Taylor series for $\phi(u)$ is:

$$C_\kappa = \frac{2f^{(\kappa+1)}(0)}{(\kappa+1)!} = \frac{2}{(\kappa+1)!} \frac{(2(\kappa+1))!}{(\kappa+1)!4^{\kappa+1}} = \frac{2}{4^{\kappa+1}} \binom{2\kappa+2}{\kappa+1}$$

For any $\kappa \geq 1$, this coefficient is less than 1. For example, for $\kappa = 1$, $C_1 = \frac{2}{16} \binom{4}{2} = \frac{12}{16} = \frac{3}{4} < 1$. In general, using Stirling's approximation $\binom{2n}{n} \leq \frac{4^n}{\sqrt{\pi n}}$, we have:

$$C_\kappa = \frac{2}{4^{\kappa+1}} \binom{2\kappa+2}{\kappa+1} \leq \frac{2}{4^{\kappa+1}} \frac{4^{\kappa+1}}{\sqrt{\pi(\kappa+1)}} = \frac{2}{\sqrt{\pi(\kappa+1)}} < 1 \quad \text{for } \kappa \geq 1.$$

Since $\phi(u) = C_\kappa u^{\kappa+1} + \mathcal{O}(u^{\kappa+2})$ and the leading coefficient C_κ is less than 1, there exists a sufficiently small $\rho_\kappa \in (0, 1)$ such that $\phi(u) \leq u^{\kappa+1}$ for all $u \in [0, \rho_\kappa]$. Specifically, you can choose any $\theta \in (C_\kappa, 1)$ and take ρ_κ so that $|\phi(u) - C_\kappa u^{\kappa+1}| \leq (\theta - C_\kappa)u^{\kappa+1}$ on $[0, \rho_\kappa]$. Then

$$\phi(u) \leq \theta u^{\kappa+1} \leq u^{\kappa+1} \quad (40)$$

which makes the "exists ρ_κ " completely explicit.

Combining the recursion Eq.(34) and Eq.(40), we have

$$\delta_{t,j+1} = \phi(\delta_{t,j}) \leq (\delta_{t,j})^{\kappa+1}$$

After q steps, we obtain the final result by repeated application:

$$\delta_{t,q} \leq (\delta_{t,q-1})^{\kappa+1} \leq ((\delta_{t,q-2})^{\kappa+1})^{\kappa+1} \leq \dots (\delta_{t,0})^{(\kappa+1)^q}$$

Finally, we conclude:

$$\delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}$$

This establishes that the orthogonality residual decays with an order of $\kappa + 1$ at each step of the NEWTON-SCHULZ if NEWTON-SCHULZ polynomial is of degree κ . $\square$

Corollary 1 (Final constant factor bound). *Let $\delta_0 := \sup_t \delta_{t,0} < 1$ (Remark 1). Then for all t and $q \geq 1$,*

$$\delta_{t,q} \leq \delta_0^{(\kappa+1)^q}, \quad \varepsilon_q \leq 1 - \sqrt{1 - \delta_0^{(\kappa+1)^q}}, \quad \chi_q = (1 - \varepsilon_q)^{-1} \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}.$$

Proof. Combine Lemma 3 with Lemma 1. $\square$

Corollary 2 (Case when $\kappa \in \{1, 2\}$). *For the polynomials $p_1(\lambda) = \frac{3}{2} - \frac{1}{2}\lambda$ and $p_2(\lambda) = \frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2$, the residual recursion specializes to $\delta_{t,j+1} \leq \delta_{t,j}^2$ and $\delta_{t,j+1} \leq \delta_{t,j}^3$ for all $\delta_{t,j} \in [0, 1]$. These are concrete instances of the general residual map in Lemma 2.*

Proof. Direct substitution into $\phi(u) = 1 - (1-u)[p_\kappa(1-u)]^2$ gives $\delta_{j+1} = \frac{\delta_j^2(3+\delta_j)}{4} \leq \delta_j^2$ since $3 + \delta \leq 4$ on $[0, 1]$ for $\kappa = 1$. Also, $\delta_{j+1} = \frac{\delta_j^3(40+15\delta_j+9\delta_j^2)}{64} \leq \delta_j^3$ since $40 + 15\delta + 9\delta^2 \leq 64$ on $[0, 1]$ for $\kappa = 2$ (check at $\delta = 1$). $\square$

其中 $R_\kappa(u)$ 是余项, 其阶数为 $\mathcal{O}(u^{\kappa+1})$ 。将等式(39)代入等式(35)中 $\phi(u)$ 的表达式:

$$\begin{aligned}\phi(u) &= 1 - (1-u)[f(u) - R_\kappa(u)]^2 \\ &= 1 - (1-u)[f(u)^2 - 2f(u)R_\kappa(u) + R_\kappa(u)^2] \\ &= 1 - (1-u) \left[\frac{1}{1-u} - \frac{2R_\kappa(u)}{\sqrt{1-u}} + R_\kappa(u)^2 \right] \\ &= 1 - [1 - 2\sqrt{1-u}R_\kappa(u) + (1-u)R_\kappa(u)^2] \\ &= 2\sqrt{1-u}R_\kappa(u) - (1-u)R_\kappa(u)^2\end{aligned}$$

余项 $R_\kappa(u)$ 由 $R_\kappa(u) = \frac{f^{(\kappa+1)}(c)}{(\kappa+1)!} u^{\kappa+1}$ 给出, 其中 $c \in (0, u)$ 。作为 $u \rightarrow 0$, $\phi(u)$ 的主导项由 $2\sqrt{1-u} \cdot \frac{f^{(\kappa+1)}(0)}{(\kappa+1)!} u^{\kappa+1}$ 确定。 $f(u) = (1-u)^{-1/2}$ 在 $u = 0$ 处的导数为 $f^{(s)}(0) = \frac{(2s)!}{s!4^s}$ 。因此, $\phi(u)$ 的泰勒级数的主系数为:

$$C_\kappa = \frac{2f^{(\kappa+1)}(0)}{(\kappa+1)!} = \frac{2}{(\kappa+1)!} \frac{(2(\kappa+1))!}{(\kappa+1)!4^{\kappa+1}} = \frac{2}{4^{\kappa+1}} \binom{2\kappa+2}{\kappa+1}$$

对于任何 $\kappa \geq 1$, 该系数都小于1。例如, 对于 $\kappa = 1$, $C_1 = \frac{2}{16} \binom{4}{2} = \frac{12}{16} = \frac{3}{4} < 1$ 。通常, 使用斯特林近似 $\binom{2n}{n} \leq \frac{4^n}{\sqrt{\pi n}}$, 我们有:

$$C_\kappa = \frac{2}{4^{\kappa+1}} \binom{2\kappa+2}{\kappa+1} \leq \frac{2}{4^{\kappa+1}} \frac{4^{\kappa+1}}{\sqrt{\pi(\kappa+1)}} = \frac{2}{\sqrt{\pi(\kappa+1)}} < 1 \quad \text{for } \kappa \geq 1.$$

由于 $\phi(u) = C_\kappa u^{\kappa+1} + \mathcal{O}(u^{\kappa+2})$ 且首项系数 C_κ 小于1, 存在一个足够小的 $\rho_\kappa \in (0, 1)$, 使得 $\phi(u) \leq u^{\kappa+1}$ 对所有 $u \in [0, \rho_\kappa]$ 成立。具体来说, 你可以选择任何 $\theta \in (C_\kappa, 1)$, 并取 ρ_κ , 使得 $|\phi(u) - C_\kappa u^{\kappa+1}| \leq (\theta - C_\kappa)u^{\kappa+1}$ 在 $[0, \rho_\kappa]$ 上成立。然后

$$\phi(u) \leq \theta u^{\kappa+1} \leq u^{\kappa+1} \quad (40)$$

这使得 "存在 ρ_κ " 完全明确。

结合递推公式 Eq.(34) 和 Eq.(40), 我们得到

$$\delta_{t,j+1} = \phi(\delta_{t,j}) \leq (\delta_{t,j})^{\kappa+1}$$

经过 q 步, 通过反复应用, 我们得到最终结果:

$$\delta_{t,q} \leq (\delta_{t,q-1})^{\kappa+1} \leq ((\delta_{t,q-2})^{\kappa+1})^{\kappa+1} \leq \dots (\delta_{t,0})^{(\kappa+1)^q}$$

最后, 我们得出结论:

$$\delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}$$

这表明, 如果 NEWTON-SCHULZ 多项式是 κ 次的, 那么 NEWTON-SCHULZ 算法每一步的正交残差都以 $\kappa + 1$ 的阶数衰减。 $\square$

推论1 (最终常数因子界限)。设 $\delta_0 := \sup_t \delta_{t,0} < 1$ (注释1)。那么对于所有 t 和 $q \geq 1$,

$$\delta_{t,q} \leq \delta_0^{(\kappa+1)^q}, \quad \varepsilon_q \leq 1 - \sqrt{1 - \delta_0^{(\kappa+1)^q}}, \quad \chi_q = (1 - \varepsilon_q)^{-1} \leq [1 - \delta_0^{(\kappa+1)^q}]^{-1/2}.$$

证明。将引理3与引理1结合起来。 $\square$

推论2 (当 $\kappa \in \{1, 2\}$)。对于多项式 $p_1(\lambda) = \frac{3}{2} - \frac{1}{2}\lambda$ 和 $p_2(\lambda) = \frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2$, 残差递归专门化为 $\delta_{j+1} \leq \delta_j^2$ 和 $\delta_{j+1} \leq \delta_j^3$, 对于所有 $\delta_j \in [0, 1]$ 。这些都是引理2中一般残差映射的具体实例。

证明。将直接代入 $\phi(u) = 1 - (1-u)[p_\kappa(1-u)]^2$ 得到 $\delta_{j+1} = \frac{\delta_j^2(3+\delta_j)}{4} \leq \delta_j^2$, 因为 $3 + \delta \leq 4$ 在 $[0, 1]$ 上对于 $\kappa = 1$ 。此外, $\delta_{j+1} = \frac{\delta_j^3(40+15\delta_j+9\delta_j^2)}{64} \leq \delta_j^3$, 因为 $40 + 15\delta + 9\delta^2 \leq 64$ 在 $[0, 1]$ 上对于 $\kappa = 2$ (在 $\delta = 1$ 处检查)。 $\square$

D.1 DETAILED PROOFS FOR CASE WHEN $\kappa \in \{1, 2\}$

Residual Decay by 1st-order NEWTON–SCHULZ Polynomial. Define the polynomial $p(\lambda)$ for NEWTON–SCHULZ steps as

$$p(\lambda) = \frac{3}{2} - \frac{1}{2}\lambda$$

Then for any fixed t and any $j \geq 0$, the residual $\delta_{t,j}$ satisfies

$$\delta_{t,j+1} \leq (\delta_{t,j})^2$$

Consequently, if $\delta_{t,0} \leq \rho < 1$ is sufficiently small for all t , then $\delta_{t,q} \leq \rho^{2^q}$ for all t .

Proof. Deriving the function $\tau(\lambda)$ in Lemma 2,

$$\tau(\lambda) = \lambda[p(\lambda)]^2 = \lambda \left(\frac{3}{2} - \frac{1}{2}\lambda \right)^2 = \frac{9}{4}\lambda - \frac{3}{2}\lambda^2 + \frac{1}{4}\lambda^3$$

Then the derivative of $\tau(\lambda)$ is

$$\tau'(\lambda) = \frac{9}{4} - 3\lambda + \frac{3}{4}\lambda^2 = \frac{3}{4}(\lambda - 1)(\lambda - 3)$$

Since $\tau'(\lambda) \geq 0$ for $\lambda \in [0, 1]$ and $\tau(1) = 1$, we can apply Lemma 2. The orthogonality residual $\delta_{t,j}$ is updated by one NEWTON–SCHULZ step as

$$\begin{aligned} \delta_{t,j+1} &= \phi(\delta_{t,j}) = 1 - (1 - \delta_{t,j})[p(1 - \delta_{t,j})]^2 \\ &= 1 - (1 - \delta_{t,j}) \left(\frac{3}{2} - \frac{1}{2}(1 - \delta_{t,j}) \right)^2 = \frac{\delta_{t,j}^2(3 + \delta_{t,j})}{4} \leq \delta_{t,j}^2 \end{aligned}$$

By induction, after q steps, we get the relationship, $\delta_{t,q} \leq (\delta_{t,0})^{2^q}$. Given the assumption that $\delta_{t,0} \leq \rho < 1$ for all t , the conclusion follows directly:

$$\delta_{t,q} \leq \rho^{2^q}$$

This shows that the orthogonality residual decreases quadratically with each iteration, which is a very rapid rate of convergence. $\square$

Residual Decay by 2nd-order NEWTON–SCHULZ Polynomial. Define the polynomial $p(\lambda)$ for NEWTON–SCHULZ steps as

$$p(\lambda) = \frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2$$

Then for any fixed t and any $j \geq 0$, the residual $\delta_{t,j}$ satisfies

$$\delta_{t,j+1} \leq (\delta_{t,j})^3$$

Consequently, if $\delta_{t,0} \leq \rho < 1$ is sufficiently small for all t , then $\delta_{t,q} \leq \rho^{3^q}$ for all t .

Proof. Deriving the function $\tau(\lambda)$ in Lemma 2,

$$\tau(\lambda) = \lambda[p(\lambda)]^2 = \lambda \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right)^2$$

Then the derivative of $\tau(\lambda)$ is

$$\begin{aligned} \tau'(\lambda) &= \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right)^2 + 2\lambda \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right) \\ &= \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right) \left(\frac{15}{8} + \frac{3}{4}\lambda + \frac{3}{8}\lambda^2 \right) = \frac{1}{64} (20 + (5 - 3\lambda)^2) (4 + (1 + \lambda)^2) \end{aligned}$$

D.1 $\kappa \in \{1, 2\}$ 情况下的详细证明

一阶 NEWTON–SCHULZ 多项式残差衰减。 EWTON–SCHULZ 多项式。定义 NEWTON–SCHULZ 步骤的多项式 $p(\lambda)$ 为

$$p(\lambda) = \frac{3}{2} - \frac{1}{2}\lambda$$

然后对于任何固定的 t 和任何 $j \geq 0$, 残差 $\delta_{t,j}$ 满足

$$\delta_{t,j+1} \leq (\delta_{t,j})^2$$

因此, 如果对于所有 t , $\delta_{t,0} \leq \rho < 1$ 足够小, 那么对于所有 t , $\delta_{t,q} \leq \rho^{2^q}$

证明。 在引理2中推导函数 $\tau(\lambda)$,

$$\tau(\lambda) = \lambda[p(\lambda)]^2 = \lambda \left(\frac{3}{2} - \frac{1}{2}\lambda \right)^2 = \frac{9}{4}\lambda - \frac{3}{2}\lambda^2 + \frac{1}{4}\lambda^3$$

那么 $\tau(\lambda)$ 的导数是

$$\tau'(\lambda) = \frac{9}{4} - 3\lambda + \frac{3}{4}\lambda^2 = \frac{3}{4}(\lambda - 1)(\lambda - 3)$$

由于 $\tau'(\lambda) \geq 0$ 对于 $\lambda \in [0, 1]$ 和 $\tau(1) = 1$, 我们可以应用引理2。正交残差 $\delta_{t,j}$ 通过一个 NEWTON–SCHULZ 步骤更新为

$$\begin{aligned} \delta_{t,j+1} &= \phi(\delta_{t,j}) = 1 - (1 - \delta_{t,j})[p(1 - \delta_{t,j})]^2 \\ &= 1 - (1 - \delta_{t,j}) \left(\frac{3}{2} - \frac{1}{2}(1 - \delta_{t,j}) \right)^2 = \frac{\delta_{t,j}^2(3 + \delta_{t,j})}{4} \leq \delta_{t,j}^2 \end{aligned}$$

归纳地, 经过 q 步后, 我们得到关系式, $\delta_{t,q} \leq (\delta_{t,0})^{2^q}$ 。给定假设 $\delta_{t,0} \leq \rho < 1$ 对于所有 t , 结论直接成立:

$$\delta_{t,q} \leq \rho^{2^q}$$

这表明正交残差在每次迭代中二次减小, 这是一个非常快的收敛速度。 $\square$

残差通过2阶 NEWTON衰减EWTON–SCHULZ 多项式。定义 NEWTON–SCHULZ 步骤的多项式 $p(\lambda)$ 为

$$p(\lambda) = \frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2$$

然后对于任何固定的 t 和任何 $j \geq 0$, 残差 $\delta_{t,j}$ 满足

$$\delta_{t,j+1} \leq (\delta_{t,j})^3$$

因此, 如果 $\delta_{t,0} \leq \rho < 1$ 对于所有 t 足够小, 那么 $\delta_{t,q} \leq \rho^{3^q}$ 对于所有 t 。

证明。 在引理2中推导函数 $\tau(\lambda)$,

$$\tau(\lambda) = \lambda[p(\lambda)]^2 = \lambda \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right)^2$$

那么 $\tau(\lambda)$ 的导数是

$$\begin{aligned} \tau'(\lambda) &= \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right)^2 + 2\lambda \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right) \\ &= \left(\frac{15}{8} - \frac{5}{4}\lambda + \frac{3}{8}\lambda^2 \right) \left(\frac{15}{8} + \frac{3}{4}\lambda + \frac{3}{8}\lambda^2 \right) = \frac{1}{64} (20 + (5 - 3\lambda)^2) (4 + (1 + \lambda)^2) \end{aligned}$$

Since $\tau'(\lambda) \geq 0$ for $\lambda \in [0, 1]$ and $\tau(1) = 1$, we can apply Lemma 2. The orthogonality residual $\delta_{t,j}$ is updated by one NEWTON-SCHULZ step as

$$\begin{aligned}\delta_{t,j+1} &= \phi(\delta_{t,j}) = 1 - (1 - \delta_{t,j})[p(1 - \delta_{t,j})]^2 \\ &= 1 - (1 - \delta_{t,j}) \left(\frac{15}{8} - \frac{5}{4}(1 - \delta_{t,j}) + \frac{3}{8}(1 - \delta_{t,j})^2 \right)^2 \\ &= 1 - (1 - \delta_{t,j}) \left(1 + \frac{1}{2}\delta_{t,j} + \frac{3}{8}\delta_{t,j}^2 \right)^2 \\ &= 1 - (1 - \delta_{t,j}) \left(1 + \delta_{t,j} + \delta_{t,j}^2 + \frac{3}{8}\delta_{t,j}^3 + \frac{9}{64}\delta_{t,j}^4 \right) \\ &= \frac{\delta_{t,j}^3(40 + 15\delta_{t,j} + 9\delta_{t,j}^2)}{64} \leq \delta_{t,j}^3\end{aligned}$$

By induction, after q steps, we get the relationship, $\delta_{t,q} \leq (\delta_{t,0})^{3^q}$. Given the assumption that $\delta_{t,0} \leq \rho < 1$ for all t , the conclusion follows directly:

$$\delta_{t,q} \leq \rho^{3^q}$$

This shows that the orthogonality residual decreases cubically with each iteration, which is a very rapid rate of convergence. $\square$

由于 $\tau'(\lambda) \geq 0$ 对于 $\lambda \in [0, 1]$ 和 $\tau(1) = 1$, 我们可以应用引理2。正交残差 $\delta_{t,j}$ 通过一个 NEWTON-SCHULZ 步骤更新为

$$\begin{aligned}\delta_{t,j+1} &= \phi(\delta_{t,j}) = 1 - (1 - \delta_{t,j})[p(1 - \delta_{t,j})]^2 \\ &= 1 - (1 - \delta_{t,j}) \left(\frac{15}{8} - \frac{5}{4}(1 - \delta_{t,j}) + \frac{3}{8}(1 - \delta_{t,j})^2 \right)^2 \\ &= 1 - (1 - \delta_{t,j}) \left(1 + \frac{1}{2}\delta_{t,j} + \frac{3}{8}\delta_{t,j}^2 \right)^2 \\ &= 1 - (1 - \delta_{t,j}) \left(1 + \delta_{t,j} + \delta_{t,j}^2 + \frac{3}{8}\delta_{t,j}^3 + \frac{9}{64}\delta_{t,j}^4 \right) \\ &= \frac{\delta_{t,j}^3(40 + 15\delta_{t,j} + 9\delta_{t,j}^2)}{64} \leq \delta_{t,j}^3\end{aligned}$$

归纳地, 经过 q 步, 我们得到关系式, $\delta_{t,q} \leq (\delta_{t,0})^{3^q}$ 。在假设 $\delta_{t,0} \leq \rho < 1$ 对所有 t 成立的前提下, 结论直接得证:

$$\delta_{t,q} \leq \rho^{3^q}$$

这表明正交残差随着每次迭代呈三次方减少, 这是一个非常快的收敛速度。 $\square$

E WALL-CLOCK VIA COMPUTATIONAL COMPLEXITY.

At each iteration t , MUON performs an orthogonalization of the momentum matrix M_t via either SVD or NEWTON–SCHULZ (NS). We write Φ_{gemm} for the effective GEMM throughput (FLOP/s), and Φ_{svd} for the effective throughput of the SVD routine. In practice $\Phi_{\text{gemm}} \gg \Phi_{\text{svd}}$ due to far higher hardware utilization of GEMM.

Per-iteration Orthogonalization FLOPs. For a single layer index by ℓ with $m \leq n$ (for $m > n$, apply to $M_t^\top$ and transpose-trick):

- **MUON with SVD.** A thin SVD of $M_t \in \mathbb{R}^{m \times n}$ and extracting the polar factor $U_t V_t^\top$ costs (Golub & Reinsch, 1971):

$$\text{FLOPs}_{\text{svd}}^{(\ell)}(m, n) = \Theta(4m^2n + 8m^3)$$

Wall-clock time per layer is $t_{\text{svd}}^{(\ell)} = \text{FLOPs}_{\text{svd}}^{(\ell)}(m, n) / \Phi_{\text{svd}}$.

- **MUON with NS (q -steps, κ -degree).** NEWTON–SCHULZ follows Horner’s rule when recursively updating the scaled momentum matrix using the NEWTON–SCHULZ polynomial. NEWTON–SCHULZ forms $A = XX^\top \in \mathbb{R}^{m \times m}$ and applies the degree- κ polynomial to X via Horner’s rule. Each NS step needs one $m \times n$ by $n \times m$ GEMM to build A and κ multiplies AY (each $m \times m$ by $m \times n$). Hence,

$$\text{FLOPs}_{\text{ns}}^{(\ell)}(m, n; q, \kappa) = \Theta(2q(\kappa + 1)m^2n)$$

Wall-clock time per layer is $t_{\text{ns}}^{(\ell)} = \text{FLOPs}_{\text{ns}}^{(\ell)}(m, n; q, \kappa) / \Phi_{\text{gemm}}$.

Lemma 11. For a layer with $m \leq n$, the wall-clock time ratio between MUON with SVD and MUON with NEWTON–SCHULZ (q -steps, κ -degree) is,

$$\frac{t_{\text{svd}}^{(\ell)}}{t_{\text{ns}}^{(\ell)}} = \frac{\Theta(4m^2n + 8m^3)}{\Theta(2q(\kappa + 1)m^2n)} \cdot \frac{\Phi_{\text{gemm}}}{\Phi_{\text{svd}}} = \Theta\left(\frac{2 + 4\frac{m}{n}}{q(\kappa + 1)}\right) \cdot \underbrace{\frac{\Phi_{\text{gemm}}}{\Phi_{\text{svd}}}}_{\text{efficiency ratio} \gg 1}$$

Discussion of Lemma 11. With the practical setting $q \in \{2, 3\}$ and $\kappa \in \{1, 2\}$, the algebraic factor $\frac{2+4(m/n)}{q(\kappa+1)}$ is $\mathcal{O}(0.3 \sim 1)$, so the wall-clock speedup is essentially the GEMM/SVD efficiency ratio. On modern GPUs, $\Phi_{\text{gemm}}/\Phi_{\text{svd}}$ is often 4~10, so NS typically yields a multi- $\times$ speedup per iteration over SVD, matching empirical observations in Figure 1.

Practical Interpretation. NEWTON–SCHULZ scales linearly in q (accuracy knob) and uses only GEMMs, which map efficiently to GPUs. Exact SVD pays an additional m^3 term and typically incurs larger constants. Hence for small q and modest κ , NEWTON–SCHULZ is substantially cheaper per update than a full SVD while achieving near-exact orthogonalization in practice.

E 墙上时钟时间 通过 计算复杂度。

在每次迭代 t 中，MUON 通过 SVD 或 NEWTON–SCHULZ (NS) 对动量矩阵 M_t 进行正交化。我们用 Φ_{gemm} 表示有效 GEMM 吞吐量 (FLOP/s)，用 Φ_{svd} 表示 SVD 例程的有效吞吐量。在实践中 $\Phi_{\text{gemm}} \gg \Phi_{\text{svd}}$ 由于 GEMM 的硬件利用率远高于其他操作。

每迭代 正交化 FLOPs. 对于索引为 ℓ 的单个层，使用 $m \leq n$ (对于 $m > n$, 应用于 $M_t^\top$ 并转置技巧)：

- **MUON 与 SVD.** $M_t \in \mathbb{R}^{m \times n}$ 的薄 SVD 和提取极因子 $U_t V_t^\top$ 的成本 (Golub & Reinsch, 1971)：

$$\text{FLOPs}_{\text{svd}}^{(\ell)}(m, n) = \Theta(4m^2n + 8m^3)$$

每层的墙上时钟时间是 $t_{\text{svd}}^{(\ell)} = \text{FLOPs}_{\text{svd}}^{(\ell)}(m, n) / \Phi_{\text{svd}}$

- **MUON 与 NS (q -步, κ -度).** NEWTON–SCHULZ 在使用 NEWTON–SCHULZ 多项式递归更新缩放动量矩阵时遵循 Horner 规则。NEWTON–SCHULZ 形成 $A = XX^\top \in \mathbb{R}^{m \times m}$ 并通过 Horner 规则将度数为 κ 的多项式应用于 X 。每个 NS 步需要一次 $m \times n$ by $n \times m$ GEMM 来构建 A ，并且 κ 乘以 AY (每个 $m \times m$ by $m \times n$)。因此，

$$\text{FLOPs}_{\text{ns}}^{(\ell)}(m, n; q, \kappa) = \Theta(2q(\kappa + 1)m^2n)$$

每层墙上时钟时间为 $t_{\text{ns}}^{(\ell)} = \text{FLOPs}_{\text{ns}}^{(\ell)}(m, n; q, \kappa) / \Phi_{\text{gemm}}$

引理 11. 对于具有 $m \leq n$ 的层，MUON 与 SVD 和 MUON 与 NEWTON–SCHULZ (q -步, κ -度) 之间的墙上时钟时间比率是，

$$\frac{t_{\text{svd}}^{(\ell)}}{t_{\text{ns}}^{(\ell)}} = \frac{\Theta(4m^2n + 8m^3)}{\Theta(2q(\kappa + 1)m^2n)} \cdot \frac{\Phi_{\text{gemm}}}{\Phi_{\text{svd}}} = \Theta\left(\frac{2 + 4\frac{m}{n}}{q(\kappa + 1)}\right) \cdot \underbrace{\frac{\Phi_{\text{gemm}}}{\Phi_{\text{svd}}}}_{\text{efficiency ratio} \gg 1}$$

引理 11 的讨论. 在实际设置 $q \in \{2, 3\}$ 和 $\kappa \in \{1, 2\}$ 下，代数因子 $2+4(m/n)/q(\kappa+1)$ 是 $\mathcal{O}(0.3 \sim 1)$ ，因此墙上时钟加速本质上就是 GEMM/SVD 效率比率。在现代 GPU 上， $\Phi_{\text{gemm}}/\Phi_{\text{svd}}$ 通常是 4~10，因此 NS 通常在每次迭代中比 SVD 实现多 $\times$ 倍加速，这与图 1 中的经验观察结果一致。

实际应用. NEWTON–SCHULZ 在 q (精度调节) 上线性缩放，并且仅使用 GEMMs，这些操作能高效映射到 GPU 上。精确 SVD 需要额外支付 m^3 项，并且通常导致更大的常数。因此对于小 q 和适度 κ ，NEWTON–SCHULZ 每次更新成本远低于 fullSVD，同时在实践中实现了近乎精确的正交化。

F NUMERICAL EXPERIMENTS DETAIL

F.1 EXPERIMENTAL SETTING

Task and metric. All experiments are conducted on CIFAR-10 (50k train / 10k test) with the standard channel-wise normalization mean = (0.4914, 0.4822, 0.4465) and std = (0.2470, 0.2435, 0.2616).

We report cross-entropy train loss and test loss. Plots show the ($mean \pm 1 \cdot std$) over 5 independent runs, and we also report the wall-clock time (in seconds) accumulated from the beginning of training.

Model. We use a compact convolutional network (CIFARNET, approximately $2M$ parameters): a fixed whitening 2×2 convolution (weights frozen, bias trainable), followed by three ConvGroups (each: 3×3 conv $\rightarrow$ MaxPool2d(2) $\rightarrow$ GELU $\rightarrow 3 \times 3$ conv $\rightarrow$ GELU) with widths (64, 256, 256) and a linear head. BatchNorm layers keep affine weights frozen (biases are trainable). For stability, the first portion of each convolution is Dirac-initialized, and the linear head is variance-normalized.

Hyperparameters and schedules. Unless stated otherwise, we train for 50 epochs with a batch size of $B = 512$ on the GPU (the CPU fallback uses $B = 256$). Each run uses a different seed.

- **Main optimizer (one of MUON (q), MUON (SVD), or SGD-M)** on convolutional filters: learning rate $\eta = 0.0860632$, momentum $\beta = 0.730778$.
- **Auxiliary SGD** (whitening bias, BatchNorm biases, linear head): learning rates $\eta_{\text{other}} = 1.4949 \times 10^{-3}$ and $\eta_{\text{head}} = 1.72446$ with momentum 0.989703 (Nesterov).
- **Label smoothing** 0.2, and **gradient clipping** $\|\cdot\|_2 \leq 1.0$.
- **Schedule:** The same warm-up + cosine schedule is applied to *all* optimizers: a 5% linear warm-up of total steps followed by cosine decay to 0. Schedulers step once per update.

Evaluation. At the end of each epoch, we evaluate the test loss using the same normalization (no augmentation). We record (i) epoch-wise training loss, (ii) test loss, and (iii) the cumulative wall-clock time. Results are aggregated across 5 runs. Figures report ($mean \pm 1 \cdot std$) and include both epoch-aligned and time-aligned views to disentangle statistical and systemic effects.

F.2 NEWTON-SCHULZ STEPS (q) ABLATIONS.

Optimizers under comparison. We compare:

1. **MUON q -step:** SGD with momentum followed by an orthogonalization of the momentum matrix using q NEWTON-SCHULZ steps (Algorithm 1); $q \in \{0, 1, 2, 3\}$. The case $q = 0$ is a normalization-only ablation (no orthogonalization).
2. **MUON (SVD):** MUON with an exact polar step $UV^\top$ via SVD.
3. **SGD-M:** SGD with momentum baseline with identical schedules.

All methods share identical schedules and auxiliary updates (Appendix F.1).

Orthogonalization details. For MUON with q -NEWTON-SCHULZ steps, we use NEWTON-SCHULZ polynomial, p_κ

$$p(\lambda) = a + b\lambda + c\lambda^2, \quad (a, b, c) = (15/8, -5/4, 3/8),$$

applied as $X \leftarrow aX + (bA + cA^2)X$ with $A = XX^\top$ and $X = M/\|M\|_F$; if the matrix is *tall*, we employ the transpose trick. The SVD variant normalizes M by its Frobenius norm and returns $UV^\top$. In all MUON variants, each weight tensor is re-normalized once per step to the Frobenius norm $\sqrt{\text{out_channels}}$ to stabilize scales.

Ablations on q . Our main comparison varies the number of NEWTON-SCHULZ steps $q \in \{1, 2, 3\}$ while holding the polynomial fixed to p_κ , and contrasts these with MUON with SVD and SGD-M. All other components (architecture, schedules, augmentation) are kept identical across methods to ensure a fair comparison.

F 数值实验细节

F.1 实验设置

任务和指标。 所有实验均在 CIFAR-10 (50k 训练集 / 10k 测试集) 上进行，采用标准通道方向归一化均值 = (0.4914, 0.4822, 0.4465) 和标准差 = (0.2470, 0.2435, 0.2616)。

我们报告了交叉熵训练损失和测试损失。图表显示了 ($mean \pm 1 \cdot std$) 在5次独立运行中的变化，我们还报告了从训练开始累积的墙上时钟时间（秒）。

模型。 我们使用一个紧凑的卷积网络（CIFARNET，大约 $2M$ 参数）：一个固定的白化 2×2 卷积（权重冻结，偏差可训练），然后是三个ConvGroups（每个： 3×3 卷积 $\rightarrow$ MaxPool2d(2) $\rightarrow$ GELU $\rightarrow 3 \times 3$ 卷积 $\rightarrow$ GELU），宽度为 (64, 256, 256)，以及一个线性头。BatchNorm层保持仿射权重冻结（偏差可训练）。为了稳定性，每个卷积的第一部分使用狄拉克初始化，线性头进行方差归一化。

超参数和调度。 除非另有说明，我们在 GPU 上使用批量大小 $B = 512$ 训练 50 个周期（CPU 回退使用 $B = 256$ ）。每次运行都使用不同的种子。

- **主优化器 (MUON (q), MUON (SVD), or SGD-M)** 对卷积滤波器：学习率 $\eta = 0.0860632$ ，动量 $\beta = 0.730778$ 。
- **辅助 SGD** (白化偏差，BatchNorm 偏差，线性头): 学习率 $\eta_{\text{其他}} = 1.4949 \times 10^{-3}$ 和 $\eta_{\text{头}} = 1.72446$ ，动量 0.989703 (Nesterov)。
- **标签平滑** 0.2，和 **梯度裁剪** $\|\cdot\|_2 \leq 1.0$ 。
- **计划：** 相同的预热 +余弦计划应用于所有 优化器：总步数的5%线性预热，然后余弦衰减至 0。调度器每次更新时步进一次。

评估。 在每个周期结束时，我们使用相同的归一化（不进行增强）来评估测试损失。我们记录 (i) 按周期划分的训练损失，(ii) 测试损失，以及 (iii) 累计墙上时间。结果是在 5 次运行中汇总的。图表报告 ($mean \pm 1 \cdot std$) 并包括按周期对齐和按时间对齐的视图，以分离统计和系统效应。

F.2 NEWTON-SCHULZ步数(q)消融

优化器正在比较。 我们比较：

1. **MUON q -步：** 带有动量的SGD，随后使用 q NEWTON-SCHULZ步数（算法1）； $q \in \{0, 1, 2, 3\}$ 。该情况 $q = 0$ 是仅归一化的消融（无正交化）。 2. **MUON (SVD)：** 通过SVD进行精确极步的MUON。 3. **SGD-M：** 带有动量基线的SGD，具有相同调度。

所有方法共享相同调度和辅助更新（附录F.1）。

正交化细节。 对于MUON与 q -牛顿-舒尔茨步数，我们使用牛顿-舒尔茨多项式, p_κ

$$p(\lambda) = a + b\lambda + c\lambda^2, \quad (a, b, c) = (15/8, -5/4, 3/8),$$

应用于 $X \leftarrow aX + (bA + cA^2)X$ ，使用 $A = XX^\top$ 和 $X = M/\|M\|_F$ ；如果矩阵是 高维的，我们采用转置技巧。SVD变体通过其Frobenius范数对 M 进行归一化，并返回 $UV^\top$ 。在所有 MUON变体中，每个权重张量每步都会重新归一化到Frobenius范数 $\sqrt{\text{out_channels}}$ ，以稳定 _ 通道。

对 q 。 我们的主要比较在保持多项式固定为 p_κ 的同时变化 NEWTON-SCHULZ 步数 $q \in \{1, 2, 3\}$ ，并将这些与带SVD的μ子和SGD-M进行对比。所有其他组件（架构、调度、增强）在所有方法中保持相同，以确保公平比较。

G ADDITIONAL NUMERICAL EXPERIMENTS

Optimizers compared:

- SGD with Momentum
- Muon (Newton–Schulz) with 1, 2, 3 steps per update
- Muon with exact SVD (polar factor)

Dataset and Model:

- MNIST (60K) / MLP (0.5M)
- CIFAR-10 (50K) / CifarNet (2M) : Main text
- CIFAR-100 (50K) / ResNet-18 (11.2M)
- Tiny-ImageNet (100K) / WideResNet-28-10 (36.6M)
- FineWeb (10M tokens from sample-10BT) / NanoGPT (124.2M) & GPT-2 (1.3B)
 - block_size = 1024
 - num_blocks = (10M - 1) // 1024 = 9,765 sequences (for causal LM)
 - n_layer: 12 & 24 (transformer blocks)
 - n_head: 12 & 16
 - n_embd: 768 & 2048
 - Vocab size: tokenizer.vocab_size from GPT2TokenizerFast (50257)
 - Token embedding: nn.Embedding(vocab_size, 768)
 - Positional embedding: nn.Embedding(block_size, 768)
 - Each of the 12 transformer blocks
 - * LayerNorm → multi-head causal self-attention (QKV + output projection) → residual
 - * LayerNorm → 4×-wide MLP (3072 hidden) → residual
 - Final LayerNorm
 - Total n_params: 124.2M & 1313.63M (1.31B)

Hyper-parameters

- **MLP on MNIST:**
 - model: 784 → 512 → 256 → 10
 - learning rate: 0.08; momentum: 0.7
 - 256 batch size; 50 epochs, 5 runs
- **ResNet-18 on CIFAR-100:**
 - model: torchvision.models.resnet18
 - learning rate: 0.08; momentum: 0.7
 - 512 batch size; 50 epochs, 5 runs
- **WideResNet-28-10 on Tiny-ImageNet:**
 - model: WideResNet(depth=28, widen_factor=10, num_classes=200, drop_rate=0.0)
 - learning rate: 0.08; momentum: 0.7
 - 128 batch size; 30 epochs, 3 runs
- **NanoGPT & GPT-1.3B on FineWeb:**
 - model: NanoGPT & GPT-1.3B
 - learning rate: 0.02; momentum: 0.95; batch size: 8
 - 10M training tokens, 1M validation tokens
 - 8 RTX-3090 GPUs
 - max steps: 6000

G 额外的数值实验

比较的优化器:

- 带动量的SGD
- μ子（Newton–Schulz）每次更新1、2、3步
- μ子使用精确SVD（极因子）

数据集和模型:

- MNIST (60K) / MLP (0.5M)
- CIFAR-10 (50K) / CifarNet (2M) : 主文本
- CIFAR-100 (50K) / ResNet-18 (11.2M)
- Tiny-ImageNet (100K) / WideResNet-28-10 (36.6M)
- FineWeb (10M tokens from sample-10BT)/ NanoGPT(124.2M) & GPT-2 (1.3B)– 块大小 = 1024– 块数 _ = (10M- 1)// 1024 = 9,765 序列 (因果LM)– n_层: 12 & 24 (transformer块)– n 头: 12 & 16– n 嵌入: 768 & 2048– 词汇量: tokenizer.vocab_大小fromGPT2TokenizerFast (50257)– 词嵌入: nn.Embedding(vocab_大小, 768)– 位置嵌入: nn.Embedding块_大小,768)– 12 个 transformer 块 * LayerNorm → 多头因果自注意力 (QKV + 输出投影) →残差 * LayerNorm → 4×-宽 MLP (3072 隐藏) → 残差– 最终 LayerNorm– 总 n_参数: 124.2M & 1313.63M(1.31B)

超参数

- **MLP on MNIST:**
 - 模型: 784 → 512 → 256 → 10– 学习率: 0.08; 动量: 0.7– 256 批量大小; 50 个周期, 5 次运行
- **ResNet-18 on CIFAR-100:** – 模型: torchvision.models.resnet18– 学习率: 0.08; 动量: 0.7– 512 批量大小; 50 个周期, 5 次运行
- **WideResNet-28-10 on Tiny-ImageNet:** – 模型: WideResNet(depth=28, widen_factor=10, num_classes=200, drop_rate=0.0)– 学习率: 0.08; 动量: 0.7– 128 批量大小; 30 个周期, 3 次运行
- **NanoGPT & GPT-1.3B on FineWeb:** – 模型: NanoGPT & GPT-1.3B– 学习率: 0.02; 动量: 0.95; 批量大小: 8– 10M 训练 token, 1M 验证 token– 8 RTX-3090 GPU– 最大步数: 6000

G.1 MLP ON MNIST

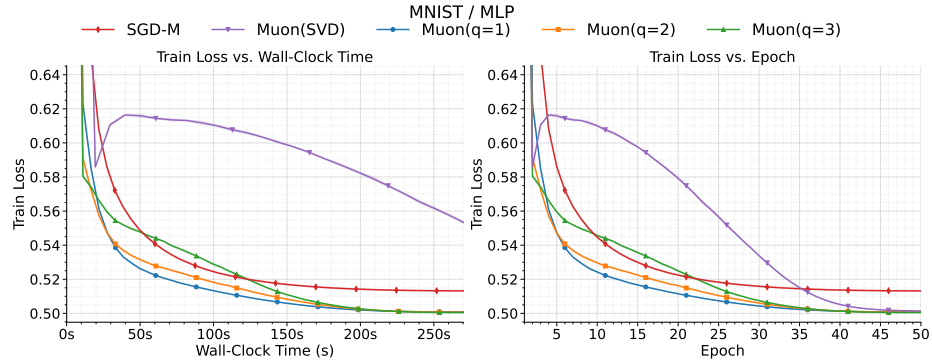


Figure 2: Train losses of MLP on MNIST across wall-clock time and epochs

Table 2: Wall-clock time training performance of MLP (0.5M) on MNIST dataset

Optimizer	Wall-clock time train loss				
	50 sec	100 sec	150 sec	200 sec	250 sec
SGD-M	0.550	0.525	0.517	0.514	0.513
Muon with SVD	0.616	0.612	0.600	0.583	0.565
Muon ($q=1$)	0.526	0.514	0.506	0.502	0.501
Muon ($q=2$)	0.532	0.518	0.509	0.503	0.501
Muon ($q=3$)	0.548	0.529	0.511	0.503	0.501

G.2 CIFARNet ON CIFAR-10

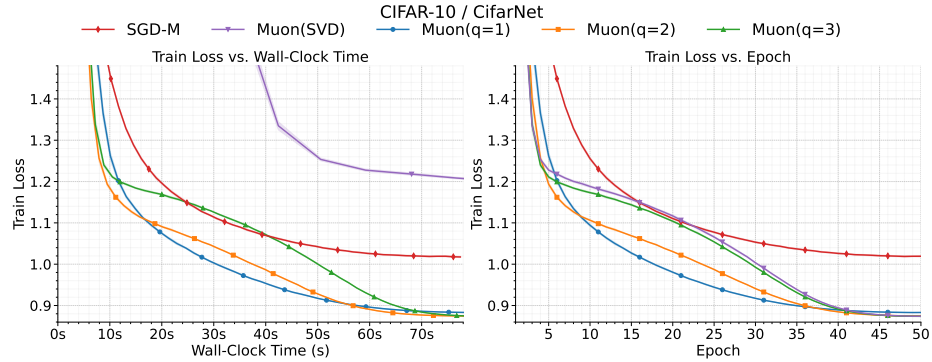


Figure 3: Train losses of CifarNet on CIFAR-10 across wall-clock time and epochs

Table 3: Wall-clock time training performance of CifarNet (2M) on CIFAR-10

Optimizer	Wall-clock time train loss					
	10 sec	20 sec	30 sec	40 sec	50 sec	60 sec
SGD-M	1.366	1.157	1.084	1.045	1.021	1.010
Muon with SVD	2.211	1.334	1.227	1.207	1.193	1.182
Muon ($q=1$)	1.130	1.014	0.945	0.905	0.888	0.884
Muon ($q=2$)	1.129	1.064	1.001	0.932	0.888	0.876
Muon ($q=3$)	1.191	1.145	1.079	0.990	0.905	0.876

G.1 MLP ON MNIST

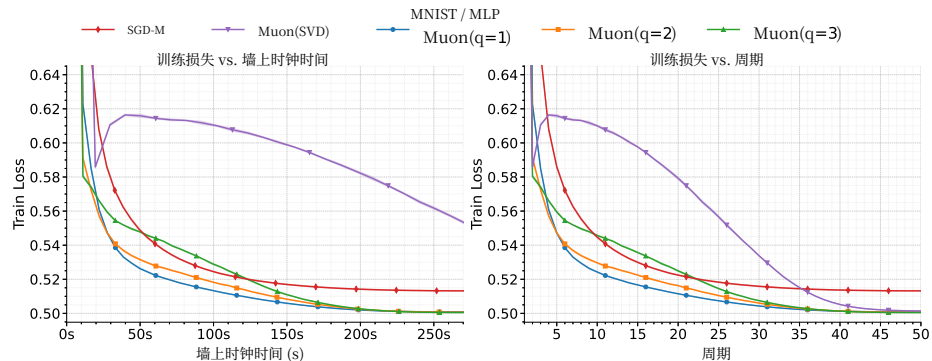


图2：在MNIST上，MLP的训练损失随墙上时钟时间和周期的变化

表2：MLP (0.5M) 在MNIST数据集上的墙上时钟时间训练性能

优化器	时钟时间训练损失				
	50秒	100秒	150秒	200秒	250秒
SGD-M	0.550	0.525	0.517	0.514	0.513
带SVD的 μ 子	0.616	0.612	0.600	0.583	0.565
μ 子 ($q=1$)	0.526	0.514	0.506	0.502	0.501
μ 子 ($q=2$)	0.532	0.518	0.509	0.503	0.501
μ 子 ($q=3$)	0.548	0.529	0.511	0.503	0.501

G.2 CifarNet CIFAR-10

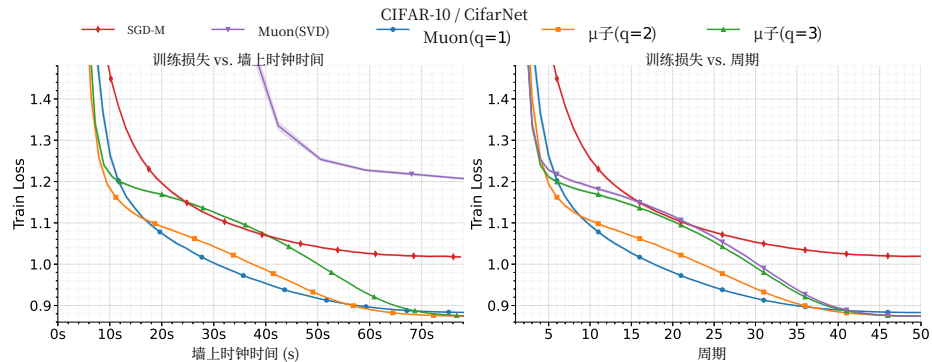


图3：CifarNet在CIFAR-10上的训练损失随墙上时钟时间和周期变化

表3：CifarNet (2M) 在CIFAR-10上的墙上时钟时间训练性能

优化器	墙上时钟时间训练损失					
	10秒	20秒	30秒	40秒	50秒	60秒
SGD-M	1.366	1.157	1.084	1.045	1.021	1.010
带SVD的 μ 子	2.211	1.334	1.227	1.207	1.193	1.182
μ 子 ($q=1$)	1.130	1.014	0.945	0.905	0.888	0.884
μ 子 ($q=2$)	1.129	1.064	1.001	0.932	0.888	0.876
μ 子 ($q=3$)	1.191	1.145	1.079	0.990	0.905	0.876

G.3 RESNET-18 ON CIFAR-100

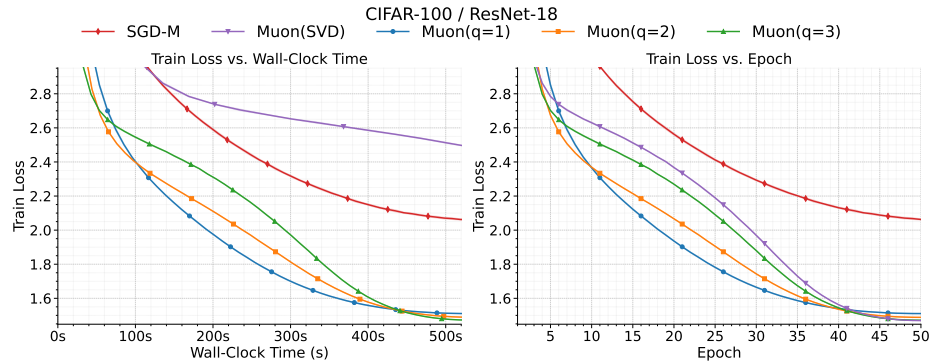


Figure 4: Train losses of ResNet-18 on CIFAR-100 across wall-clock time and epochs

Table 4: Wall-clock time training performance of ResNet-18 (11.2M) on CIFAR-100

Optimizer	Wall-clock time train loss				
	100 sec	200 sec	300 sec	400 sec	500 sec
SGD-M	3.096	2.612	2.324	2.157	2.072
Muon with SVD	3.202	2.767	2.668	2.594	2.522
Muon ($q=1$)	2.427	1.986	1.716	1.563	1.514
Muon ($q=2$)	2.407	2.127	1.826	1.584	1.495
Muon ($q=3$)	2.554	2.329	1.985	1.624	1.481

G.4 WIDERESNET-28-10 ON TINY-IMAGENET

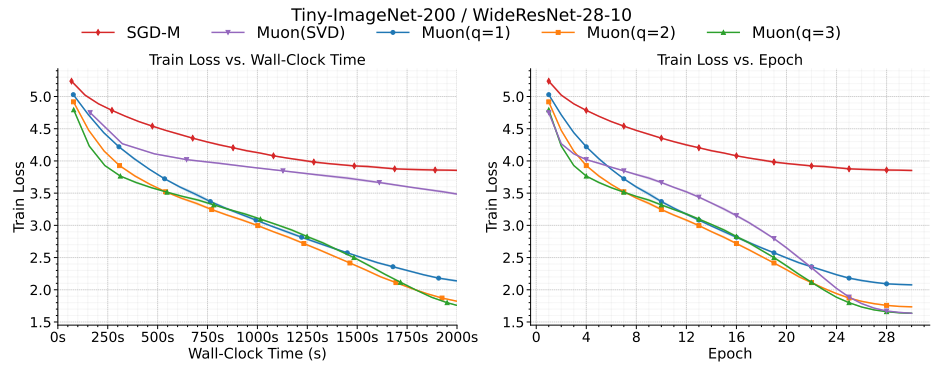


Figure 5: Train losses of WideResNet-28-10 on Tiny-ImageNet across wall-clock time and epochs

Table 5: Wall-clock time training performance of WideResNet-28-10 (36.6M) on Tiny-ImageNet

Optimizer	Wall-clock time train loss				
	400 sec	800 sec	1200 sec	1600 sec	2000 sec
SGD-M	4.694	4.299	4.045	3.914	3.857
Muon with SVD	4.267	4.020	3.846	3.734	3.520
Muon ($q=1$)	4.035	3.369	2.903	2.495	2.144
Muon ($q=2$)	3.766	3.246	2.815	2.311	1.856
Muon ($q=3$)	3.666	3.317	2.929	2.372	1.800

G.3 RESNET-18 ON CIFAR-100

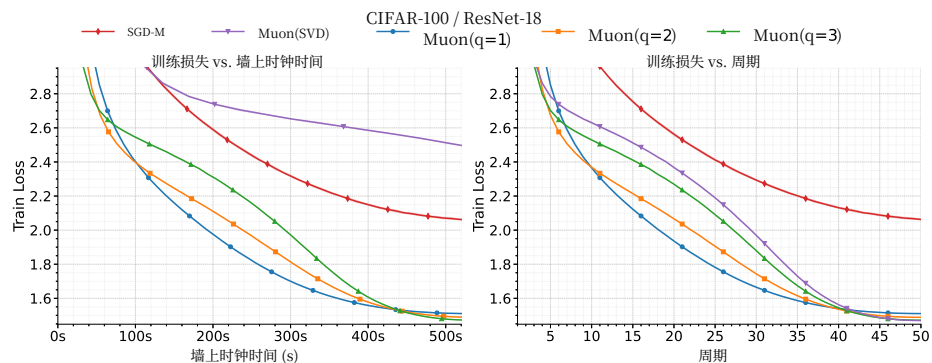


图4: ResNet-18在CIFAR-100上的训练损失随墙上时钟时间和周期变化

表4: ResNet-18 (11.2M) 在 CIFAR-100 上的墙上时钟时间训练性能

优化器	时钟时间训练损失				
	100秒	200秒	300秒	400秒	500秒
SGD-M	3.096	2.612	2.324	2.157	2.072
带SVD的μ子	3.202	2.767	2.668	2.594	2.522
μ子 ($q=1$)	2.427	1.986	1.716	1.563	1.514
μ子 ($q=2$)	2.407	2.127	1.826	1.584	1.495
MUON ($q=3$)	2.554	2.329	1.985	1.624	1.481

G.4 WideResNet-28-10ON Tiny-ImageNet

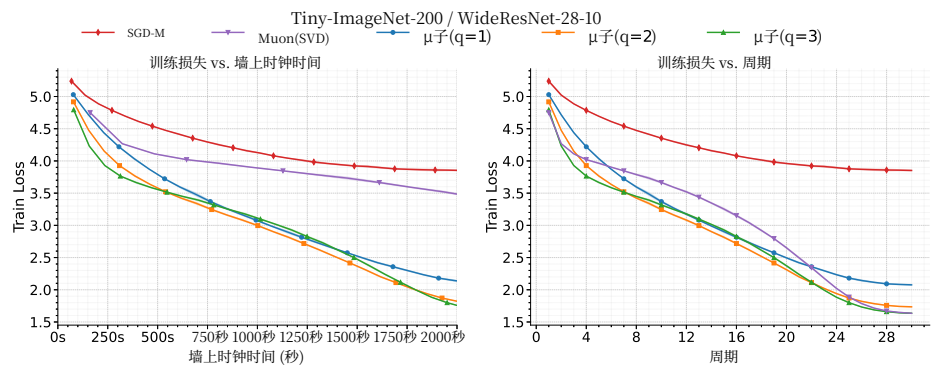


图 5: WideResNet-28-10 在 Tiny-ImageNet 上的训练损失随墙上时钟时间和周期变化

Ta表 5: WideResNet-28-10 (36.6M) 在 Tiny-ImageNet 上的墙上时钟时间训练性能

优化器	墙上时钟时间训练损失				
	400秒	800秒	1200秒	1600秒	2000秒
SGD-M	4.694	4.299	4.045	3.914	3.857
带SVD的μ子	4.267	4.020	3.846	3.734	3.520
μ子 ($q=1$)	4.035	3.369	2.903	2.495	2.144
μ子 ($q=2$)	3.766	3.246	2.815	2.311	1.856
μ子 ($q=3$)	3.666	3.317	2.929	2.372	1.800

G.5 NANO GPT ON FINEWEB

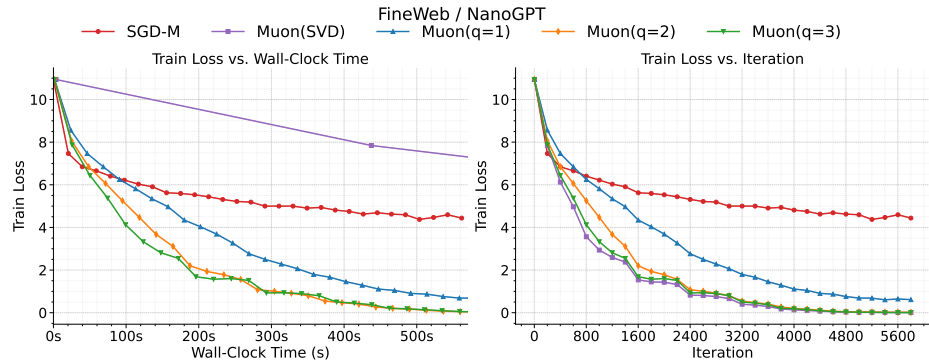


Figure 6: Train losses of NanoGPT on FineWeb across wall-clock time and epochs

Table 6: Wall-clock time training performance of NanoGPT (124M) on FineWeb

Optimizer	Wall-clock time train loss					
	100 sec	200 sec	300 sec	400 sec	500 sec	600 sec
SGD-M	10.938	7.656	6.000	5.375	4.938	4.594
Muon with SVD	10.250	9.188	8.625	8.062	7.594	7.125
Muon ($q=1$)	5.969	4.094	2.469	1.367	0.879	0.637
Muon ($q=2$)	5.000	2.344	1.047	0.395	0.157	0.030
Muon ($q=3$)	3.984	2.125	1.180	0.523	0.142	0.026

G.6 GPT-2 BASED MODEL (1.3B) ON FINEWEB

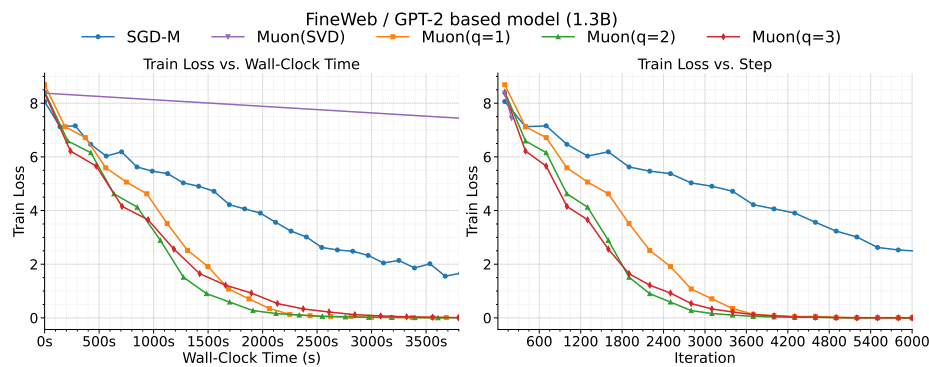


Figure 7: Train losses of GPT-2 based model (1.3B) on FineWeb across wall-clock time and epochs

Table 7: Wall-clock time training performance of GPT-2 based model (1.3B) on FineWeb

Optimizer	Wall-clock time train loss					
	600 sec	1200 sec	1800 sec	2400 sec	3000 sec	3600 sec
SGD-M	6.4062	5.3750	4.2812	3.2969	2.4219	2.0156
Muon with SVD	8.2280	8.0810	7.9340	7.7870	7.6400	7.4930
Muon ($q=1$)	6.0625	3.5156	1.0781	0.1436	0.1338	0.0109
Muon ($q=2$)	5.1562	2.8906	0.5898	0.1133	0.0349	0.0113
Muon ($q=3$)	5.6562	3.2031	1.2188	0.4609	0.1201	0.0286

G.5 NanoGPT FineWeb

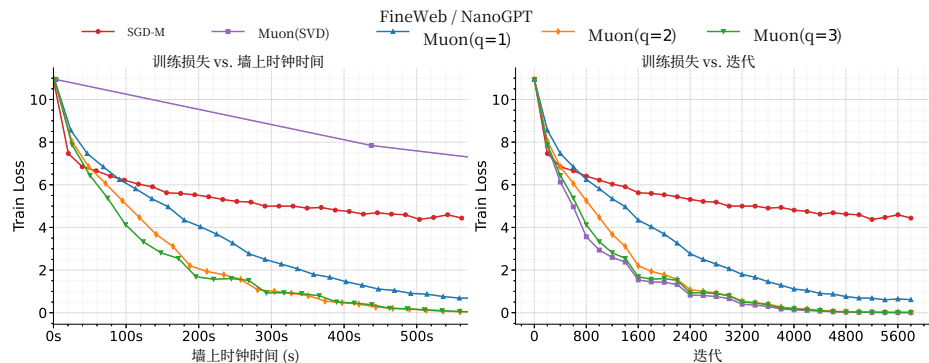


图6: NanoGPT在FineWeb上按墙上时钟时间和周期变化的训练损失

表6: NanoGPT（124M）在FineWeb上的墙上时钟时间训练性能

优化器	墙上时钟时间训练损失					
	100秒	200秒	300秒	400秒	500秒	600秒
SGD-M	10.938	7.656	6.000	5.375	4.938	4.594
带SVD的μ子	10.250	9.188	8.625	8.062	7.594	7.125
μ子 ($q=1$)	5.969	4.094	2.469	1.367	0.879	0.637
μ子 ($q=2$)	5.000	2.344	1.047	0.395	0.157	0.030
μ子 ($q=3$)	3.984	2.125	1.180	0.523	0.142	0.026

G.6 基于 GPT-2 的模型(1.3B) FineWeb

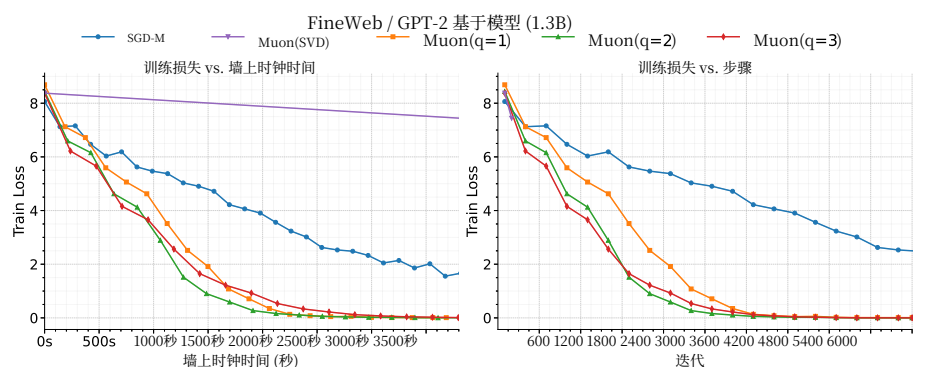


Figure 7: 基于 GPT-2 的模型 (1.3B) FineWeb 在墙上时钟时间和周期上的训练损失

表 7: 基于 GPT-2 的模型 (1.3B) FineWeb 的墙上时钟时间训练性能

优化器	墙上时钟时间训练损失					
	600秒	1200秒	1800秒	2400秒	3000秒	3600秒
SGD-M	6.4062	5.3750	4.2812	3.2969	2.4219	2.0156
带SVD的μ子	8.2280	8.0810	7.9340	7.7870	7.6400	7.4930
μ子 ($q=1$)	6.0625	3.5156	1.0781	0.1436	0.1338	0.0109
μ子 ($q=2$)	5.1562	2.8906	0.5898	0.1133	0.0349	0.0113
μ子 ($q=3$)	5.6562	3.2031	1.2188	0.4609	0.1201	0.0286

H ADDITIONAL ABLATION EXPERIMENTS

H.1 NEWTON-SCHULZ-POLYNOMIAL DEGREE- κ ABLATIONS.

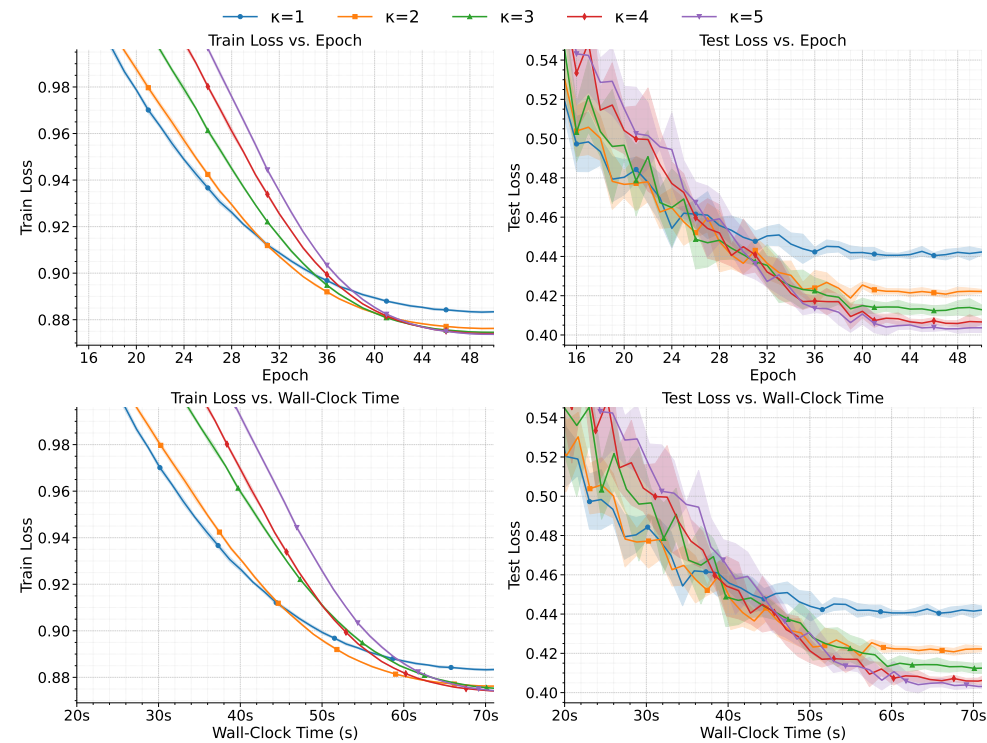


Figure 8: **Degree κ sweep.** NEWTON-SCHULZ polynomial degree $\kappa \in \{1, \dots, 5\}$ at fixed $q = 3$. Larger κ improves train/test loss but increases time per step.

Beyond the step-sweep in Fig. 1, we perform a controlled ablation that varies the *degree* κ of the NEWTON-SCHULZ polynomial while fixing the number of NEWTON-SCHULZ steps to $q = 3$ for all variants. Concretely, we instantiate a family of MUON with degree- κ polynomial optimizers whose orthogonalization step applies, per layer, the update

$$X \leftarrow p_\kappa(X X^\top) X, \quad X = \frac{M}{\|M\|_F}$$

where M is the momentum matrix, and $p_\kappa(\lambda) = \sum_{m=0}^{\kappa} c_m \lambda^m$ is the degree- κ NEWTON-SCHULZ polynomial that matches the derivatives of $\lambda^{-1/2}$ at $\lambda = 1$ up to order κ . The coefficients are generated analytically from the Taylor series of $\lambda^{-1/2}$ at 1, ensuring the residual recursion $\delta_{j+1} \leq (\delta_j)^{\kappa+1}$ (Lemma 3). Tall matrices are handled using the transpose trick.

Ablation on degree κ . Fixing $q = 3$, increasing the degree $\kappa \in \{1, \dots, 5\}$ improves optimization (loss drops faster at a fixed epoch) but lengthens each step, yielding a clear accuracy-time trade-off (see Fig. 8). (This mirrors the theory that the residual contracts as $\delta_{j+1} \leq \delta_j^{\kappa+1}$ (Lemma 3), while computation scales with polynomial evaluations.)

Compared optimizers. We evaluate $\kappa \in \{1, 2, 3, 4, 5\}$ under a fixed iteration budget $q = 3$. This ablation isolates the effect of the polynomial degree κ at a fixed step q , directly testing the theory-driven prediction that larger κ accelerates orthogonality residual decay (see Table 1).

H 额外消融实验

H.1 牛顿-舒尔茨多项式阶- κ 消融。

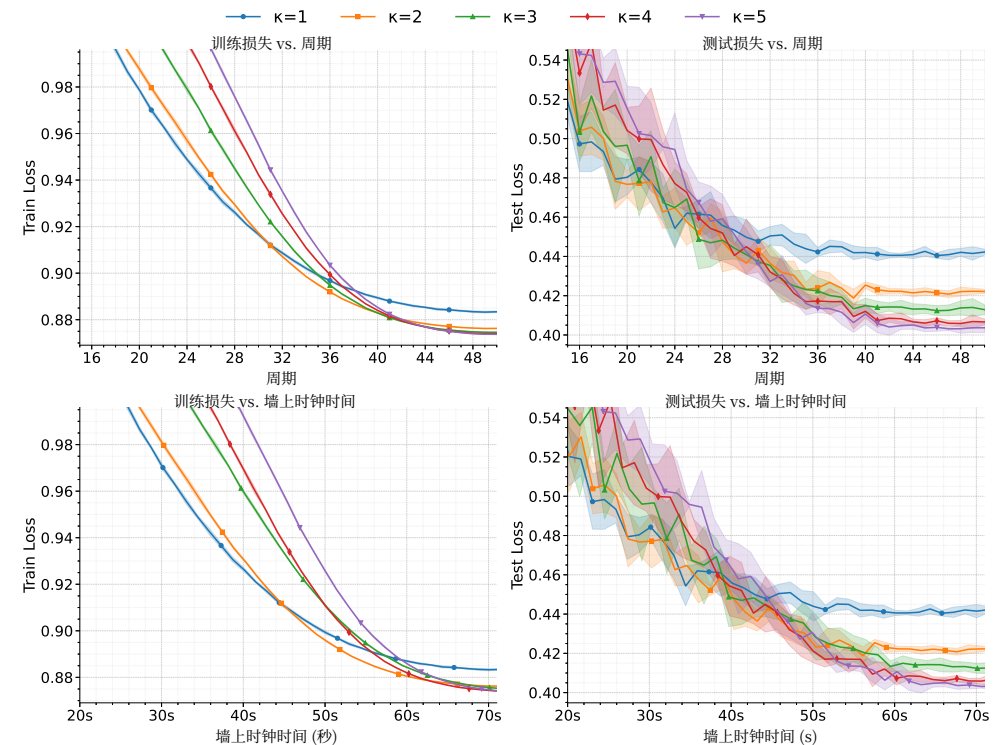


图 8: **阶 κ 扫描。** 牛顿-舒尔茨多项式阶 $\kappa \in \{1, \dots, 5\}$ 在固定 $q = 3$ 时。更大的 κ 提高了训练/测试损失，但增加了每步时间。

在图1所示的步进扫描之外，我们进行了一种受控的消融实验，该实验会改变NEWTON-SCHULZ多项式的阶数 κ ，同时将NEWTON-SCHULZ步数固定为 $q = 3$ ，所有变体均如此。具体来说，我们实例化了一族MUON，其阶数为- κ 的多项式优化器，其正交化步骤按层应用更新

$$X \leftarrow p_\kappa(X X^\top) X, \quad X = \frac{M}{\|M\|_F}$$

其中 M 是动量矩阵，而 $p_\kappa(\lambda) = \sum_{m=0}^{\kappa} c_m \lambda^m$ 是阶数为- κ 的NEWTON-SCHULZ多项式，该多项式与 $\lambda^{-1/2}$ 在 $\lambda = 1$ 处的导数（最高阶为 κ ）相匹配。系数通过从1处的 $\lambda^{-1/2}$ 泰勒级数解析生成，确保残差递归 $\delta_{j+1} \leq (\delta_j)^{\kappa+1}$ （引理3）。对于大矩阵，使用转置技巧进行处理。

消融实验针对阶数 κ 。 固定 $q = 3$ ，增加阶数 $\kappa \in \{1, \dots, 5\}$ 可以提高优化效果（在固定周期下损失下降更快），但会延长每一步的执行时间，从而产生明显的精度-时间权衡（参见图8）。（这与理论一致：残差会随着 $\delta_{j+1} \leq \delta_j^{\kappa+1}$ 收缩（引理3），而计算量会随着多项式求值次数增加。）

对比优化器。 我们在固定的迭代预算 $q = 3$ 下评估 $\kappa \in \{1, 2, 3, 4, 5\}$ 。这项消融实验隔离了多项式阶数 κ 在固定步长 q 时的效果，直接验证了理论预测：更大的 κ 加速正交残差衰减（见表1）。

H.2 RANK-DEPENDENCE.

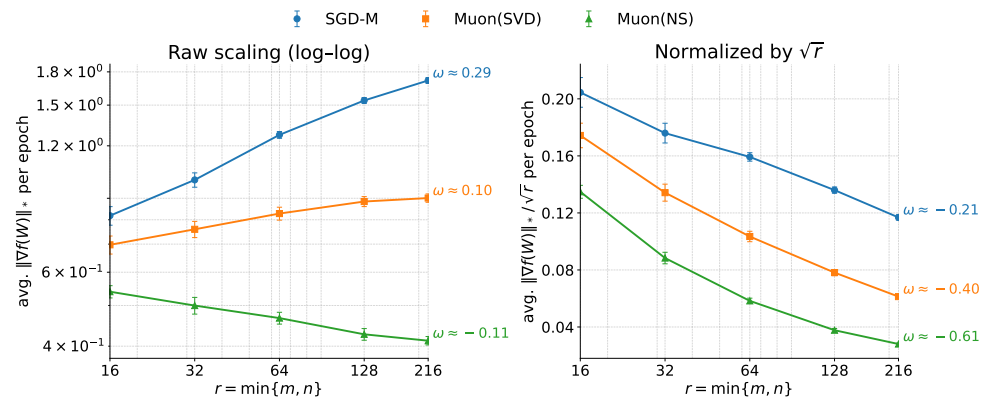


Figure 9: **Rank dependence.** Epoch-averaged gradient norm vs. rank r (left: raw log-log; right: normalized by $\sqrt{r}$). MUON variants are nearly r -invariant; SGD-M scales up with rank.

Table 8: **log-log slopes** (ω) from Fig. 9.

Method	ω (raw)	ω (normalized by $\sqrt{r}$)
SGD-M	0.292	-0.208
MUON (SVD)	0.102	-0.398
MUON (NS)	-0.106	-0.606

Rank dependence. We vary the monitored layer’s effective rank $r \in \{16, 32, 64, 128, 216\}$ and plot the epoch-averaged $\|\nabla f(W)\|_*$ on a log-log scale. SGD-M shows a positive slope ≈ 0.3 (grows with r), whereas MUON and its variants are nearly flat. After normalizing by $\sqrt{r}$, the two MUON variants follow the predicted $r^{-1/2}$ trend, while SGD-M remains non-flat (see Fig. 9).

Goal and prediction. We test how the nuclear-norm of the per-step gradient on a monitored layer scales with the layer’s dimension parameter $r := \min\{m, n\}$ for a weight $W \in \mathbb{R}^{m \times n}$. The theory predicts that orthogonalizing the momentum suppresses the r -growth which occurs in either MUON with NEWTON-SCHULZ or MUON with SVD, while SGD-M has $\sqrt{r}$ dependence.

Controlling r . We vary the out-channels of the first convolution in the first ConvGroup and keep all other widths fixed. For this layer, the weight tensor has shape $[\text{out}, \text{in}, 3, 3]$ with $\text{in} = 24$ (from the fixed "whitening" 2×2 conv). Flattening by rows yields a matrix of shape $m \times n$ with $m = \text{out}$ and $n = \text{in} \cdot 3 \cdot 3 = 24 \cdot 9 = 216$, so $r = \min\{\text{out}, 216\}$. We sweep $\text{out} \in \{16, 32, 64, 128, 216\}$, hence $r \in \{16, 32, 64, 128, 216\}$.

Optimizers under comparison. We compare:

- **SGD-M** (baseline),
- **MUON (SVD)** (exact polar $UV^\top$), and
- **MUON (NS)** with $q = 3$ Newton-Schulz steps.

All methods share identical schedules and auxiliary updates (Appendix F.1).

Results. Across $r \in \{16, 32, 64, 128, 216\}$, SGD-M exhibits a positive log-log slope (about 0.3), while both MUON variants are nearly flat. After dividing by $\sqrt{r}$, the MUON curves show a slope of about -0.5 (MUON with SVD: -0.4 and MUON with NEWTON-SCHULZ: -0.61), as predicted, whereas SGD-M becomes almost flat (slope with -0.21).

H.2 排名依赖性。

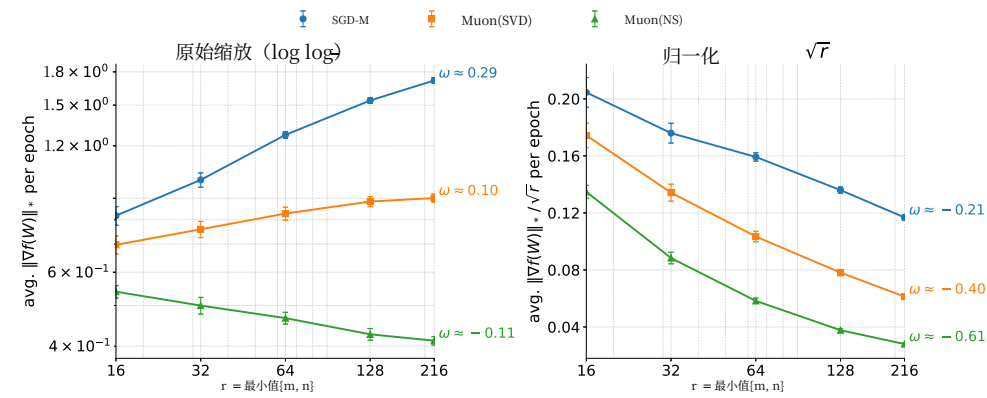


图9: **秩依赖性。** 周期平均梯度范数与秩 r 的关系 (左: 原始对数-对数; 右: 归一化于 $\sqrt{r}$)。MUON变体几乎 $\sqrt{r}$ -不变; SGD-M随秩增加而扩展。

表8: **对数-对数斜率** (ω)来自图9。

方法	ω (原始)	ω (归一化于 $\sqrt{r}$)
SGD-M	0.292	-0.208
MUON (SVD)	0.102	-0.398
MUON (NS)	-0.106	-0.606

秩依赖性。 我们改变监控层的有效秩 $r \in \{16, 32, 64, 128, 216\}$ 并绘制 epoch 平均的 $\|\nabla f(W)\|_*$ 在对数-对数尺度上。SGD-M 显示出正斜率 ≈ 0.3 (随着 r 增长), 而 MUON 及其变体则几乎平坦。在用 $\sqrt{r}$ 归一化后, 两个 MUON 变体遵循预测的 $r^{-1/2}$ 趋势, 而 SGD-M 仍然非平坦 (见图 9)。

目标与预测。 我们测试在监控层上每一步梯度的核范数如何随层的维度参数 $r := \min\{m, n\}$ 对权重 $W \in \mathbb{R}^{m \times n}$ 进行缩放。理论预测, 正交化动量会抑制在带 NEWTON-SCHULZ 的 μ 子 或带 SVD 的 μ 子 中发生的 r 增长, 而 SGD-M 具有 $\sqrt{r}$ 依赖性。

控制 r 。 我们改变第一组卷积层中第一个卷积层的输出通道, 并保持其他所有宽度固定。对于这一层, 权重张量的形状为 $[\text{out}, \text{in}, 3, 3]$, 其中 in 来自固定的“白化” 2×2 卷积。按行展平后得到一个形状为 $m \times n$ 的矩阵, 包含 $m = \text{out}$ 和 $n = \text{in} \cdot 3 \cdot 3 = 24 \cdot 9 = 216$, 因此 $r = \min\{\text{out}, 216\}$ 。我们扫描 $\text{out} \in \{16, 32, 64, 128, 216\}$, 因此 $r \in \{16, 32, 64, 128, 216\}$ 。

正在比较的优化器。 我们比较:

- **SGD-M**(基线)、
- **MUON (SVD)** (精确极 $UV^\top$), 和
- **MUON (NS)**带有 $q = 3$ Newton-Schulz 步数。

所有方法共享相同的调度和辅助更新 (附录F.1)。

结果。 在 $r \in \{16, 32, 64, 128, 216\}$ 中, SGD-M 表现出正的对数-对数斜率 (约 0.3), 而两个 MUON 变体则几乎平坦。除以 $\sqrt{r}$ 后, MUON 曲线的斜率约为 -0.5 (带 SVD 的 MUON: -0.4, 带 NEWTON-SCHULZ 的 MUON: -0.61), 正如所预测的, 而 SGD-M 则几乎平坦 (斜率为 -0.21)。

H.3 BATCH SIZE B ABLATIONS.

We sweep $B \in \{64, 128, 256, 512, 1024\}$ with MUON ($q = 3$) under identical schedules and report epoch-aligned and time-aligned views (Fig. 10). The schedule is step-based, so larger B implies fewer total steps over E epochs. We report both epoch-aligned and wall-clock-aligned curves.

From a systems perspective, increasing B improves throughput up to a regime of diminishing returns. From an optimization perspective, a larger B reduces gradient noise but also decreases the frequency of orthogonalization steps per epoch ($N_{\text{proj}} = N_{\text{train}}/B$).

In our runs, the best time-to-accuracy is achieved for a large batch size $B = 1024$, while a small B suffers from noise, taking a greater amount of time.

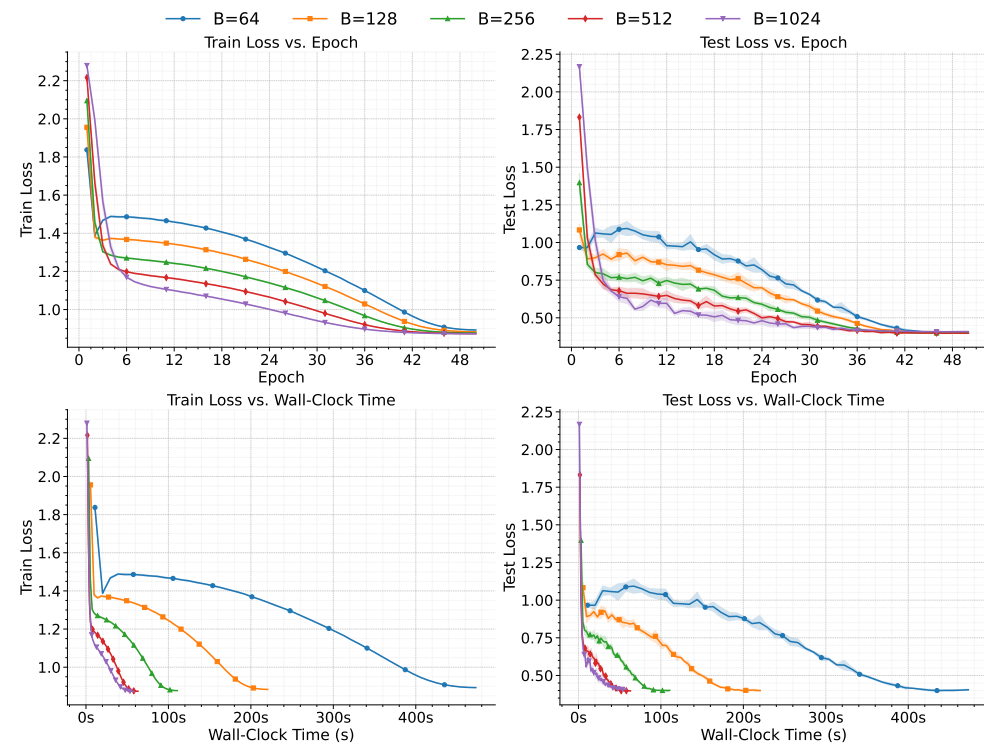


Figure 10: **Batch size.** Train/test loss vs. epoch and wall-clock for $B \in \{64, 128, 256, 512, 1024\}$.

H.3 批量大小 B 删除。

我们使用MUON ($q = 3$) 在相同调度下扫描 $B \in \{64, 128, 256, 512, 1024\}$ ，并报告了周期对齐和时间对齐的视图（图10）。调度是按步长的，因此更大的 B 意味着在 E 个周期内总步数更少。我们报告了周期对齐和时钟对齐的曲线。

从系统角度来看，增加 B 可以提高吞吐量，直到收益递减的阶段。从优化角度来看，更大的 B 减少了梯度噪声，但也降低了每个周期正交化步骤的频率 ($N_{\text{proj}} = N_{\text{train}}/B$)。

在我们的运行中，最佳时间-精度是在大批量大小 $B = 1024$ 下实现的，而小 B 则因噪声问题耗时更长。

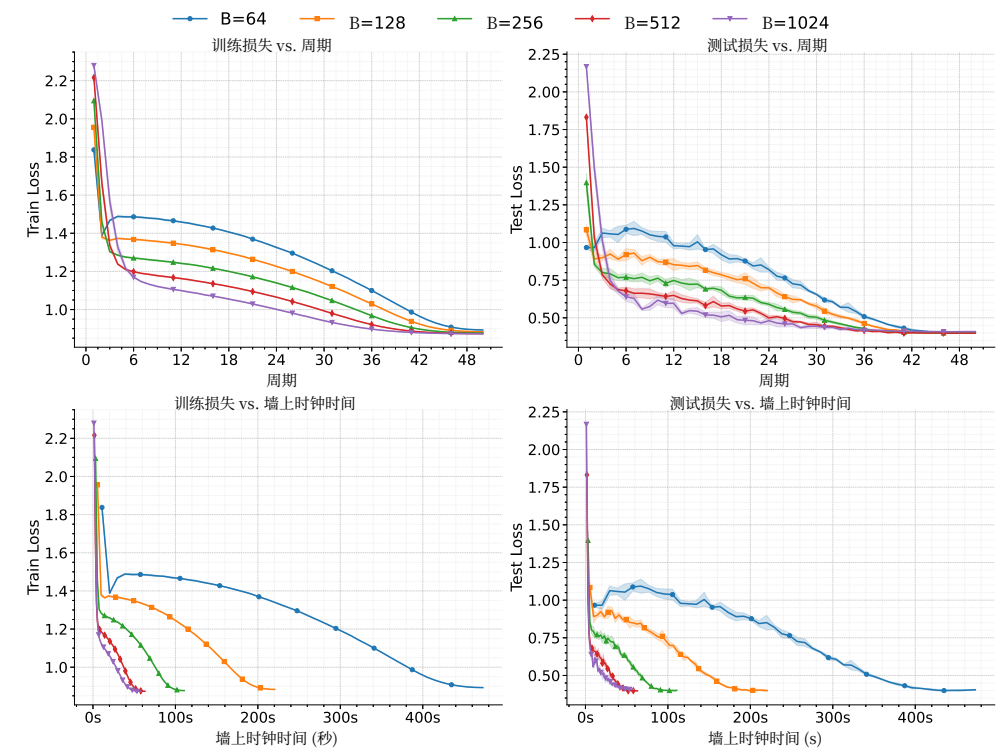


Figure 10: **批量大小。** $B \in \{64, 128, 256, 512, 1024\}$ 的训练/测试损失与周期和时钟的关系。

H.4 DEGREE-2 NS POLYNOMIAL VS. AD-HOC DEGREE-2 NS POLYNOMIAL

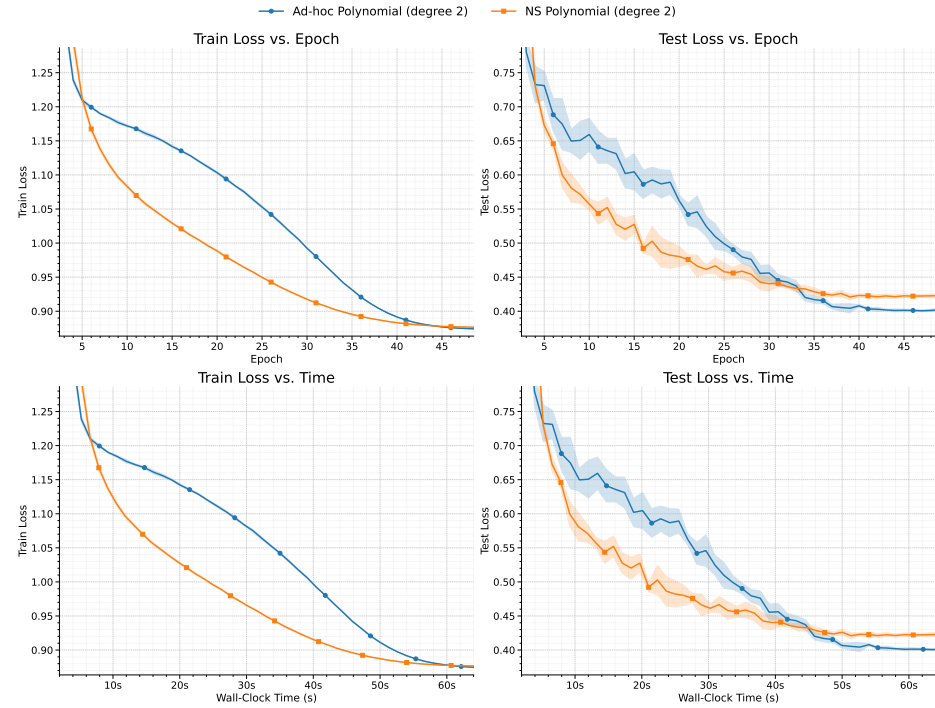


Figure 11: NS polynomial vs. Ad-hoc polynomial. Train/test loss vs. epoch and wall-clock.

Analysis of $p_{\text{ad-hoc}}$

Let $p_{\text{ad-hoc}}(\lambda) = 3.4445 - 4.7750\lambda + 2.0315\lambda^2$ and $\tau(\lambda) = \lambda p(\lambda)^2$.

The statement that $\tau_{\text{ad-hoc}}(\lambda)$ is monotone non-decreasing on $[0, 1]$ is false, and the underlying condition that $p_{\text{ad-hoc}}(\lambda) \in [0, 1]$ is also not met.

On the interval $[0, 1]$, the range of $p(\lambda)$ is $[0.701, 3.4445]$. Since this range is not contained within $[0, 1]$, the premise $p(\lambda) \in [0, 1]$ is false.

A function is monotone non-decreasing if its derivative is greater than or equal to zero over the entire interval. The derivative of $\tau(\lambda)$ is:

$$\tau'_{\text{ad-hoc}}(\lambda) = \frac{d}{d\lambda} \tau_{\text{ad-hoc}}(\lambda) = 20.635\lambda^4 - 77.5824\lambda^3 + 110.3556\lambda^2 - 65.781\lambda + 11.8641$$

To check for monotonicity, we can evaluate the derivative at the endpoints of the interval $[0, 1]$:

- At $\lambda = 0$: $\tau'_{\text{ad-hoc}}(0) = 11.8641$
- At $\lambda = 1$: $\tau'_{\text{ad-hoc}}(1) = 20.635 - 77.5824 + 110.3556 - 65.781 + 11.8641 = -0.5087$

Since $\tau'_{\text{ad-hoc}}(0) > 0$ and $\tau'_{\text{ad-hoc}}(1) < 0$, the derivative changes from positive to negative within the interval. This means the function $\tau_{\text{ad-hoc}}(\lambda)$ increases for a portion of the interval and then decreases.

Therefore, the claim that $\tau_{\text{ad-hoc}}(\lambda)$ is monotone non-decreasing on $[0, 1]$ is false. The function has a local maximum at approximately $\lambda \approx 0.308$.

Hence, the monotonicity premise of Lemma 2 fails for $p_{\text{ad-hoc}}$, even though $\tau(1) = p(1)^2 = 0.701^2 < 1$ holds. This explains why our guarantees apply to p_κ but not to the ad-hoc quadratic, which remains an empirical heuristic.

H.4 次数-2 NS 多项式 vs. 自举次数-2 NS 多项式

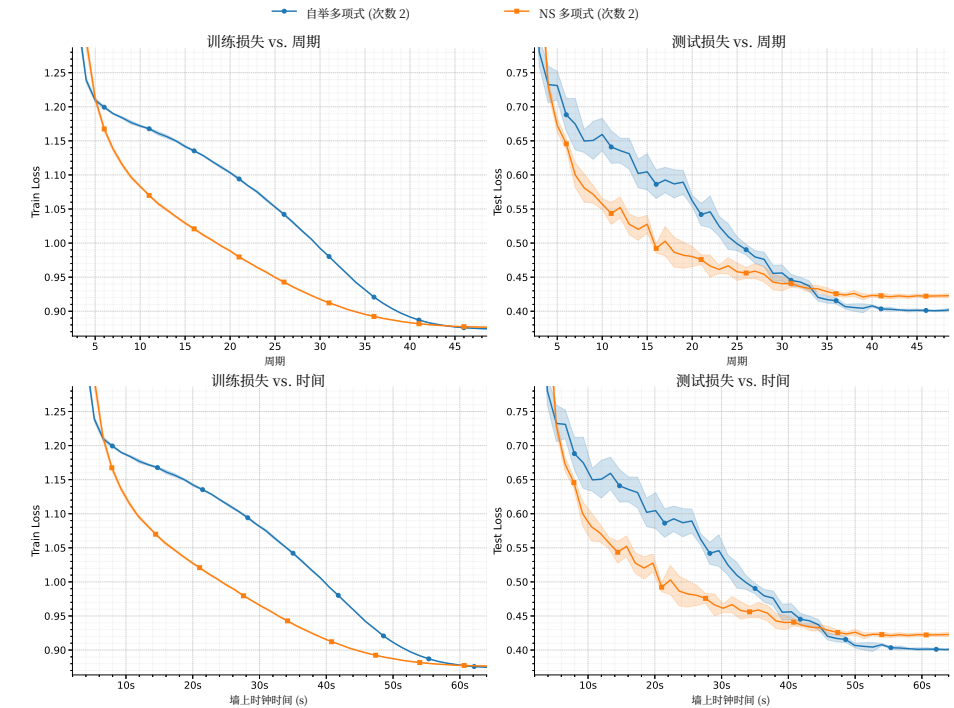


图11: NS多项式与Ad-hoc多项式。训练/测试损失与周期及时钟时间。

对 $p_{\text{ad-hoc}}$ 的分析

Let $p_{\text{ad-hoc}}(\lambda) = 3.4445 - 4.7750\lambda + 2.0315\lambda^2$ and $\tau(\lambda) = \lambda p(\lambda)^2$.

The 声明 $\tau_{\text{ad-hoc}}(\lambda)$ 在 $[0, 1]$ 上单调非递减是错误的, 并且底层协在 $p_{\text{ad-hoc}}(\lambda) \in [0, 1]$ 也不满足。

在区间 $[0, 1]$ 上, $p(\lambda)$ 的范围是 $[0.701, 3.4445]$ 。由于这个范围不包含在 $[0, 1]$, 前提 $p(\lambda) \in [0, 1]$ 中, 所以该前提是错误的。

如果函数在整个区间上的导数大于或等于零, 则该函数是单调非减的。 $\tau(\lambda)$ 的导数是:

$$\tau'_{\text{ad-hoc}}(\lambda) = \frac{d}{d\lambda} \tau_{\text{ad-hoc}}(\lambda) = 20.635\lambda^4 - 77.5824\lambda^3 + 110.3556\lambda^2 - 65.781\lambda + 11.8641$$

要检查单调性, 我们可以在区间的端点 $[0, 1]$ 处计算导数:

- At $\lambda = 0$: $\tau'_{\text{ad-hoc}}(0) = 11.8641$
- At $\lambda = 1$: $\tau'_{\text{ad-hoc}}(1) = 20.635 - 77.5824 + 110.3556 - 65.781 + 11.8641 = -0.5087$

由于 $\tau'_{\text{ad-hoc}}(0) > 0$ 且 $\tau'_{\text{ad-hoc}}(1) < 0$, 导数在区间内从正变为负。这意味着函数 $\tau_{\text{ad-hoc}}(\lambda)$ 在区间的一部分内增加, 然后减少。

因此, $\tau_{\text{ad-hoc}}(\lambda)$ 在 $[0, 1]$ 上单调非减的断言是错误的。该函数在约 $\lambda \approx 0.308$ 处有一个局部最大值。

因此, 引理 $\tau(1) = p(1)^2 = 0.701^2 < 1$ 的单调性前提对 $p_{\text{ad-hoc}}$ 不成立, 尽管 $(\cdot)^2$ 成立。这解释了为什么我们的保证适用于 p_κ , 但不适用于 ad-hoc 二次函数, 后者仍然是一个经验启发式方法。